Volumes

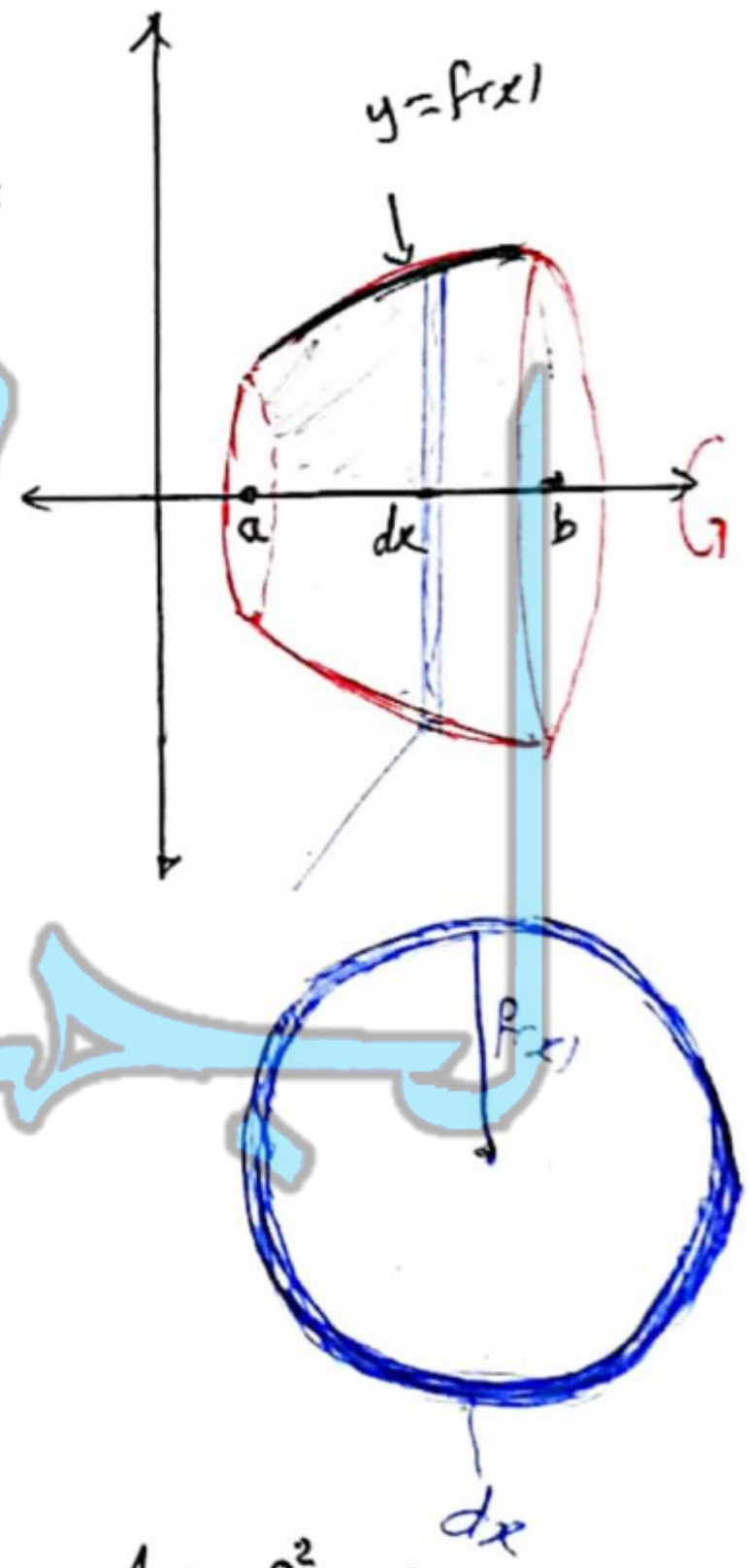
* Disks and Washers Method:

Rotation about x-axis's

Disks

If the region that is bounded by the curve of $y=f(x)$, the x-axis's and the lines $x=a$ and $x=b$ is revolved about the x-axis's complete revolution, then the volume of the solid that results is given by

$$V=\pi\int_a^b f^2(x)\,dx$$



$$dV=\pi f^2(x)\,dx$$

$$V=\pi\int_a^b f^2(x)\,dx$$

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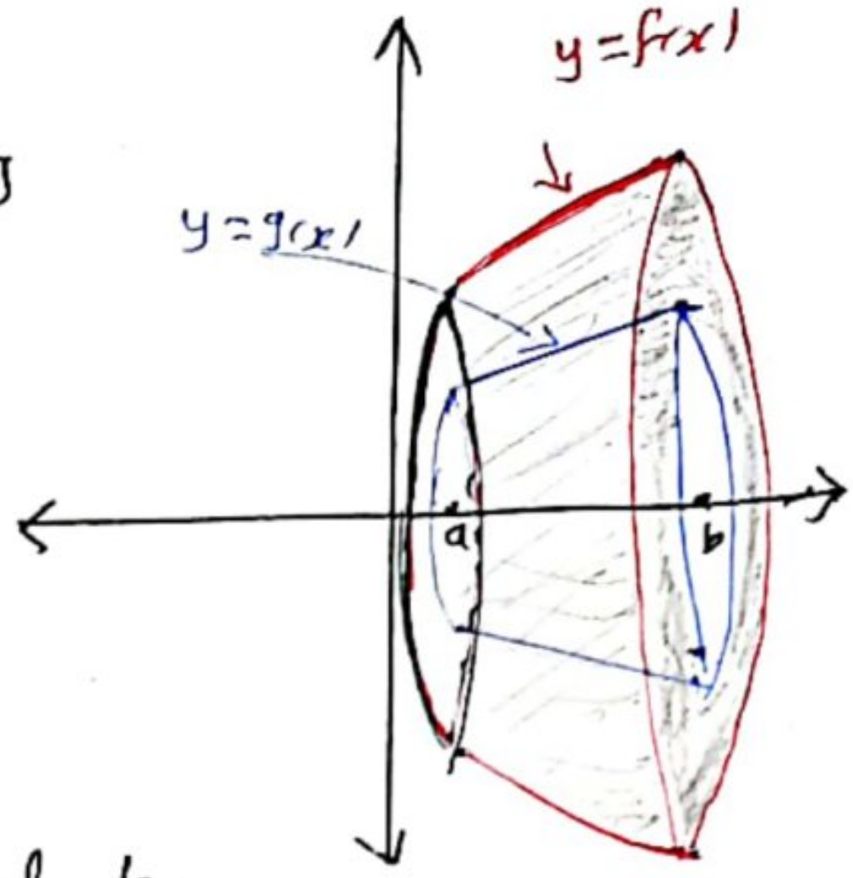
Rotation about x-axis's

Washers

If $f(x) \geq g(x) \geq 0$ or $f(x) \leq g(x) \leq 0$ $\Big\}\ \forall x \in [a,b]$

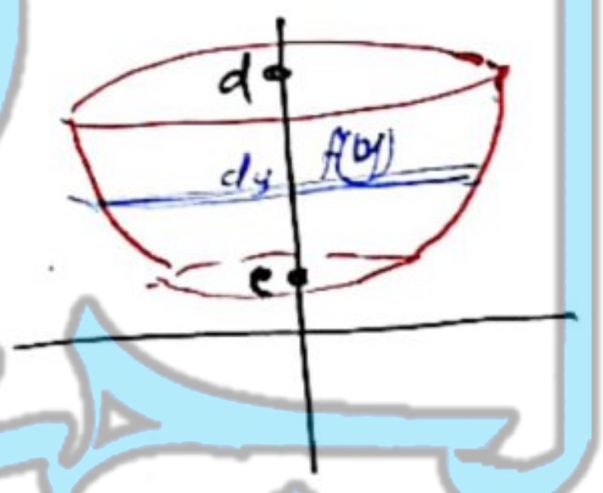


then if the region that is bounded by the curves of $f(x)$ and $g(x)$, $x \in [a,b]$ is rotated about the x-axis's then the volume of the solid that is generated by this method is given by

$$V = \pi \int_a^b (f^2(x) - g^2(x))\,dx$$

* Rotation about the y-axis's

Disks



If the region enclosed by the curve $x = f(y) \geq 0$ or $x = f(y) \leq 0$ $\Big\}\ \forall y \in [c,d]$ is rotated about the y-axis's then the solid that is generated by this method is given by

$$V = \pi \int_c^d f^2(y)\,dy$$

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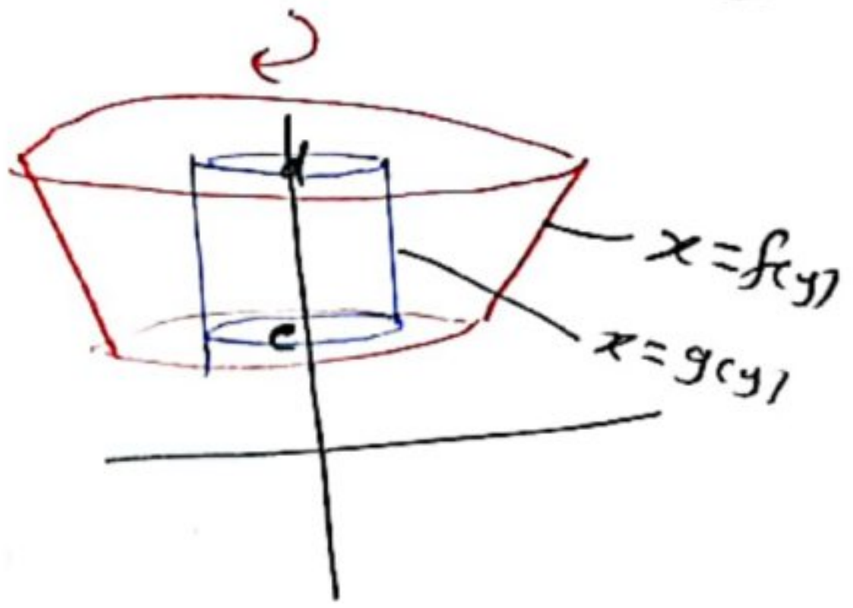
about y

Washers

If $\left.\begin{matrix} f(y) \geq g(y) \geq 0 \\ \text{or } f(y) \leq g(y) \leq 0 \end{matrix}\right\} \forall\, y \in [c,d]$



then the volume of the solid that we obtained when the region enclosed by the curves of $f(y)$, and $g(y)$ $y \in [c,d]$ is rotated a complete revolution about the y-axis, is given by

$$V = \pi \int_c^d (f^2(y) - g^2(y))\, dy$$

e.g.: Find the volume of the solid that is generated when the region of the given curves is rotated about the given ~~curve~~ axis.

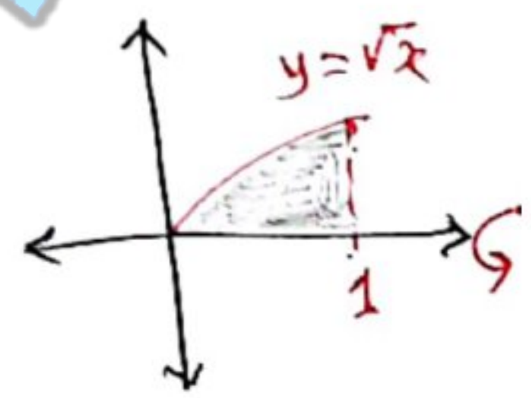
1) $y = \sqrt{x}$, $y = 0$ and $x = 1$ about the x-axis.

$$V = \pi \int_0^1 (\sqrt{x})^2 dx$$

$$V = \pi \int_0^1 x\, dx$$

$$= \pi \left[\frac{x^2}{2}\right]_0^1$$

$$= \frac{1}{2}\pi \text{ c.u} \quad \text{(cubic units)}$$



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[4]

2) $y = \sin x$, $y = \cos x$, $x \in [0, \frac{\pi}{4}]$ about x-axis

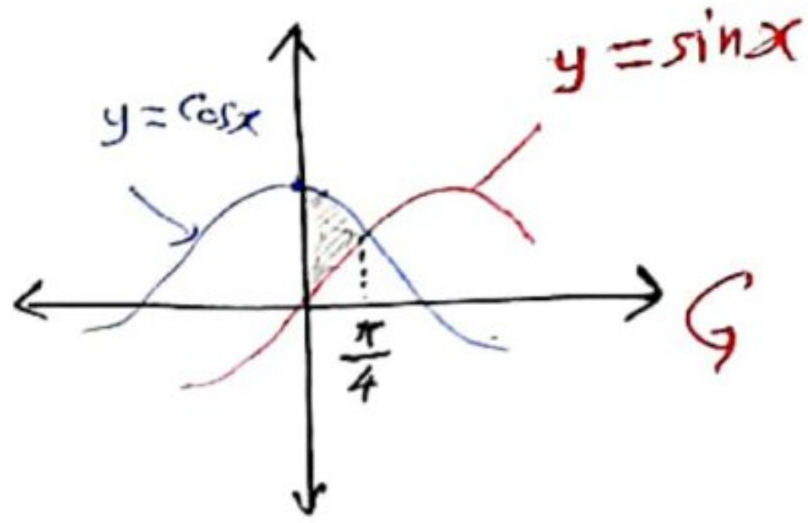
$$V = \pi \int_0^{\pi/4} (\cos^2 x - \sin^2 x)\, dx$$

$$V = \pi \int_0^{\pi/4} \cos 2x\, dx$$

$$= \frac{\pi}{2} \sin 2x \Big|_0^{\pi/4}$$

$$= \frac{\pi}{2} [\sin \frac{\pi}{2} - \sin 0]$$

$$= \frac{\pi}{2} \text{ c.u}$$



3) $y = \frac{1}{x}$, $y = 1$, $y = 2$ and the y-axis. about y-axis

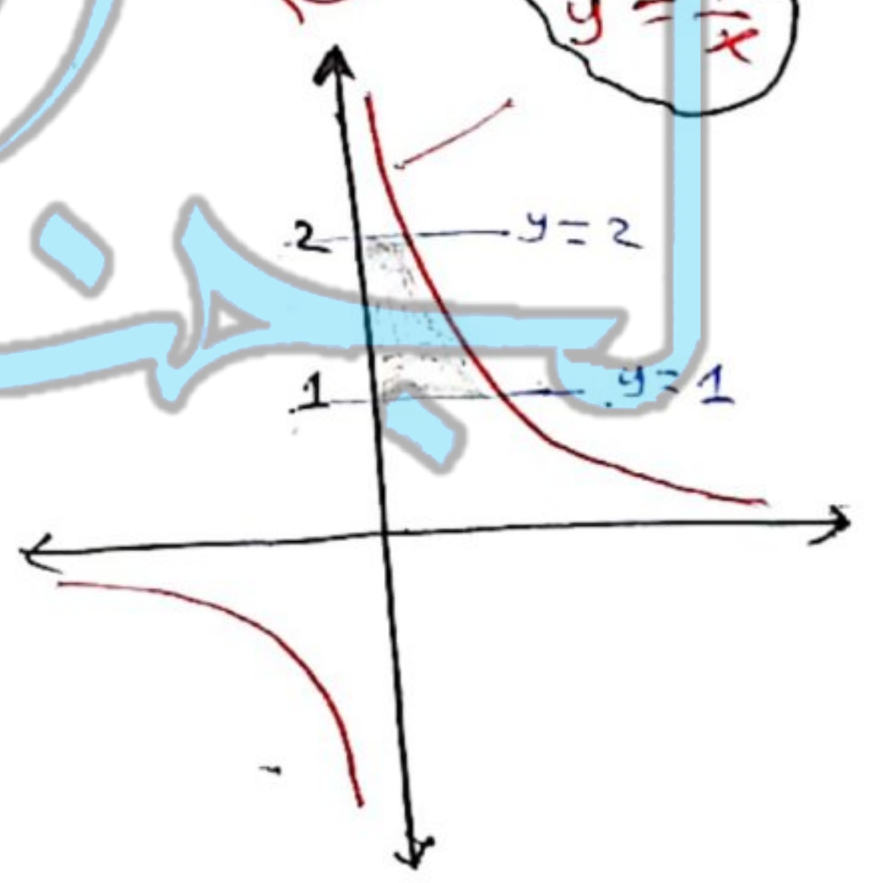
$$V = \pi \int_1^2 (\frac{1}{y})^2\, dy$$

$$V = \pi \int_1^2 \frac{1}{y^2}\, dy$$

$$= \pi (\frac{-1}{y}) \Big|_1^2$$

$$= \pi [\frac{-1}{2} + 1]$$

$$= \frac{\pi}{2} \text{ c.u}$$



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4) $x = -y^2$, $y = x+2$ about y-axis.

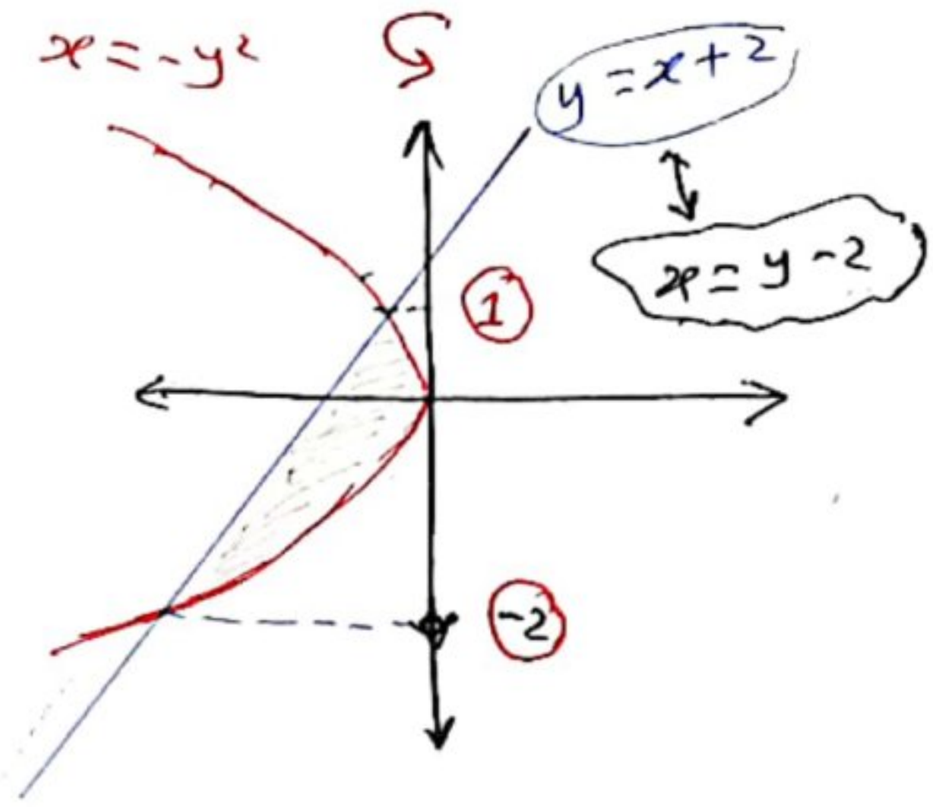
Points of intersection

$$y-2 = -y^2$$

$$\Rightarrow y^2+y-2=0$$

$$\Rightarrow (y-1)(y+2)=0$$

$$\Rightarrow y=1 \quad \text{or} \quad y=-2$$



$$\therefore V = \pi \int_{-2}^{1} \left((y-2)^2 - (-y^2)^2 \right) dy$$

$$= \pi \int_{-2}^{1} \left((y-2)^2 - y^4 \right) dy$$

$$= \pi \left[\frac{(y-2)^3}{3} - \frac{y^5}{5} \right]_{-2}^{1}$$

$$= \frac{72}{5}\pi \quad c.u$$

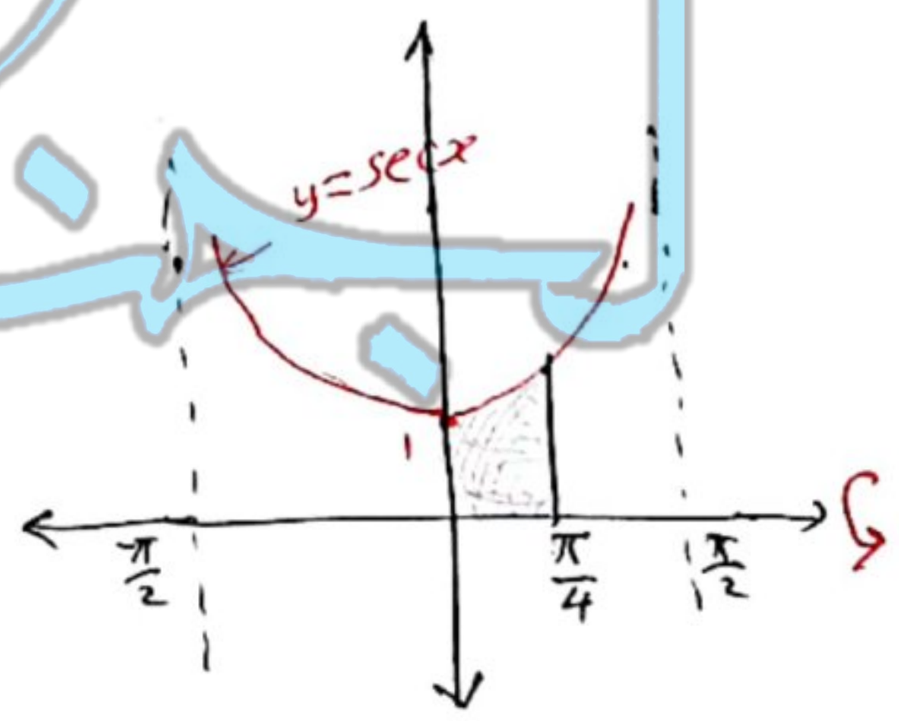
5) $y = \sec x$, $y = 0$ and $x \in [0, \frac{\pi}{4}]$ about x-axis

$$V = \pi \int_{0}^{\frac{\pi}{4}} \sec^2 x \, dx$$

$$= \pi \left[\tan x \right]_{0}^{\pi/4}$$

$$= \pi \left[1 - 0 \right]$$

$$= \pi \quad c.u$$



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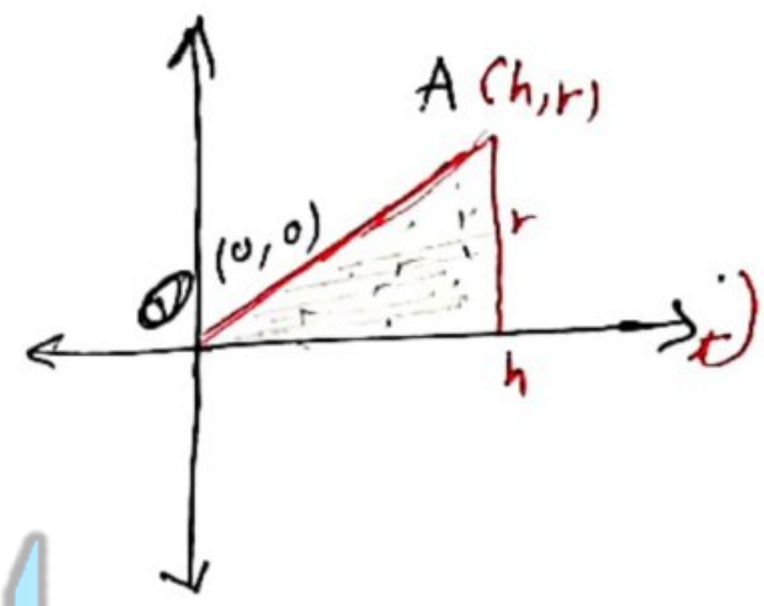
e.g: By using disks method, show that the volume of the cone whose radius (r) and height (h) is given by

$$V = \frac{1}{3}\pi r^2 h$$

Sol:

Eq. of $\overleftrightarrow{OA}$: $y = \frac{r}{h}x$



$$\therefore V = \pi \int_0^h \left(\frac{r}{h}x\right)^2 dx$$

$$= \frac{\pi r^2}{h^2} \int_0^h x^2 dx$$

$$= \frac{\pi r^2}{h^2} \cdot \frac{x^3}{3}\Big|_0^h$$

$$= \frac{\pi r^2}{h^2} \left[\frac{h^3}{3} - 0\right]$$

$$= \frac{1}{3}\pi r^2 h .$$

Slope $\overleftrightarrow{OA} = \frac{r}{h}$

the eq. of $\overleftrightarrow{OA}$ is $y = \frac{r}{h}x$

EX: By using disks method, show that the volume (V) of the sphere with radius (r) is given by

$$V = \frac{4}{3}\pi r^3$$

Hint : rotate a semi-circle.

END

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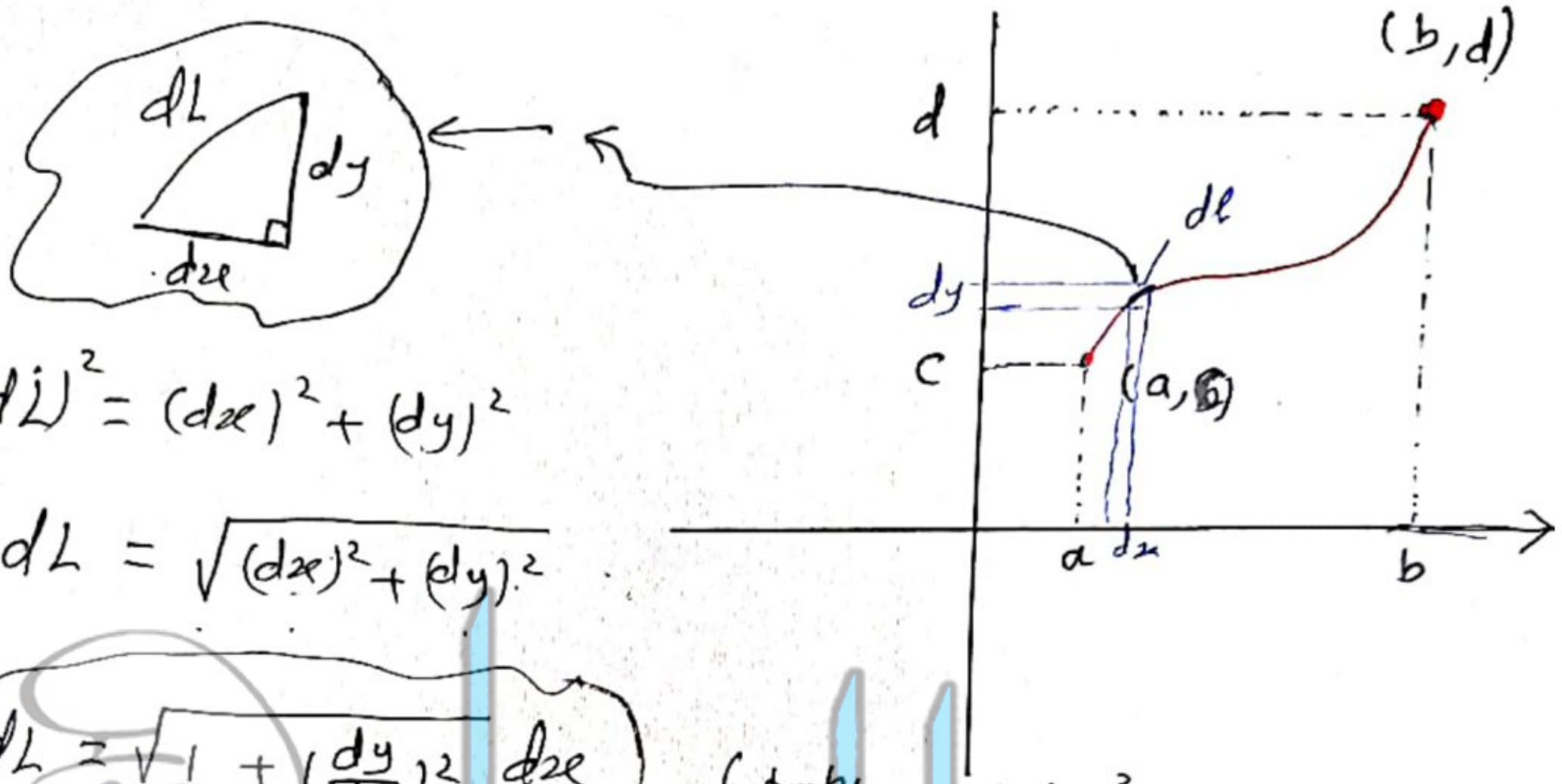
3) Length of a plane curve (Arc length)



$$(dL)^2 = (dx)^2 + (dy)^2$$

$$\Rightarrow dL = \sqrt{(dx)^2 + (dy)^2}$$

$$\Rightarrow dL = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$ (taking $(dx)^2$ as common factor)

OR

$$dL = \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\, dy$$ $(dy)^2$: common factor

OR

$$dL = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$ $(dt)^2$: common factor if $x = x(t)$, $y = y(t)$ (parametric equations)

$t_1 \le t \le t_2$ t: parameter

For example if $x = t^2$, $y = 2t$

$t = 0 \longrightarrow (0, 0)$

$t = 1 \longrightarrow (1, 2)$

$t = 2 \longrightarrow (4, 4)$

$t = 3 \longrightarrow (9, 6)$

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So we get the following rules to compute the length of the curve

1) $$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

2) $$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

3) $$L = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

e.g: Find the arc length of the given curve over the given interval.

1) $y = 2x + 1$ from $x = 0$ to $x = 1$.

Sol:

$$L = \int_0^1 \sqrt{\left(\frac{dy}{dx}\right)^2 + 1}\, dx$$

$$= \int_0^1 \sqrt{(2)^2 + 1}\, dx$$

$$= \int_0^1 \sqrt{5}\, dx$$

$$= \sqrt{5} \text{ L.U}$$

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2) $x = \frac{2}{3} y^{\frac{3}{2}} + 5$ from $y=0$ to $y=2$

Sol:

$$L = \int_0^2 \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\, dy$$

$$\frac{dx}{dy} = y^{\frac{1}{2}} \Longrightarrow \left(\frac{dx}{dy}\right)^2 + 1 = y + 1$$

$$\therefore L = \int_0^2 \sqrt{y+1}\, dy$$

$$= \frac{2}{3}\left[\sqrt{27} - 1\right] = \frac{2}{3}\left[3\sqrt{3} - 1\right] \text{ L.U.}$$

3) $x = 2\cos t$, $y = 2\sin t$ $\quad 0 \le t \le \frac{\pi}{2}$

Sol:

$$L = \int_0^{\pi/2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$= \int_0^{\pi/2} \sqrt{(-2\sin t)^2 + (2\cos t)^2}\, dt$$

$$= \int_0^{\pi/2} \sqrt{4\sin^2 t + 4\cos^2 t}\, dt$$

$$= \int_0^{\pi/2} 2\sqrt{\sin^2 t + \cos^2 t}\, dt$$

$$= \int_0^{\pi/2} 2\, dt = \pi \text{ L.U.}$$

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4) $x = \frac{1}{3}(y^2+2)^{3/2}$ from $y=0$ to $y=1$

Sol:

$$L = \int_0^1 \sqrt{\left(\frac{dx}{dy}\right)^2 + 1}\, dy$$

$$\frac{dx}{dy} = \frac{1}{2}(y^2+2)^{\frac{1}{2}}(2y) = y\sqrt{y^2+2}$$

$$\therefore \left(\frac{dx}{dy}\right)^2 + 1 = y^2(y^2+2) + 1$$

$$= y^4 + 2y^2 + 1$$

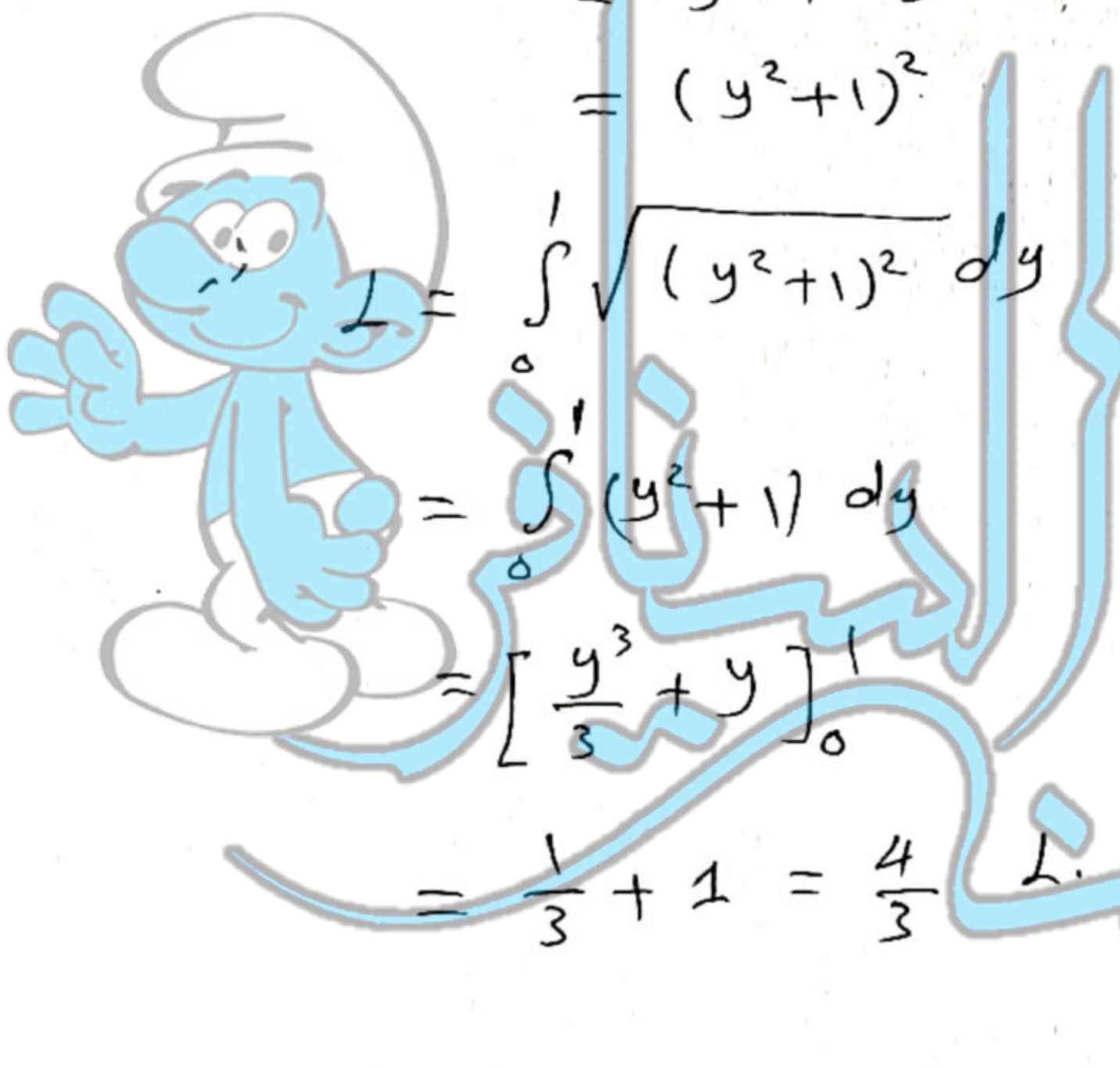
$$= (y^2+1)^2$$

$$L = \int_0^1 \sqrt{(y^2+1)^2}\, dy$$

$$= \int_0^1 (y^2+1)\, dy$$

$$= \left[\frac{y^3}{3} + y\right]_0^1$$

$$= \frac{1}{3} + 1 = \frac{4}{3} \text{ L.U}$$



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L'Hopital's Rule:

If f is differentiable and

$$\lim_{x \to \square} \frac{f(x)}{g(x)} \quad \text{gives} \quad \frac{0}{0} \text{ or } \frac{\pm\infty}{\pm\infty}$$

then $$\lim_{x \to \square} \frac{f(x)}{g(x)} = \lim_{x \to \square} \frac{f'(x)}{g'(x)}$$

inside $\square$ could be a, a^+, a^-, ∞ or $-\infty$, $a \in \mathbb{R}$

Indeterminate Forms:

1) $\frac{0}{0}$ 2) $\frac{\pm\infty}{\pm\infty}$ 3) $\infty - \infty$ 4) $0 \cdot (\pm\infty)$

5) 0^0 6) 1^∞ 7) ∞^0

assume y and take $\ln$

e.g: Find

1) $\lim_{x \to 2} \frac{x^2 - 4}{x^3 - 8}$ $\left(\frac{0}{0}\right)$

$$\lim_{x \to 2} \frac{x^2 - 4}{x^3 - 8} = \lim_{x \to 2} \frac{2x}{3x^2} = \frac{4}{12} = \frac{1}{3}$$

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2) $\lim_{x\to 0} \frac{\cos 3x - \cos 2x}{x^2}$ $\left(\frac{0}{0}\right)$

$$\lim_{x\to 0} \frac{\cos 3x - \cos 2x}{x^2} = \lim_{x\to 0} \frac{-3\sin 3x + 2\sin 2x}{2x} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x\to 0} \frac{-9\cos 3x + 4\cos 2x}{2}$$

$$= \frac{-9+4}{2}$$

$$= \frac{-5}{2}$$

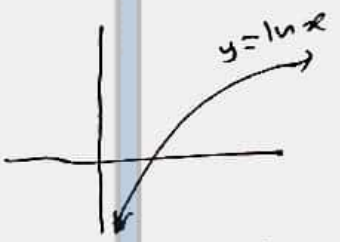
$\sin 0 = 0$

$\cos 0 = 1$

3) $\lim_{x\to\infty} \frac{\ln x}{x}$ $\left(\frac{\infty}{\infty}\right)$

$$= \lim_{x\to\infty} \frac{\frac{1}{x}}{1}$$

$$= \frac{0}{1} = 0$$



$\ln \infty = \infty$

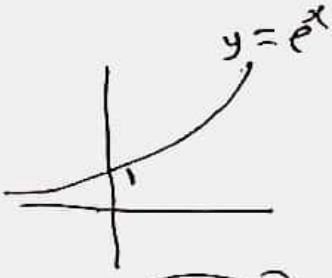
$\ln 0^+ = -\infty$

!!

4) $\lim_{x\to\infty} \frac{e^x}{x^2}$ $\left(\frac{\infty}{\infty}\right)$

$$= \lim_{x\to\infty} \frac{e^x}{2x} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x\to\infty} \frac{e^x}{2} = \infty.$$



$e^{\infty} = \infty$

$e^{-\infty} = 0$

$e^0 = 1$

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5) $\lim_{x\to\infty} \frac{x^2}{ex}$ $\left(\frac{\infty}{\infty}\right)$

$$\lim_{x\to\infty} \frac{x^2}{ex} = \lim_{x\to\infty} \frac{2x}{ex} \quad \left(\frac{\infty}{\infty}\right)$$

$$= \lim_{x\to\infty} \frac{2}{e^x}$$

$$= 0$$

If n is a natural number you can show that

$$\lim_{x\to\infty} \frac{e^x}{x^n} = \infty \quad \text{and} \quad \lim_{x\to\infty} \frac{x^n}{e^x} = 0$$

6) $\lim_{x\to\infty} \frac{x^{-2}}{\sin(\frac{1}{x})}$ $\left(\frac{0}{0}\right)$

Notice that

$\infty \xrightarrow{+ve} \infty$

$\infty \xrightarrow{-ve} 0$

$$\Rightarrow \lim_{x\to\infty} \frac{x^{-2}}{\sin(\frac{1}{x})} = \lim_{x\to\infty} \frac{-2x^{-3}}{\frac{-1}{x^2}\cos\frac{1}{x}}$$

$$= \lim_{x\to\infty} \frac{2x^{-1}}{\cos(\frac{1}{x})} = \frac{0}{1} = 0$$

7) $\lim_{x\to 0^+} \frac{-5e^x + 5}{3x^2}$ $\frac{0}{0}$

$e^0 = 1$

$$\lim_{x\to 0^+} \frac{-5e^x + 5}{3x^2} = \lim_{x\to 0^+} \frac{-5e^x}{6x} = -\infty$$

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8) $\lim_{x\to 0^+} x\ln x$ $\quad 0\cdot(-\infty)$

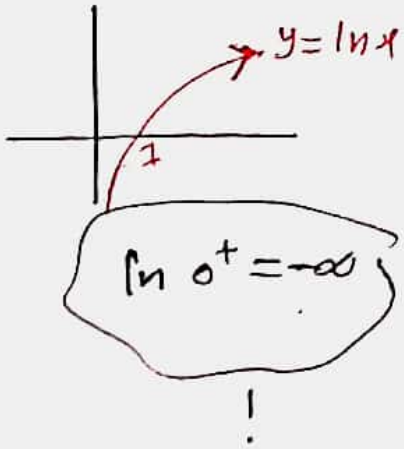
$$\lim_{x\to 0^+} x\ln x = \lim_{x\to 0^+} \frac{\ln x}{x^{-1}} \quad \frac{-\infty}{\infty}$$

$$= \lim_{x\to 0^+} \frac{\frac{1}{x}}{\frac{-1}{x^2}} =$$

$$= \lim_{x\to 0^+} (-x) = 0$$

Note

$$ab = \frac{a}{\frac{1}{b}} = \frac{b}{\frac{1}{a}}$$



$\ln 0^+ = -\infty$

!

9) $\lim_{x\to \frac{\pi}{2}^+} \tan(x-\frac{\pi}{2})\sec x$ $\quad 0\cdot(-\infty)$

$\sec x = \frac{1}{\cos x}$

$$\lim_{x\to \frac{\pi}{2}^+} \tan(x-\frac{\pi}{2})\sec x = \lim_{x\to \frac{\pi}{2}^+} \frac{\tan(x-\frac{\pi}{2})}{\cos x} \quad \frac{0}{0}$$

$$= \lim_{x\to \frac{\pi}{2}^+} \frac{\sec^2(x-\frac{\pi}{2})}{-\sin x}$$

$$= -1$$

10) $\lim_{x\to 0^+} x\csc 3x$ $\quad 0\cdot(\infty)$ $\quad \csc x = \frac{1}{\sin x}$

$\sin 0 = 0$

$$\lim_{x\to 0^+} x\csc 3x = \lim_{x\to 0^+} \frac{x}{\sin 3x} \quad \frac{0}{0}$$

$$= \lim_{x\to 0^+} \frac{1}{3\cos 3x} = \frac{1}{3}$$

Also, you can show that $\lim_{x\to 0^-} x\csc 3x = \frac{1}{3}$

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11) $\lim_{x\to 0}\left(\frac{1}{x}-\frac{1}{\sin x}\right)$

$\infty-\infty$ or $-\infty+\infty$ $\Rightarrow$ 0^+ or 0^-

5

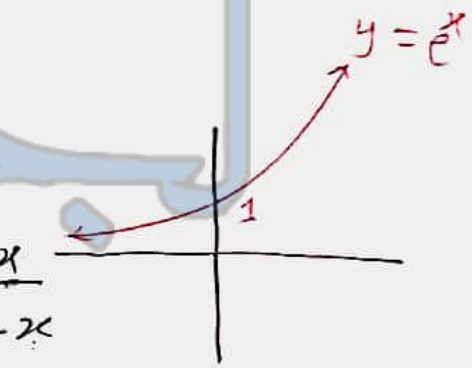
$$=\lim_{x\to 0}\left(\frac{\sin x - x}{x\sin x}\right) \quad \frac{0}{0}$$

$$=\lim_{x\to 0}\frac{\cos x - 1}{x\cos x+\sin x} \quad \frac{0}{0}$$

$$=\lim_{x\to 0}\frac{-\sin x}{-x\sin x+\cos x+\cos x}$$

$$=\frac{0}{0+1+1}=0$$

12) $\lim_{x\to 0^+}\left(\frac{1}{x}-\frac{1}{e^x-1}\right)$ $\infty-\infty$



$$\Rightarrow \lim_{x\to 0^+}\left(\frac{1}{x}-\frac{1}{e^x-1}\right)=\lim_{x\to 0^+}\frac{e^x-1-x}{xe^x-x}$$

$$\Rightarrow \lim_{x\to 0^+}\left(\frac{1}{x}-\frac{1}{e^x-1}\right)=\lim_{x\to 0^+}\frac{e^x-1}{xe^x+e^x-1} \quad \frac{0}{0}$$

$$=\lim_{x\to 0}\frac{e^x}{xe^x+e^x+e^x}=\frac{1}{2}$$

Also, you can show $\lim_{x\to 0^-}\left(\frac{1}{x}-\frac{1}{e^x-1}\right)=\frac{1}{2}$

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13) $\lim_{x \to 0^+} (\sin x)^x$ $\quad (0^0)$

Let $y = (\sin x)^x \Rightarrow \ln y = \ln (\sin x)^x$

$\Rightarrow \ln y = x \ln (\sin x)$

$\lim_{x \to 0^+} \ln y = \lim_{x \to 0^+} x \ln (\sin x)$ $\quad (0 \cdot (-\infty))$

$= \lim_{x \to 0^+} \dfrac{\ln \sin x}{\frac{1}{x}}$ $\quad \left(\dfrac{-\infty}{\infty}\right)$

$= \lim_{x \to 0^+} \dfrac{\frac{\cos x}{\sin x}}{\frac{-1}{x^2}}$ $\quad (\div)$

$= \lim_{x \to 0^+} \left(\dfrac{\cos x}{\sin x} \cdot \dfrac{x^2}{-1} \right)$

$= \lim_{x \to 0^+} \left(\dfrac{x}{\sin x} \cdot (-x \cos x) \right)$

$\lim_{x \to 0^+} \ln y = 1 (0) = 0$

$\therefore \quad e^{\lim_{x \to 0^+} \ln y} = e^0$

$\Rightarrow \lim_{x \to 0^+} e^{\ln y} = 1 \Rightarrow \lim_{x \to 0^+} y = 1$

$\therefore \lim_{x \to 0^+} (\sin x)^x = 1$

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14) $\lim_{x \to 0^+} (\cos x)^{\frac{1}{x}}$ $\quad 1^{\infty}$

Let $y = (\cos x)^{\frac{1}{x}} \Rightarrow \ln y = \ln (\cos x)^{\frac{1}{x}}$

$\Rightarrow \ln y = \frac{\ln \cos x}{x}$

$\therefore \lim_{x \to 0^+} \ln y = \lim_{x \to 0^+} \frac{\ln \cos x}{x}$ $\quad \frac{0}{0}$

$= \lim_{x \to 0} \frac{\frac{-\sin x}{\cos x}}{1} = \frac{0}{1} = 0$

$\therefore \lim_{x \to 0^+} (\cos x)^{\frac{1}{x}} = e^0 = 1$

15) $\lim_{x \to 0} (1+2x)^{\frac{-4}{x}}$ $\quad 0^{\infty}$ $\quad 0^+ \; -\infty \quad 0^- \; \infty$

Let $y = (1+2x)^{-4/x}$

$\Rightarrow \ln y = \ln (1+2x)^{-\frac{4}{x}}$

$\Rightarrow \ln y = \frac{-4 \ln (1+2x)}{x}$ $\quad \frac{0}{0}$ $\quad \frac{-\infty}{+\infty}$ $\quad \frac{0}{0}$

$\Rightarrow \lim_{x \to 0} \ln y = \lim_{x \to 0} \frac{-4 \ln (1+2x)}{x}$

$= \lim_{x \to 0} \frac{\frac{-8}{1+2x}}{1} = -8$

$\therefore \lim_{x \to 0} (1+2x)^{-4/x} = e^{-8}$

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16) $\lim_{x \to \infty} \left(1 + \frac{a}{x}\right)^{x}$ $\quad a \neq 0$ $\quad 1^{\infty}$

Let $y = \left(1 + \frac{a}{x}\right)^{x} \Rightarrow \ln y = \ln\left(1 + \frac{a}{x}\right)^{x}$

$\Rightarrow \ln y = x \ln\left(1 + \frac{a}{x}\right)$

$\infty \cdot 0$

$\lim_{x \to \infty} \ln y = \lim_{x \to \infty} x \ln\left(1 + \frac{a}{x}\right)$

$= \lim_{x \to \infty} \frac{\ln\left(1 + \frac{a}{x}\right)}{\frac{1}{x}}$ $\quad \frac{0}{0}$

$= \lim_{x \to \infty} \frac{\frac{-\frac{a}{x^2}}{1 + \frac{a}{x}}}{\frac{-1}{x^2}}$ $\quad \div$

$= -\lim_{x \to \infty} \left(\frac{\frac{-a}{x^2}}{1 + \frac{a}{x}} \cdot \frac{x^2}{-1}\right)$

$= \lim_{x \to \infty} \left(\frac{a}{1 + \frac{a}{x}}\right) = a$

$\therefore \lim_{x \to \infty} \left(1 + \frac{a}{x}\right)^{x} = e^{a}$

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Rule:

$$\lim_{x \to \infty} \left(1 + \frac{a}{x}\right)^x = e^a$$

e.g: Use the above rule or else, find the following Limits

1) $\lim_{x \to \infty} \left(1 + \frac{3}{x}\right)^x$ $\quad 1^\infty$

$$\lim_{x \to \infty} \left(1 + \frac{3}{x}\right)^x = e^3 \quad \text{(by the rule)}$$

2) $\lim_{x \to \infty} \left(1 - \frac{2}{5x}\right)^x$

$$\lim_{x \to \infty} \left(1 - \frac{2}{5x}\right)^x = \lim_{x \to \infty} \left(1 + \frac{-2/5}{x}\right)^x = e^{-2/5}$$

3) $\lim_{x \to \infty} \left(1 - \frac{10}{x}\right)^{4x}$

$$\lim_{x \to \infty} \left(1 - \frac{10}{x}\right)^{4x} = \left[\lim_{x \to \infty} \left(1 + \frac{-10}{x}\right)^x\right]^4$$

$$= (e^{-10})^4 = e^{-40}$$

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4) $\lim_{x\to\infty}\left(\frac{x+6}{x+2}\right)^x$ $\quad 1^\infty$

$$\lim_{x\to\infty}\left(\frac{x+6}{x+2}\right)^x = \lim_{x\to\infty}\left(\frac{\frac{x}{x}+\frac{6}{x}}{\frac{x}{x}+\frac{2}{x}}\right)^x$$

$$= \lim_{x\to\infty}\frac{\left(1+\frac{6}{x}\right)^x}{\left(1+\frac{2}{x}\right)^x}$$

$$= \frac{e^6}{e^2} = e^4$$

5) $\lim_{x\to\infty}\left(\frac{5x+1}{4x+1}\right)^x = \infty$ (direct)

6) $\lim_{x\to\infty}\left(\frac{2x+7}{3x+5}\right)^x = 0$ (direct)

7) $\lim_{x\to\infty}\left(x - \ln(e^x+1)\right)$ $\quad \infty-\infty$

$$= \lim_{x\to\infty}\left(\ln e^x - \ln(e^x+1)\right) = \lim_{x\to\infty}\ln\left(\frac{e^x}{e^x+1}\right)$$

$\left(\frac{\infty}{\infty}\right)$

$$= \ln\left(\lim_{x\to\infty}\frac{e^x}{e^x+1}\right) = \ln\left(\lim_{x\to\infty}\left(\frac{e^x}{e^x}\right)\right)$$

$$= \ln 1 = 0.$$

END

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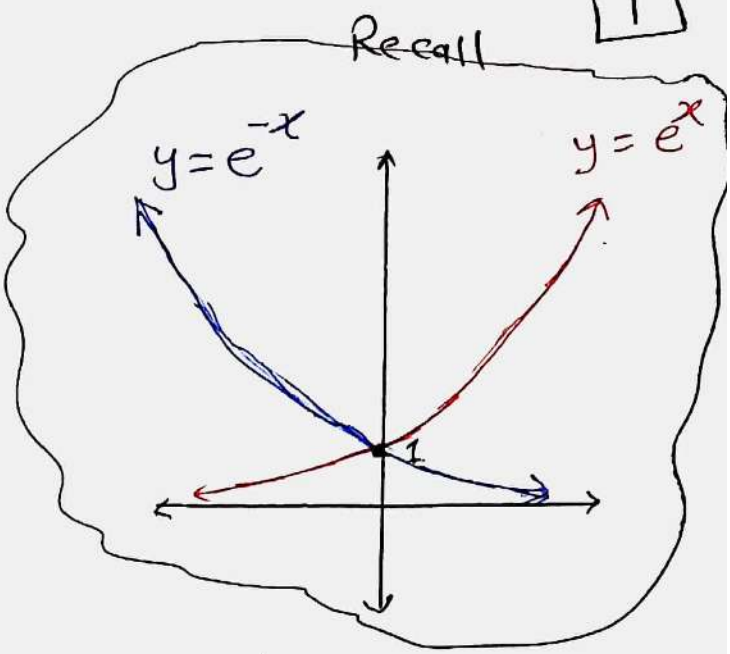
Hyperbolic Functions:



Def:

1) Hyperbolic Sine

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

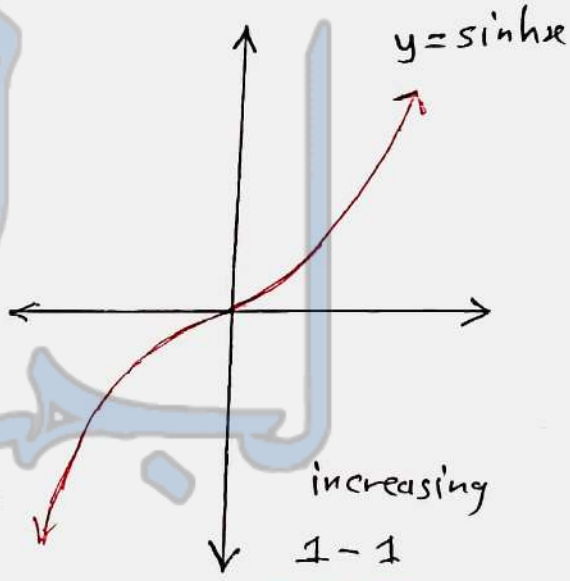
The function

$y = \sinh x$

Domain $= \mathbb{R}$

Range $= \mathbb{R}$

$y(0) = 0 \Rightarrow \sinh 0 = 0$



odd fun.

$\sinh(-x) = -\sinh x$
$\forall x \in R$

e.g.: Solve for x ?

$\sinh x = 2$

sol:

$$\sinh x = 2 \Rightarrow \frac{e^x - e^{-x}}{2} = 2$$

$$\Rightarrow e^x - e^{-x} = 4$$

$$* e^x \Rightarrow e^{2x} - 1 = 4e^x$$

$$\Rightarrow (e^x)^2 - 4e^x - 1 = 0$$

$$e^x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(-1)}}{2(1)}$$

$$e^x = \frac{4 \pm \sqrt{16+4}}{2}$$

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$$\Rightarrow e^x = \frac{4 \pm 2\sqrt{5}}{2}$$

$$\Rightarrow e^x = 2 \pm \sqrt{5}$$

rejected since $e^x > 0$

$$\Rightarrow e^x = 2 + \sqrt{5}$$

$$\Rightarrow x = \ln(2+\sqrt{5})$$

e.g: Simplify $\sinh(\ln a)$, $a > 0$

Sol:

$$\sinh(\ln a) = \frac{e^{\ln a} - e^{-\ln a}}{2}$$

$$= \frac{a - e^{\ln a^{-1}}}{2}$$

$$= \frac{a - a^{-1}}{2}$$ ✓

$$* \xrightarrow{a} \sinh(\ln a) = \frac{a^2 - 1}{2a}$$ ✓

2 Hyperbolic Cosine

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

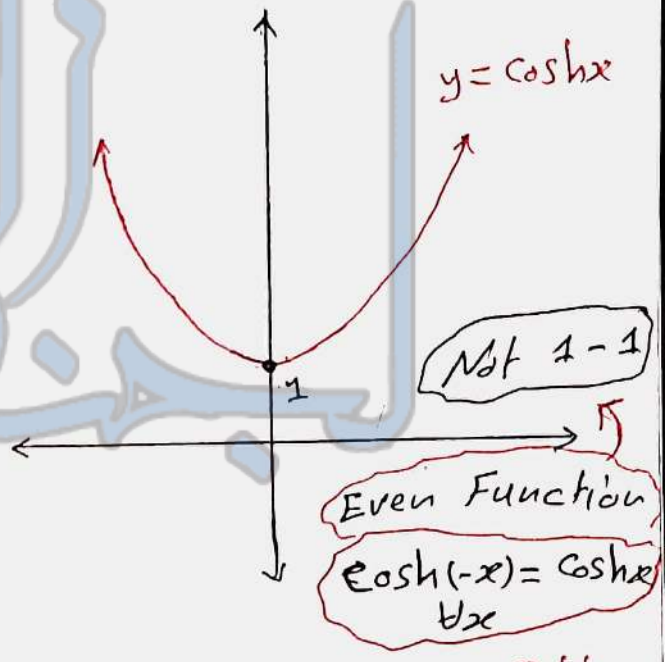
The function

$y = \cosh x$

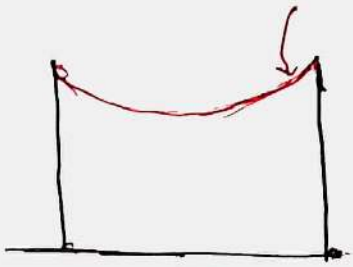
The domain $= \mathbb{R}$

The Range $= [1, \infty)$

$y(0) = 1 \Rightarrow \cosh 0 = 1$.



Cable



Catenary

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e.g: Find the solution sets of the following hyperbolic equations

1) $\cosh x = 3$ 2) $\cosh x = 0$ 3) $\cosh(2x+1) = 1$

Sol:

1) $\cosh x = 3 \Rightarrow \frac{e^x + e^{-x}}{2} = 3$

$\Rightarrow e^x + e^{-x} = 6$

$* e^x \Rightarrow e^{2x} + 1 = 6e^x$

$\Rightarrow (e^x)^2 - 6e^x + 1 = 0$

$\Rightarrow e^x = \frac{6 \pm \sqrt{36 - 4(1)(1)}}{2}$

$\Rightarrow e^x = \frac{6 \pm \sqrt{32}}{2}$

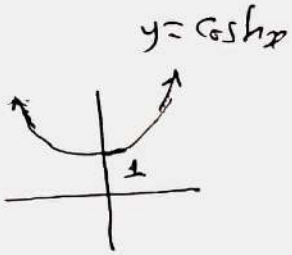
$\Rightarrow e^x = \frac{6 \pm 4\sqrt{2}}{2}$

$e^x = 3 \pm 2\sqrt{2}$

$e^x = 3 + 2\sqrt{2}$ or $e^x = 3 - 2\sqrt{2}$

$\Rightarrow x = \ln(3 \mp 2\sqrt{2})$

$S.S = \{\ln(3 \mp 2\sqrt{2})\}$



2) $\cosh x = 0 \Rightarrow$ has no solution

$\Rightarrow S.S = \phi$.

3) $\cosh(2x+1) = 1$

$\Rightarrow 2x + 1 = 0 \Rightarrow x = -\frac{1}{2}$

$S.S = \{-\frac{1}{2}\}$

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Hyperbolic tangent

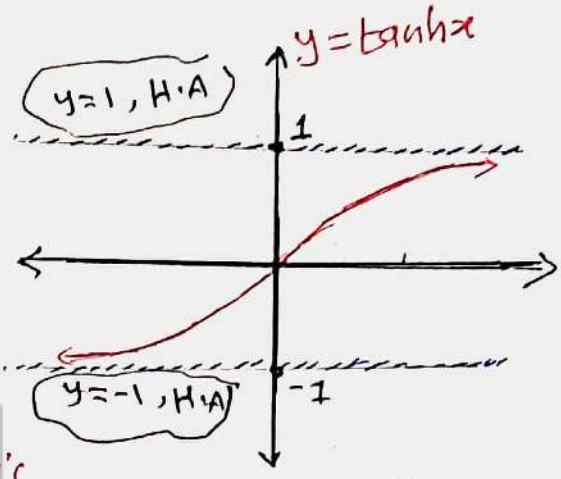
$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

The Function $y = \tanh x$

Its domain $= \mathbb{R}$

Its Range $= (-1, 1)$

$\tanh 0 = 0$



Increasing

1 - 1

odd fun.

$\tanh(-x) = -\tanh x \quad \forall x$

$\lim_{x \to \infty} \tanh x = 1$

$\lim_{x \to -\infty} \tanh x = -1$

e.g: Solve the following hyperbolic equations

1) $\tanh x = 2$

2) $\tanh x = \frac{3}{4}$

Sol:

1) $\tanh x = 2 \Rightarrow$ S.S $= \phi$ (why??)

2) $\tanh x = \frac{3}{4} \Rightarrow \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{3}{4}$

$$\Rightarrow \frac{e^{2x} - 1}{e^{2x} + 1} = \frac{3}{4}$$

$$\Rightarrow 4e^{2x} - 4 = 3e^{2x} + 3$$

$$\Rightarrow e^{2x} = 7$$

$$\Rightarrow 2x = \ln 7 \Rightarrow x = \frac{1}{2} \ln 7$$

or $x = \ln \sqrt{7}$

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Hyperbolic Cotangent:

$$\coth x = \frac{1}{\tanh x} = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}, \quad x \neq 0$$

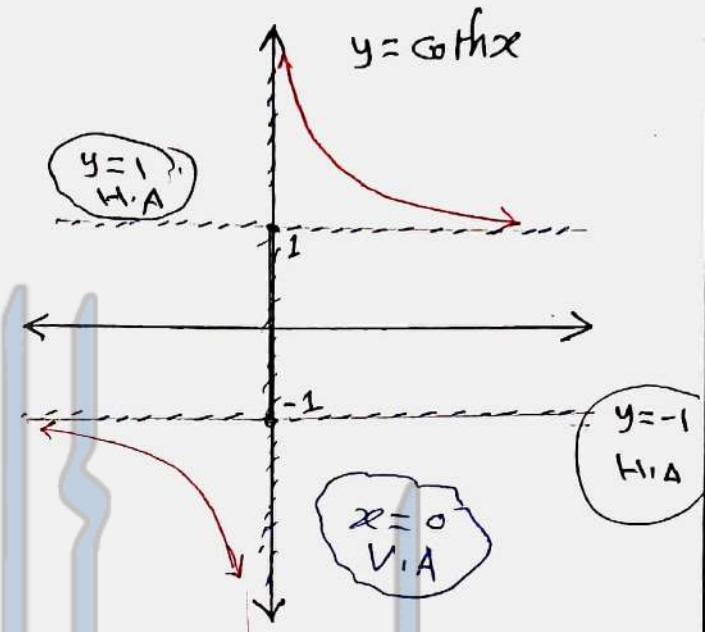
The Function

$y = \coth x$

Its domain $= \mathbb{R} \setminus \{0\}$

Its Range $= (-\infty, -1) \cup (1, \infty)$

$= \mathbb{R} \setminus [-1, 1]$



e.g: Find the solution of

$\coth x = \frac{1}{2}$

Sol:

S.S = ϕ

Decreasing on $(-\infty, 0)$ and on $(0, \infty)$

1 - 1

$\lim_{x \to \infty} \coth x = 1$

$\lim_{x \to -\infty} \coth x = -1$

$\lim_{x \to 0^+} \coth x = \infty$

$\lim_{x \to 0^-} \coth x = -\infty$

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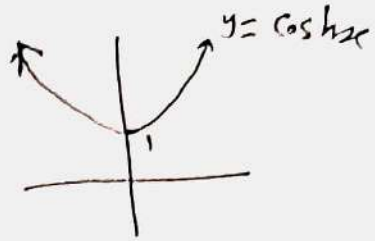
5 Hyperbolic secant:

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^{x}+e^{-x}}$$



The Function $y = \operatorname{sech} x$,

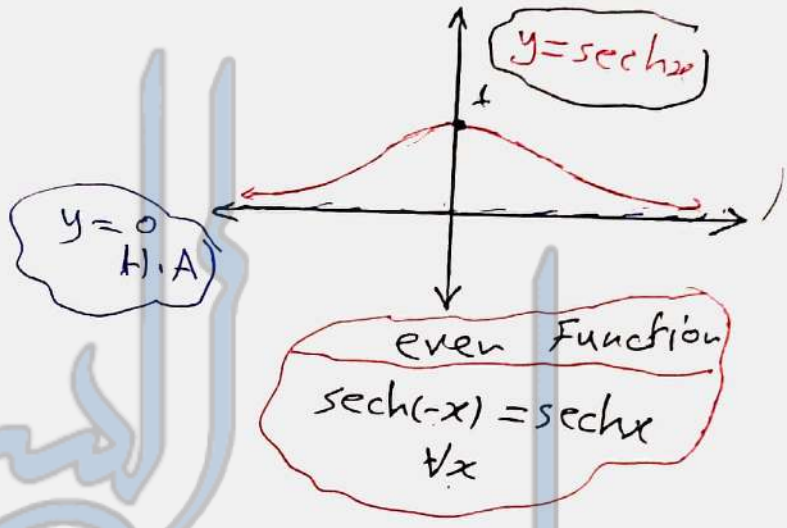
Its domain = $\mathbb{R}$

Its Range = $(0, 1]$

$\lim_{x \to \infty} \operatorname{sech} x = 0$

$\lim_{x \to -\infty} \operatorname{sech} x = 0$

$\operatorname{sech} 0 = 1$



even Function

$\operatorname{sech}(-x) = \operatorname{sech} x \quad \forall x$

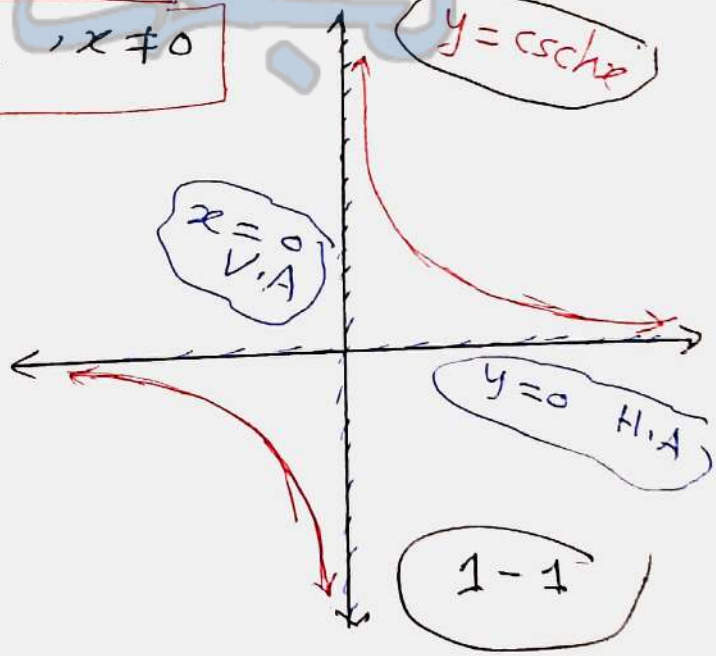
6 Hyperbolic cosecant:

$$\operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^{x}-e^{-x}}, \quad x \neq 0$$

The Function $y = \operatorname{csch} x$

Its domain = $\mathbb{R} \setminus \{0\}$

Its Range = $\mathbb{R} \setminus \{0\}$



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Hyperbolic Identities:

e.g: show that $\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$

Sol:

$$L.H.S = \frac{e^x - e^{-x}}{2} \cdot \frac{e^y + e^{-y}}{2} + \frac{e^x + e^{-x}}{2} \cdot \frac{e^y - e^{-y}}{2}$$

(Left hand side)

$$= \frac{e^{x+y} + e^{x-y} - e^{-x+y} - e^{-x-y} + e^{x+y} - e^{x-y} + e^{-x+y} - e^{-x-y}}{4}$$

$$= \frac{2e^{x+y} - 2e^{-x-y}}{4}$$

$$= \frac{e^{x+y} - e^{-(x+y)}}{2}$$

$$= \sinh(x+y) \qquad R.H.S$$

Important Identities

1) $\cosh x + \sinh x = e^x$

2) $\cosh x - \sinh x = e^{-x}$

3) $\cosh^2 x - \sinh^2 x = 1$

4) $\sinh(2x) = 2\sinh x \cosh x$

5) $\cosh(2x) = \begin{cases} \cosh^2 x + \sinh^2 x \\ 2\sinh^2 x + 1 \\ 2\cosh^2 x - 1 \end{cases}$

6) $\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$

7) $\sinh(x-y) = \sinh x \cosh y - \cosh x \sinh y$

8) $\cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$

9) $\cosh(x-y) = \cosh x \cosh y - \sinh x \sinh y$

Osborn's Rule (optional)

To convert Trig. Identity to Hyperbolic Identity

e.g

$1 + \tan^2 A = \sec^2 A$

↓

$1 - \tanh^2 A = \operatorname{sech}^2 A$

$1 + \cot^2 A = \csc^2 A$

$1 - \coth^2 A = -\operatorname{csch}^2 A$

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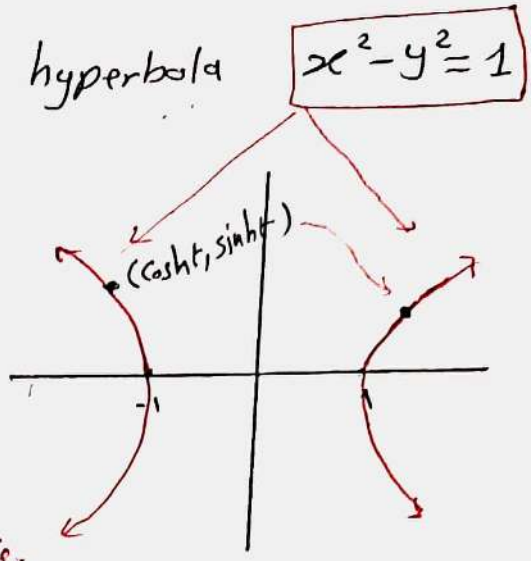
Note

The parametric equations

$x = \cosh t$, $y = \sinh t$, $t \in \mathbb{R}$

satisfy the equation of the hyperbola $x^2 - y^2 = 1$



e.g : show that

1) $y = \sinh x \longrightarrow \frac{dy}{dx} = \cosh x$

2) $y = \cosh x \Longrightarrow \frac{dy}{dx} = \sinh x$ and hence,

3) $y = \tanh x \Longrightarrow \frac{dy}{dx} = \operatorname{sech}^2 x$

Sol:

1) $y = \sinh x \Longrightarrow y = \frac{e^x - e^{-x}}{2}$

$\Rightarrow \frac{dy}{dx} = \frac{e^x + e^{-x}}{2} = \cosh x$ ✓

2) $y = \cosh x \Longrightarrow y = \frac{e^x + e^{-x}}{2}$

$\Rightarrow \frac{dy}{dx} = \frac{e^x - e^{-x}}{2} = \sinh x$ ✓

3) $y = \tanh x \Longrightarrow y = \frac{\sinh x}{\cosh x}$

$\Rightarrow \frac{dy}{dx} = \frac{\cosh x \cosh x - \sinh x \sinh x}{\cosh^2 x}$

$= \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x}$

$= \frac{1}{\cosh^2 x}$

$\therefore \quad \frac{dy}{dx} = \operatorname{sech}^2 x$ ✓

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Differentiation Rules of Hyperbolic Functions:

1) $y = \sinh g(x) \Rightarrow \frac{dy}{dx} = g'(x)\cosh g(x)$

2) $y = \cosh g(x) \Rightarrow \frac{dy}{dx} = g'(x)\sinh g(x)$

3) $y = \tanh g(x) \Rightarrow \frac{dy}{dx} = g'(x)\operatorname{sech}^2 g(x)$

4) $y = \coth g(x) \Rightarrow \frac{dy}{dx} = -g'(x)\operatorname{csch}^2 g(x)$

5) $y = \operatorname{sech} g(x) \Rightarrow \frac{dy}{dx} = -g'(x)\operatorname{sech} g(x)\tanh g(x)$

6) $y = \operatorname{csch} g(x) \Rightarrow \frac{dy}{dx} = -g'(x)\operatorname{csch} g(x)\coth g(x)$

e.g: Find $\frac{dy}{dx}$

1) $y = \sinh x^3 + \cosh 4x \Rightarrow \frac{dy}{dx} = 3x^2\cosh x^3 + 4\sinh 4x$

2) $y = \tanh(5^{-x}) \Rightarrow \frac{dy}{dx} = -5^{-x}(\ln 5)\operatorname{sech}^2(5^{-x})$

3) $y = \operatorname{sech}^4(2x) \Rightarrow \frac{dy}{dx} = 4\operatorname{sech}^3(2x)(-2\operatorname{sech} 2x \tanh 2x)$

$= -8\operatorname{sech}^4 2x \tanh 2x$

e.g: Show that $y = \sinh(ax) + \cosh(ax)$, satisfies the differential equation $y'' - a^2 y = 0$

Sol:

$y = \sinh(ax) + \cosh ax \rightarrow y' = a\cosh ax + a\sinh ax$

$\rightarrow y'' = a^2\sinh ax + a^2\cosh ax$

Now substitute in the D.E, to get

$y'' - a^2 y = a^2\sinh ax + a^2\cosh ax - a^2(\sinh ax + \cosh ax)$

$= a^2\sinh ax + a^2\cosh ax - a^2\sinh ax - a^2\cosh ax$

$= 0$ ✓

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Integration Rules of Hyperbolic Functions:

Recall

$\int f(x)\,dx = F(x) + C$

$\Rightarrow \int f(ax+b)\,dx = \frac{1}{a} F(ax+b) + C$

1) $\int \sinh x\,dx = \cosh x + C$

2) $\int \cosh x\,dx = \sinh x + C$

3) $\int \operatorname{sech}^2 x\,dx = \tanh x + C$

4) $\int \operatorname{csch}^2 x\,dx = -\coth x + C$

5) $\int \operatorname{sech} x \tanh x\,dx = -\operatorname{sech} x + C$

6) $\int \operatorname{csch} x \coth x\,dx = -\operatorname{csch} x + C$

e.g: Evaluate

1) $\int \cosh 3x\,dx = \frac{1}{3}\sinh 3x + C$

2) $\int 5\operatorname{csch}^2(4x+3)\,dx = -\frac{5}{4}\coth(4x+3) + C$

3) $\int \operatorname{sech} 2x \tanh 2x\,dx = -\frac{1}{2}\operatorname{sech} 2x + C$

4) $\int e^x \sinh x\,dx = \int e^x \cdot \frac{e^x - e^{-x}}{2}\,dx$

$= \frac{1}{2}\int (e^{2x} - 1)\,dx = \frac{1}{2}\left[\frac{1}{2}e^{2x} - x\right] + C$

$= \frac{1}{4}e^{2x} - \frac{1}{2}x + C$

5) $\int (\cosh x + \sinh x)^7\,dx$

$= \int (e^x)^7\,dx$

$= \int e^{7x}\,dx$

$= \frac{1}{7}e^{7x} + C$

$\cosh x + \sinh x = e^x$

$\cosh x - \sinh x = e^{-x}$

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6) $\int_0^{\ln 5} \cosh 2x\, dx = \frac{1}{2}\sinh 2x \Big]_0^{\ln 5}$ → 0

$= \frac{1}{2}[\sinh(2\ln 5) - \sinh 0]$

$= \frac{1}{2}\sinh \ln(25)$ ✓

$= \frac{1}{2}\left[\frac{e^{\ln 25} - e^{-\ln(25)}}{2}\right]$

$= \frac{1}{2}\left[\frac{25 - 25^{-1}}{2}\right]$

$= \frac{1}{2}\left[\frac{625 - 1}{50}\right]$

$= \frac{624}{100} = 6.24$

e.g: Evaluate

$\int x^2 \operatorname{csch} x^3 \coth x^3\, dx$

Sol:

Let $u = x^3 \Rightarrow \frac{du}{dx} = 3x^2 \Rightarrow dx = \frac{du}{3x^2}$

$\therefore \int x^2 \operatorname{csch} x^3 \coth x^3\, dx = \int x^2 \operatorname{csch} u \coth u \cdot \frac{du}{3x^2}$

$= -\frac{1}{3}\operatorname{csch} u + C$

$= -\frac{1}{3}\operatorname{csch} x^3 + C$

-END-

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Ch. 7
Principles of Integral Evaluations

Integration Rules: (Revision)

1) $\int A\,dx = Ax + C$, A: Constant

2) $\int x^n dx = \frac{x^{n+1}}{n+1} + C$, $n \neq -1$

3) $\int \frac{1}{x} dx = \ln|x| + C$

4) $\int \frac{g'(x)}{g(x)} dx = \ln|g(x)| + C$

---

5) $\int e^x dx = e^x + C$

6) $\int a^x dx = \frac{a^x}{\ln a} + C$, $a \in R^+ \setminus \{1\}$

---

7) $\int \sin x\,dx = -\cos x + C$

8) $\int \cos x\,dx = \sin x + C$

9) $\int \sec^2 x\,dx = \tan x + C$

10) $\int \csc^2 x\,dx = -\cot x + C$

11) $\int \sec x \tan x\,dx = \sec x + C$

12) $\int \csc x \cot x\,dx = -\csc x + C$

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13) $\int \sinh x\,dx = \cosh x + C$

14) $\int \cosh x\,dx = \sinh x + C$

15) $\int \operatorname{sech}^2 x\,dx = \tanh x + C$

16) $\int \operatorname{csch}^2 x\,dx = -\coth x + C$

17) $\int \operatorname{sech} x \tanh x\,dx = -\operatorname{sech} x + C$

18) $\int \operatorname{csch} x \coth x\,dx = -\operatorname{csch} x + C$

---

19) $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$

20) $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$

21) $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{x}{a}\right| + C$

$a > 0$

Also:

* $\int \sec x\,dx = \ln|\sec x + \tan x| + C$

* $\int \csc x\,dx = \ln|\csc x - \cot x| + C$
$= -\ln|\csc x + \cot x| + C$

* $\int \cot x\,dx = \ln|\sin x| + C$

* $\int \tan x\,dx = -\ln|\cos x| + C$
$= \ln|\sec x| + C$

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e.g: Evaluate

1) $\int \frac{dx}{4x^2+4x+10}$

2) $\int \frac{dx}{\sqrt{9-4x^2}}$

3) $\int \frac{dx}{x(\ln x)^3}$

4) $\int \frac{x^3}{x^8+1}dx$

5) $\int \frac{x}{\sqrt{1-x^2}}dx$

Sol:

1) $\int \frac{dx}{4x^2+4x+10} = \int \frac{dx}{(2x+1)^2+9}$

$= \frac{1}{2}\cdot\frac{1}{3}\tan^{-1}\left(\frac{2x+1}{3}\right)+C$

$= \frac{1}{6}\tan^{-1}\left(\frac{2x+1}{3}\right)+C$

$4x^2+4x+10$

$= 4x^2+4x+1-1+10$

$= (2x+1)^2+9$

2) $\int \frac{dx}{\sqrt{9-4x^2}} = \int \frac{dx}{\sqrt{9-(2x)^2}}$

$= \frac{1}{2}\sin^{-1}\left(\frac{2x}{3}\right)+C$

3) $\int \frac{dx}{x(\ln x)^3} = \int \frac{1}{x}(\ln x)^{-3}dx$

$= \frac{(\ln x)^{-2}}{-2}+C$

$u = \ln x \Rightarrow \frac{du}{dx} = \frac{1}{x}$

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4) $$\int \frac{x^3}{x^8+1}\,dx = \int \frac{x^3}{(x^4)^2+1}\,dx$$

$$= \frac{1}{4}\int \frac{4x^3}{(x^4)^2+1}\,dx \qquad u = x^4$$

$$= \frac{1}{4}\tan^{-1}(x^4) + C$$

5) $$\int \frac{x}{\sqrt{1-x^2}}\,dx = \int x(1-x^2)^{-\frac{1}{2}}\,dx$$

$u = 1-x^2$

$du = -2x\,dx$

$$= -\frac{1}{2}\int (-2x)(1-x^2)^{-\frac{1}{2}}\,dx$$

$$= -\frac{1}{2}\,\frac{(1-x^2)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= -(1-x^2)^{1/2} + C$$

$$= -\sqrt{1-x^2} + C$$

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e.g: Evaluate

1) $\int \sinh^2 3x\,dx$ 2) $\int \tanh^2 5x\,dx$

Sol:

1) $\int \sinh^2 3x\,dx = \frac{1}{2}\int(\cosh 6x - 1)\,dx$

$= \frac{1}{2}\left(\frac{1}{6}\sinh 6x - x\right) + C$

$= \frac{1}{12}\sinh 6x - \frac{1}{2}x + C$

$\cosh 2A = 2\sinh^2 A - 1$

$\Downarrow$ $\sinh^2 A = ??$

$\sin^2 A = \frac{1}{2}(1 - \cos 2A)$

$-\sinh^2 A = \frac{1}{2}(1 - \cosh 2A)$

$\Rightarrow \sinh^2 A = \frac{1}{2}(\cosh 2A - 1)$

---

2) $\int \tanh^2 5x\,dx$

$= \int (1 - \operatorname{sech}^2 5x)\,dx$

$= x - \frac{1}{5}\tanh 5x + C$

$1 + \tan^2 A = \sec^2 A$

$1 - \tanh^2 A = \operatorname{sech}^2 A$

$\tanh^2 A = 1 - \operatorname{sech}^2 A$

- END -

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Integration by Parts:

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

Integrate w.r.t x $\Longrightarrow$ $$uv = \int u\frac{dv}{dx}dx + \int v\frac{du}{dx}dx$$

$$\Rightarrow uv = \int u\,dv + \int v\,du$$

$$\Rightarrow \boxed{\int u\,dv = uv - \int v\,du} \text{ Rule .}$$

e.g: Compute $\int x\sin x\,dx$

sol:

$u = x \Rightarrow du = dx$

$dv = \sin x\,dx \Rightarrow v = -\cos x$

$$\Rightarrow \int x\sin x\,du = -x\cos x - \int -\cos x\,dx$$

$$= -x\cos x + \int \cos x\,dx$$

$$= -x\cos x + \sin x + C$$

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Group 1: (Tabular Integration)

$\int$ Polynomial $*$ $\begin{cases} e^{Ax+B} \\ a^{Ax+B} \\ (ax+b)^n \quad , n \neq -1 \\ \sin(Ax+B) \\ \cos(Ax+B) \\ \sinh(Ax+B) \\ \cosh(Ax+B) \end{cases}$

e.g : Evaluate

1) $\int x \sin x \, dx$

Sol:

$\int x \sin x \, dx = -x\cos x + \sin x + C$

Differentiate ↓ | Integrate ↓

x	+	$\sin x$
1	−	$-\cos x$
0		$-\sin x$

2) $\int x^2 e^x dx$

Sol:

x^2	+	e^x
$2x$	−	e^x
2	+	e^x
0		e^x

$\Rightarrow \int x^2 e^x dx = x^2 e^x - 2x e^x + 2e^x + C$

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3) $\int (2x+1)\sqrt{1-x}\,dx$

Sol:

$$\int (2x+1)\sqrt{1-x}\,dx = \int (2x+1)(1-x)^{\frac{1}{2}}dx$$

	$(1-x)^{\frac{1}{2}}$
$2x+1$	
$+$	$-\frac{2}{3}(1-x)^{\frac{3}{2}}$
2	
$-$	$+\frac{4}{15}(1-x)^{\frac{5}{2}}$
0	

$$= -\frac{2}{3}(2x+1)(1-x)^{3/2} - \frac{8}{15}(1-x)^{5/2} + C$$

$$= -\frac{2}{3}(2x+1)\sqrt{(1-x)^3} - \frac{8}{15}\sqrt{(1-x)^5} + C$$

4) $\int x^2 (2^x)\,dx$

Sol:

$$\int x^2 2^x dx = \frac{x^2}{\ln 2}(2^x) - \frac{2x}{(\ln 2)^2}(2^x) + \frac{2^x}{(\ln 2)^3} + C$$

	2^x
x^2	
$+$	$\frac{1}{\ln 2}2^x$
$2x$	
$-$	$\frac{1}{(\ln 2)^2}2^x$
2	
$+$	$\frac{1}{(\ln 2)^3}2^x$
0	

5) $\int \frac{5x^2+1}{ex}\,dx$

Sol:

$$\int \frac{5x^2+1}{ex}dx = \int (5x^2+1)e^{-x}dx$$

	e^{-x}
$5x^2+1$	
$+$	$-e^{-x}$
$10x$	
$-$	e^{-x}
10	
$+$	$-e^{-x}$
0	

$$= -(5x^2+1)e^{-x} - 10xe^{-x} - 10e^{-x} + C$$

$$= -(5x^2+10x+11)e^{-x} + C$$

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6) $\int x \sinh 3x\,dx$

Sol:

$$\int x \sinh 3x\,dx = \frac{1}{3}x\cosh 3x - \frac{1}{9}\sinh 3x + C$$

x		$\sinh 3x$
	$+$	
1		$\frac{1}{3}\cosh 3x$
	$-$	
0		$\frac{1}{9}\sinh 3x$

7) $\int \frac{5x+1}{\sec 3x}\,dx$

Sol:

$$\int \frac{5x+1}{\sec 3x}\,dx = \int (5x+1)\cos 3x\,dx$$

$$= \frac{1}{3}(5x+1)\sin 3x + \frac{5}{9}\cos 3x + C$$

$5x+1$		$\cos 3x$
	$+$	
5		$\frac{1}{3}\sin 3x$
	$-$	
0		$\frac{-1}{9}\cos 3x$

Group 2:

If the integrand contains logarithmic function or inverse trig. function, first try by substitution

- works → done
- does not work → try by part assuming the log. fun. or the inverse trig. u and the rest dv and apply

$$\int u\,dv = uv - \int v\,du$$

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e.g: Evaluate

1) $\int \ln x \, dx$

Sol:

$u = \ln x \longrightarrow du = \frac{1}{x} dx$

$dv = dx \Longrightarrow v = x$

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} dx$$

$$= x \ln x - \int dx$$

$$= x \ln x - x + C$$

Note $\int \ln x^n dx = n \int \ln x \, dx$

2) $\int (\ln x)^2 dx = \int \ln^2 x \, dx$

Sol:

$u = (\ln x)^2 \Longrightarrow du = 2(\ln x)(\frac{1}{x}) dx$

$dv = dx \longrightarrow v = x$

$$\Rightarrow \int (\ln x)^2 dx = x(\ln x)^2 - 2\underbrace{\int \ln x \, dx}_{\text{by parts}}$$

$$= x(\ln x)^2 - 2(x \ln x - x) + C \quad \text{(by the previous example)}$$

$$= x(\ln x)^2 - 2x \ln x + 2x + C$$

Ex: assume $y = \ln x \Rightarrow \frac{dy}{dx} = \frac{1}{x} \Rightarrow dx = x \, dy$

and solve $\int (\ln x)^2 dx$

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3) $\int \frac{\ln x}{x^2}\,dx$

Sol:

$$\int \frac{\ln x}{x^2}\,dx = \int x^{-2}\ln x\,dx$$

$u = \ln x \longrightarrow du = \frac{1}{x}\,dx$

$dv = x^{-2}dx \Longrightarrow v = -\frac{1}{x}$

$$\Rightarrow \int \frac{\ln x}{x^2}\,dx = -\frac{\ln x}{x} + \int \frac{1}{x^2}\,dx$$

$$= -\frac{\ln x}{x} - \frac{1}{x} + C.$$

Similarly you can integrate $\int \frac{\ln x}{x^n}\,dx$, $n \neq +1$

4) $\int x \log_2(x+1)\,dx$

Sol:

$$\int x \log_2(x+1)\,dx$$

$u = \log_2(x+1) \Longrightarrow du = \frac{1}{(x+1)\ln 2}\,dx$

$dv = x\,dx \Longrightarrow v = \frac{x^2}{2}$

$$\Rightarrow \int x \log_2(x+1)\,dx = \frac{x^2}{2}\log_2(x+1) - \frac{1}{2\ln 2}\int \frac{x^2}{x+1}\,dx$$

long division

$$x+1 \;\big|\; x^2 \quad (x-1)$$
$$\mp x^2 \mp x$$
$$-x$$
$$\pm x \pm 1$$
$$1$$

$$= \frac{1}{2}x^2\log_2(x+1) - \frac{1}{2\ln 2}\left[\int (x-1)\,dx + \int \frac{1}{x+1}\,dx\right]$$

$$= \frac{1}{2}x^2\log_2(x+1) - \frac{1}{2\ln 2}\left[\frac{x^2}{2} - x + \ln(x+1)\right] + C$$

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5) $\int_0^1 \tan^{-1} x \, dx$

Sol:

$u = \tan^{-1} x \Rightarrow du = \frac{1}{1+x^2} dx$

$dv = dx \Rightarrow v = x$

$$\int_0^1 \tan^{-1} x \, dx = x \tan^{-1} x \Big|_0^1 - \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} dx$$

$$= \tan^{-1}(1) - 0 \tan^{-1} 0 - \frac{1}{2} \ln(1+x^2) \Big|_0^1$$

$$= \frac{\pi}{4} - 0 - \frac{1}{2} [\ln 2 - \ln 1]$$

$$= \frac{\pi}{4} - \frac{\ln 2}{2} = \frac{\pi}{4} - \ln \sqrt{2}$$

6) $\int \frac{\cot^{-1} \sqrt{x}}{\sqrt{x}} dx$

Sol:

$u = \cot^{-1} \sqrt{x} \Rightarrow du = -\frac{\frac{1}{2\sqrt{x}}}{1+x} dx$

$dv = \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}} dx \Rightarrow v = 2\sqrt{x}$

$$\Rightarrow \int \frac{\cot^{-1} \sqrt{x}}{\sqrt{x}} dx = 2\sqrt{x} \cot^{-1} \sqrt{x} + \int \frac{1}{1+x} dx$$

$$= 2\sqrt{x} \cot^{-1} \sqrt{x} + \ln(1+x) + C$$

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7) $\int \sin^{-1} x \, dx$

Sol:

$u = \sin^{-1} x \Longrightarrow du = \frac{1}{\sqrt{1-x^2}} dx$

$dv = dx \Longrightarrow v = x$

$\therefore \int \sin^{-1} x \, dx = x \sin^{-1} x - \left(\frac{-1}{2}\right) \int \frac{-2x}{\sqrt{1-x^2}} dx$

$= x \sin^{-1} x + \frac{1}{2} \int (-2x)(1-x^2)^{-\frac{1}{2}} dx$ $\quad (u = 1-x^2)$

$= x \sin^{-1} x + \frac{1}{2} \cdot \frac{2}{1} (1-x^2)^{\frac{1}{2}} + C$

$= x \sin^{-1} x + \sqrt{1-x^2} + C$

Group 3:

$\int e^{\text{Linear}} \cos(\text{Linear}) \, , \int e^{\text{Linear}} \sin(\text{Linear}) \, ,$

$\int a^{\text{Linear}} \cos(\text{Linear}) \, , \int a^{\text{Linear}} \sin(\text{Linear})$

$\int \sin(\ln x) \, dx \, , \int \cos(\ln x) \, dx$

all these can be solved by using integration by parts twice.

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e.g: Evaluate

1) $\int e^{2x}\cos 3x\,dx$

$u = e^{2x} \Rightarrow du = 2e^{2x}\,dx$

$dv = \cos 3x\,dx \Rightarrow v = \frac{1}{3}\sin 3x$

$\Rightarrow \int e^{2x}\cos 3x\,dx = \frac{1}{3}e^{2x}\sin 3x - \frac{2}{3}\int e^{2x}\sin 3x\,dx \quad \cdots (*)$

by Parts

$u = e^{2x} \Rightarrow du = 2e^{2x}\,dx$

$dv = \sin 3x\,dx \Rightarrow v = -\frac{1}{3}\cos 3x$

$\int e^{2x}\sin 3x\,dx = -\frac{1}{3}e^{2x}\cos 3x + \frac{2}{3}\int e^{2x}\cos 3x\,dx$

Now sub. in (*) to get

$\int e^{2x}\cos 3x\,dx = \frac{1}{3}e^{2x}\sin 3x - \frac{2}{3}\left(-\frac{1}{3}e^{2x}\cos 3x + \frac{2}{3}\int e^{2x}\cos 3x\,dx\right)$

$\Rightarrow \int e^{2x}\cos 3x\,dx = \frac{1}{3}e^{2x}\sin 3x + \frac{2}{9}e^{2x}\cos 3x - \frac{4}{9}\int e^{2x}\cos 3x\,dx$

$\Rightarrow \left(1+\frac{4}{9}\right)\int e^{2x}\cos 3x\,dx = \frac{1}{3}e^{2x}\sin 3x + \frac{2}{9}e^{2x}\cos 3x$

$\Leftrightarrow \frac{13}{9}\int e^{2x}\cos 3x\,dx = \frac{1}{3}e^{2x}\sin 3x + \frac{2}{9}e^{2x}\cos 3x$

$\times \frac{9}{13} \Rightarrow \int e^{2x}\cos 3x\,dx = \frac{3}{13}e^{2x}\sin 3x + \frac{2}{13}e^{2x}\cos 3x + C$

Ex : Evaluate

1) $\int e^{ax}\cos bx\,dx$ 2) $\int e^{ax}\sin bx\,dx$

a and b are non-zero constants

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2) $\int \sin(\ln x)\,dx$

Sol:

$u = \sin(\ln x) \Rightarrow du = \frac{1}{x}\cos(\ln x)\,dx$

$dv = dx \Rightarrow v = x$

$\int \sin(\ln x)\,dx = x\sin(\ln x) - \int \cos(\ln x)\,dx \quad \cdots (*)$

by parts

$u = \cos(\ln x) \Rightarrow du = -\frac{1}{x}\sin(\ln x)\,dx$

$dv = dx \Rightarrow v = x$

$\therefore \int \cos(\ln x)\,dx = x\cos(\ln x) + \int \sin(\ln x)\,dx$

Now sub. in (*)

$\Rightarrow \int \sin(\ln x)\,dx = x\sin(\ln x) - x\cos(\ln x) - \int \sin(\ln x)\,dx$

$\Rightarrow 2\int \sin(\ln x)\,dx = x\sin(\ln x) - x\cos(\ln x)$

$\div 2 \Rightarrow \int \sin(\ln x)\,dx = \frac{1}{2}x\sin(\ln x) - \frac{1}{2}x\cos(\ln x) + C$

Ex: Solve this example by assuming $y = \ln x$ first.

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More Examples

e.g: Evaluate

1) $\int x \sec^2 x \, dx$

$u = x \Rightarrow du = dx$

$dv = \sec^2 x \, dx \Rightarrow v = \tan x$

$\Rightarrow \int x \sec^2 x \, dx = x \tan x - \int \tan x \, dx$

$= x \tan x + \int \frac{-\sin x}{\cos x} dx$

$= x \tan x + \ln|\cos x| + C$

2) $\int x^3 e^{x^2} dx$

Let $t = x^2 \Rightarrow dt = 2x \, dx \Rightarrow dx = \frac{dt}{2x}$

$\therefore \int x^3 e^{x^2} dx = \int x^3 e^t \cdot \frac{dt}{2x}$

$= \frac{1}{2} \int x^2 e^t dt$

$= \frac{1}{2} \int t e^t dt$

t		e^t
	+	
1		e^t
	−	
0		e^t

$= \frac{1}{2}(t e^t - e^t) + C$

$= \frac{1}{2} x^2 e^{x^2} - \frac{1}{2} e^{x^2} + C$

3) $\int \frac{x e^x}{(x+1)^2} dx$

$u = x e^x \Rightarrow du = [x e^x + e^x] dx$

$dv = \frac{1}{(x+1)^2} dx \Rightarrow v = \frac{-1}{x+1}$

$\therefore \int \frac{x e^x}{(x+1)^2} dx = \frac{-x e^x}{x+1} + \int \frac{+e^x (x+1)}{x+1} dx$

$= \frac{-x e^x}{x+1} + e^x + C$

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e.g: Given that $f(1)=3$, $f(2)=5$, $f'(1)=4$ and $f'(2)=10$ → evaluate

$$\int_1^2 x\, f''(x)\,dx$$

sol:

$u = x \Rightarrow du = dx$

$dv = f''(x)\,dx \Rightarrow v = f'(x)$

$$\Rightarrow \int_1^2 x f''(x)\,dx = x f'(x)\Big]_1^2 - \int_1^2 f'(x)\,dx$$

$$= 2f'(2) - f'(1) - f(x)\Big]_1^2$$

$$= 2(10) - 4 - [f(2) - f(1)]$$

$$= 16 - (5-3)$$

$$= 14$$

The following Integrals can be solved, first by substitution then by parts.

$$\int e^{\sqrt[n]{x}}\,dx \quad , \int a^{\sqrt[n]{x}}\,dx \quad , \int \sin\sqrt[n]{x}\,dx \quad , \int \cos\sqrt[n]{x}\,dx$$

e.g: Evaluate $\int \sin\sqrt{x}\,dx$

sol:

Let $t = \sqrt{x} \Rightarrow \frac{dt}{dx} = \frac{1}{2\sqrt{x}} \rightarrow dx = 2\sqrt{x}\,dt$

$$\Rightarrow \int \sin\sqrt{x}\,dx = \int 2\sqrt{x}\sin t\,dt$$

$$= \int 2t \sin t\,dt$$

$$= -2t\cos t + 2\sin t + C$$

$$= -2\sqrt{x}\cos\sqrt{x} + 2\sin\sqrt{x} + C$$

$2t$	$\sin t$	
2	$-\cos t$	+
0	$-\sin t$	−

END

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7.3 Integrating Trigonometric Functions: (Part 1)

1 Integrals of the forms:

$$\int \sin^n x\,dx \quad , \quad \int \cos^n x\,dx \qquad n = 2,3,4,\cdots$$

Case 1 : n is an odd number

Use $\boxed{\cos^2 A + \sin^2 A = 1}$

and Integration by substitution.

e.g.: Evaluate

1) $\int \sin^3 2x\,dx$

Sol:

$$\int \sin^3 2x\,dx = \int (\sin 2x)\sin^2 2x\,dx$$

$$= \int (\sin 2x)(1 - \cos^2 2x)\,dx$$

Let $t = \cos 2x \Rightarrow dt = -2\sin 2x\,dx$

$$\Rightarrow dx = \frac{dt}{-2\sin 2x}$$

$$\therefore \int \sin^3 2x\,dx = \int (\sin 2x)(1 - t^2)\cdot \frac{dt}{-2\sin 2x}$$

$$= \frac{1}{2}\int (t^2 - 1)\,dt$$

$$= \frac{1}{2}\left(\frac{t^3}{3} - t\right) + C$$

$$= \frac{1}{6}\cos^3 2x - \frac{1}{2}\cos 2x + C$$

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2) $\int \cos^5 x \, dx$

sol:

$$\int \cos^5 x \, dx = \int \cos x (\cos^2 x)^2 dx$$

$$= \int (\cos x)(1 - \sin^2 x)^2 dx$$

Let $t = \sin x \Rightarrow dt = \cos x \, dx$

$$\Rightarrow dx = \frac{dt}{\cos x}$$

$$\therefore \int \cos^5 x \, dx = \int (\cos x)(1 - t^2)^2 \cdot \frac{dt}{\cos x}$$

$$= \int (1 - 2t^2 + t^4) dt$$

$$= t - \frac{2}{3}t^3 + \frac{1}{5}t^5 + C$$

$$= \sin x - \frac{2}{3}\sin^3 x + \frac{1}{5}\sin^3 x + C$$

Case 2: n is even

Use $\boxed{\sin^2 A = \frac{1}{2}(1 - \cos 2A)}$

or

$\boxed{\cos^2 A = \frac{1}{2}(1 + \cos 2A)}$

e.g: Evaluate

1) $\int \sin^2 3x \, dx$

sol:

$$\int \sin^2 3x \, dx = \frac{1}{2}\int (1 - \cos 6x) + C$$

$$= \frac{1}{2}\left(x - \frac{1}{6}\sin 6x\right) + C$$

$$= \frac{1}{2}x - \frac{1}{12}\sin 6x + C$$

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2) $\int \cos^2\left(\frac{x}{2}\right) dx$

Sol:

$$\int \cos^2\left(\frac{x}{2}\right) dx = \frac{1}{2}\int (1 + \cos x)\, dx$$

$$= \frac{1}{2}(x + \sin x) + C$$

$$= \frac{1}{2}x + \frac{1}{2}\sin x + C$$

3) $\int \cos^4 x\, dx$

Sol:

$$\int \cos^4 x\, dx = \int (\cos^2 x)^2 dx$$

$$= \int \left[\frac{1}{2}(1 + \cos 2x)\right]^2 dx$$

$$= \frac{1}{4}\int (1 + \cos 2x)^2 dx$$

$$= \frac{1}{4}\int (1 + 2\cos 2x + \cos^2 2x)\, dx$$

$$= \frac{1}{4}\int \left(1 + 2\cos 2x + \frac{1}{2}(1 + \cos 4x)\right) dx$$

$$= \frac{1}{4}\int \left(\frac{3}{2} + 2\cos 2x + \frac{1}{2}\cos 4x\right) dx$$

$$= \frac{1}{4}\left(\frac{3}{2}x + \sin 2x + \frac{1}{8}\sin 4x\right) + C$$

$$= \frac{3}{8}x + \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C$$

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2 Integrals of the forms:

$$\int \sin^n x \cos^m x \, dx$$

* If at least one of n or m is odd
  use case 1 of 1

* If n and m are even numbers
  use case 2 of 1

e.g: Evaluate

1) $\int \cos^7 x \sin^3 x \, dx$

$$\int \cos^7 x \sin^3 x \, dx = \int (\cos^7 x)(\sin x)(\sin^2 x) \, dx$$

$$= \int (\sin x)(\cos^7 x)(1 - \cos^2 x) \, dx$$

Let $t = \cos x \Rightarrow dt = -\sin x \, dx$

$$\Rightarrow dx = \frac{dt}{-\sin x}$$

$$\therefore \int \cos^7 x \sin^3 x \, dx = \int (\sin x)(t^7)(1 - t^2) \frac{dt}{-\sin x}$$

$$= \int t^7 (t^2 - 1) \, dt$$

$$= \int (t^9 - t^7) \, dt$$

$$= \frac{1}{10} t^{10} - \frac{1}{8} t^8 + C$$

$$= \frac{1}{10} \cos^{10} x - \frac{1}{8} \cos^8 x + C$$

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2) $\int \frac{\cos^3 x}{\sqrt{\sin x}}\,dx$

Sol:

$$\int \frac{\cos^3 x}{\sqrt{\sin x}}\,dx = \int \cos^3 x\,(\sin x)^{-\frac{1}{2}}\,dx$$

$$= \int (\cos x)\cos^2 x\,(\sin x)^{-\frac{1}{2}}\,dx$$

$$= \int (\cos x)(1-\sin^2 x)(\sin x)^{-\frac{1}{2}}\,dx$$

Let $t = \sin x \Rightarrow dt = \cos x\,dx$

$$\Rightarrow dx = \frac{dt}{\cos x}$$

$$\int \frac{\cos^3 x}{\sqrt{\sin x}}\,dx = \int (\cos x)(1-t^2)\,t^{-\frac{1}{2}}\,\frac{dt}{\cos x}$$

$$= \int (t^{-\frac{1}{2}} - t^{3/2})\,dt$$

$$= 2t^{\frac{1}{2}} - \frac{2}{5}t^{5/2} + C$$

$$= 2\sqrt{\sin x} - \frac{2}{5}\sqrt{\sin^5 x} + C$$

3) $\int \sin^2 x \cos^2 x\,dx$

Sol:

$$\int \sin^2 x \cos^2 x\,dx = \int \frac{1}{2}(1-\cos 2x)\cdot\frac{1}{2}(1+\cos 2x)\,dx$$

$$= \frac{1}{4}\int (1-\cos^2 2x)\,dx$$

$$= \frac{1}{4}\int \sin^2 2x\,dx$$

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$$\therefore \int \sin^2 x \cos^2 x = \frac{1}{4} \cdot \frac{1}{2} \int (1 - \cos 4x)\, dx$$

$$= \frac{1}{8}\left(x - \frac{1}{4}\sin 4x\right) + C$$

$$= \frac{1}{8}x - \frac{1}{32}\sin 4x + C$$

Another solution

$$\int \sin^2 x \cos^2 x\, dx = \int (\sin x \cos x)^2 dx$$

$\sin 2x = 2\sin\cos$

$$= \int \left(\frac{1}{2}\sin 2x\right)^2 dx$$

$$= \frac{1}{4}\int \sin^2 2x\, dx$$

$$\vdots$$

3 Integrals of the forms

$$\int \sec^n x \tan^m x\, dx$$

$1 + \tan^2 A = \sec^2 A$

similarly

or $\int \csc^n x \cot^m x\, dx$

$1 + \cot^2 A = \csc^2 A$

Case 1: $\int \sec^{\text{even} > 0} \tan^{\text{any real}}$

$$\int \sec^2 (\ldots \tan \ldots)$$

t

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e.g Evaluate

$$\int \sec^4 x \tan^2 x \, dx$$

Sol:

$$\int \sec^4 x \tan^2 x \, dx = \int (\sec^2 x) \sec^2 x \tan^2 x \, dx$$

$$= \int (\sec^2 x)(1 + \tan^2 x) \tan^2 x \, dx$$

Let $t = \tan x \Rightarrow dt = \sec^2 x \, dx$

$$\Rightarrow dx = \frac{dt}{\sec^2 x}$$

$$\int \sec^4 x \tan^2 x \, dx = \int (\cancel{\sec^2 x})(1 + t^2) t^2 \cdot \frac{dt}{\cancel{\sec^2 x}}$$

$$= \int (t^2 + t^4) \, dt$$

$$= \frac{1}{3} t^3 + \frac{1}{5} t^5 + C$$

$$= \frac{1}{3} \tan^3 x + \frac{1}{5} \tan^5 x + C$$

Case 2:

$$\int \sec^{\text{odd}} \tan^{\text{odd}}$$

$$= \int (\sec \tan)(\cdots \sec \cdots)$$

t

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e.g: Evaluate

1) $\int \sec^7 x \tan^3 x \, dx$

Sol:

$$\int \sec^7 x \tan^3 x \, dx = \int (\sec x \tan x) \sec^6 x \tan^2 x \, dx$$

$$= \int (\sec x \tan x) \sec^6 x (\sec^2 x - 1) \, dx$$

Let $t = \sec x \Rightarrow dt = \sec x \tan x \, dx$

$$\Rightarrow dx = \frac{dt}{\sec x \tan x}$$

$$\therefore \int \sec^7 x \tan^3 x \, dx = \int (\sec x \tan x) \, t^6 (t^2 - 1) \cdot \frac{dt}{\sec x \tan x}$$

$$= \int (t^8 - t^6) \, dt$$

$$= \frac{1}{9} t^9 - \frac{1}{7} t^7 + C$$

$$= \frac{1}{9} \sec^9 x - \frac{1}{7} \sec^7 x + C$$

2) $\int \sec x \tan^5 x \, dx$

Sol:

$$\int \sec x \tan^5 x \, dx = \int (\sec x \tan x)(\tan^2 x)^2 \, dx$$

$$= \int (\sec x \tan x)(\sec^2 x - 1)^2 \, dx$$

Let $t = \sec x \Rightarrow dt = \sec x \tan x \, dx$

$$\Rightarrow dx = \frac{dt}{\sec x \tan x}$$

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$$\therefore \int \sec x \tan^5 x \, dx = \int (\sec x \tan x)(t^2-1)^2 \cdot \frac{dt}{\sec x \tan x}$$

$$= \int (t^4 - 2t^2 + 1) dt$$

$$= \frac{1}{5}t^5 - \frac{2}{3}t^3 + t + C$$

$$= \frac{1}{5}\sec^5 x - \frac{2}{3}\sec^3 x + \sec x + C$$

Case 3: $\int \sec^{\text{odd}} \tan^{\text{even (or zero)}}$ ⟶ by Parts

e.g: $\int \sec^3 x \, dx$

Sol:

$$\int \sec^3 x \, dx = \int \sec x \sec^2 x \, dx$$

$u = \sec x \longrightarrow du = \sec x \tan x \, dx$

$dv = \sec^2 x \, dx \longrightarrow v = \tan x$

$$\therefore \int \sec^3 x \, dx = \sec x \tan x - \int \sec x \tan^2 x \, dx$$

$$= \sec x \tan x - \int \sec x (\sec^2 x - 1) dx$$

$$\int \sec^3 x \, dx = \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx$$

$$2\int \sec^3 x \, dx = \sec x \tan x + \ln|\sec x + \tan x|$$

$$\therefore \int \sec^3 x \, dx = \frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x| + C$$

- END -

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Integrating Trigonometric Functions:

(Part 2)

3 Integrals of the forms

$\int \sin A \cos B$, $\int \cos A \cos B$ & $\int \sin A \sin B$

Important Identities:

1) $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

2) $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$

3) $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$

Recall: $\sin(-x) = -\sin x$, $\cos(-x) = \cos x$

e.g: Evaluate

1) $\int \sin 3x \cos 5x \, dx$

sol:

$$\int \sin 3x \cos 5x \, dx = \frac{1}{2}\int [\sin(3x+5x) + \sin(3x-5x)]\, dx$$

$$= \frac{1}{2}\int (\sin 8x - \sin 2x)\, dx$$

$$= \frac{1}{2}\left[-\frac{1}{8}\cos 8x + \frac{1}{2}\cos 2x\right] + C$$

$$= -\frac{1}{16}\cos 8x + \frac{1}{4}\cos 2x + C$$

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2) $\int \cos 5x \cos x \, dx$

Sol:

$$\int \cos 5x \cos x \, dx = \frac{1}{2}\int [\cos 4x + \cos 6x] \, dx$$

$$= \frac{1}{2}\left[\frac{1}{4}\sin 4x + \frac{1}{6}\sin 6x\right] + C$$

$$= \frac{1}{8}\sin 4x + \frac{1}{12}\sin 6x + C$$

3) $\int \sin 2x \sin 3x \, dx$

Sol:

$$\int \sin 2x \sin 3x \, dx = \frac{1}{2}\int [\cos x - \cos 5x] \, dx$$

$$= \frac{1}{2}\left[\sin x - \frac{1}{5}\sin 5x\right] + C$$

$$= \frac{1}{2}\sin x - \frac{1}{10}\sin 5x + C$$

4) $\int_0^{\pi} \cos nx \cos mx \, dx$ n and m are two distinct integers

Sol:

$$\int_0^{\pi} \cos nx \cos mx \, dx = \frac{1}{2}\int_0^{\pi} [\cos(n-m)x + \cos(n+m)x] \, dx$$

$$= \frac{1}{2}\left[\frac{1}{n-m}\sin(n-m)x + \frac{1}{n+m}\sin(n+m)x\right]_0^{\pi}$$

$$= \frac{1}{2}[0 + 0 - (0+0)]$$

$$= 0$$

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Reduction Formulas:

$I_n = f(I_{n-1})$, $f(I_{n-2})$ or $f(I_{n-1}, I_{n-2})$, ...

n non-negative integer.

e.g: Given that

$$I_n = \int \sin^n x\,dx$$, show that

$$\int \sin^n x\,dx = -\frac{1}{n}\sin^{n-1}x\cos x + \frac{n-1}{n}\int \sin^{n-2}x\,dx$$

OR $\left(I_n = -\frac{1}{n}\sin^{n-1}x\cos x + \frac{n-1}{n}I_{n-2}\right)$

$n = 2, 3, 4, 5, \ldots$

Sol:

$$I_n = \int \sin^n x\,dx = \int \underbrace{\sin^{n-1}x \sin x\,dx}_{\text{by Parts}}$$

$u = \sin^{n-1}x \Rightarrow du = (n-1)\sin^{n-2}x\cos x\,dx$

$dv = \sin x\,dx \Rightarrow v = -\cos x$

$$\int \sin^n x\,dx = -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x\cos^2 x\,dx$$

$$\Rightarrow \int \sin^n x\,dx = -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x(1-\sin^2 x)\,dx$$

$$\Rightarrow \int \sin^n x\,dx = -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x\,dx - (n-1)\int \sin^n x\,dx$$

$$\therefore n\int \sin^n x\,dx = -\sin^{n-1}x\cos x + (n-1)\int \sin^{n-2}x\,dx$$

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$\div n \Rightarrow$ $\int \sin^n x\,dx = -\frac{1}{n}\sin^{n-1}x\cos x + \frac{n-1}{n}\int \sin^{n-2}x\,dx$ (4)

i.e $I_n = -\frac{1}{n}\sin^{n-1}x\cos x + \frac{n-1}{n}I_{n-2}$

e.g: Use the reduction formula in the above example to compute $\int \sin^5 x\,dx$

Sol:

$$\int \sin^5 x\,dx = I_5$$

$$\Rightarrow I_5 = -\frac{1}{5}\sin^{5-1}x\cos x + \frac{5-1}{5}\underbrace{\int \sin^3 x\,dx}_{I_3}$$

$$I_5 = -\frac{1}{5}\sin^4 x\cos x + \frac{4}{5}\underbrace{\int \sin^3 x\,dx}_{n=3}$$

$$= -\frac{1}{5}\sin^4 x\cos x + \frac{4}{5}\left[-\frac{1}{3}\sin^2 x\cos x + \frac{2}{3}\int \sin x\,dx\right]$$

$$\int \sin^5 x\,dx = -\frac{1}{5}\sin^4 x\cos x - \frac{4}{15}\sin^2 x\cos x - \frac{8}{15}\cos x + C$$

Ex: Prove that

$$\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1}x\sin x + \frac{n-1}{n}\int \cos^{n-2}x\,dx$$

and use it to compute

$$\int \cos^7 x\,dx .$$

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e.g! Prove that

$$\int \sec^n x\,dx = \frac{1}{n-1}\sec^{n-2}x\tan x + \frac{n-2}{n-1}\int \sec^{n-2}x\,dx$$

$n = 2, 3, 4, \dots$

Proof!

$$\int \sec^n x\,dx = \int \underbrace{\sec^{n-2}x}_{u}\ \underbrace{\sec^2 x\,dx}_{dv} \qquad \text{by Parts}$$

$u = \sec^{n-2}x \Rightarrow du = (n-2)\sec^{n-3}x\sec x\tan x\,dx$

$= (n-2)\sec^{n-2}x\tan x\,dx$

$dv = \sec^2 x\,dx \Rightarrow v = \tan x$

$$\int \sec^n x\,dx = \sec^{n-2}x\tan x - (n-2)\int \sec^{n-2}x\tan^2 x\,dx$$

$$= \sec^{n-2}x\tan x - (n-2)\int \sec^{n-2}x(\sec^2 x - 1)\,dx$$

$$\Rightarrow \int \sec^n x\,dx = \sec^{n-2}x\tan x - (n-2)\int \sec^n x\,dx - (n-2)\int \sec^{n-2}x\,dx$$

$$(n-1)\int \sec^n x\,dx = \sec^{n-2}x\tan x - (n-2)\int \sec^{n-2}x\,dx$$

$\div (n-1)$ $\Longrightarrow$

$$\int \sec^n x\,dx = \frac{1}{n-1}\sec^{n-2}x\tan x - \frac{n-2}{n-1}\int \sec^{n-2}x\,dx$$

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e.g: Use the result of the above example to evaluate $\int \sec^5 x\,dx$

Sol:

$$\int \sec^n x\,dx = \frac{1}{n-1}\sec^{n-2}x\tan x - \frac{n-2}{n-1}\int \sec^{n-2}x\,dx$$

$n = 5 \Rightarrow$

$$\int \sec^5 x\,dx = \frac{1}{4}\sec^3 x\tan x - \frac{3}{4}\int \sec^3 x\,dx$$

($n = 3$)

$$= \frac{1}{4}\sec^3 x\tan x - \frac{3}{4}\left[\frac{1}{2}\sec x\tan x - \frac{1}{2}\int \sec x\,dx\right]$$

$$\therefore \int \sec^5 x\,dx = \frac{1}{4}\sec^3 x\tan x - \frac{3}{8}\sec x\tan x + \frac{3}{8}\ln|\sec x + \tan x| + C$$

e.g: show that:

$$\int \tan^n x\,dx = \frac{1}{n-1}\tan^{n-1}x - \int \tan^{n-2}x\,dx \qquad n = 2, 3, \ldots$$

Sol:

$$\int \tan^n x\,dx = \int \tan^{n-2}x\,\tan^2 x\,dx$$

$$= \int \tan^{n-2}x\,(\sec^2 x - 1)\,dx$$

$$= \int \tan^{n-2}x\sec^2 x\,dx - \int \tan^{n-2}x\,dx$$

by substitution $u = \tan x$

$$\therefore \int \tan^n x\,dx = \frac{1}{n-1}\tan^{n-1}x - \int \tan^{n-2}x\,dx$$

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EX: Use the formula in the above example to evaluate

$$\int \tan^4 x \, dx .$$

e.g: Given that $I_n = \int x^n e^{-x} dx$, $n = 1, 2, 3$

Show that $I_n = -x^n e^{-x} + n I_{n-1}$

Proof:

$$I_n = \int x^n e^{-x}$$

$$u = x^n \Longrightarrow du = n x^{n-1} dx$$

$$dv = e^{-x} dx \Longrightarrow v = -e^{-x}$$

$$\therefore I_n = -x^n e^{-x} + n \int x^{n-1} e^{-x} dx$$

$$\therefore I_n = -x^n e^{-x} + n I_{n-1}$$

EX: Use the formula in the above example to evaluate $\int_0^1 x^2 e^{-x} dx$

- END -

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7.4 Trigonometric Substitutions

Recall

① $\int f(x)dx = F(x) + C \Rightarrow$

$$\int f(ax+b)dx = \frac{1}{a}F(ax+b) + C$$

② Important identities

1. $\cos^2\theta + \sin^2\theta = 1$
2. $1 + \tan^2\theta = \sec^2\theta$
3. $1 + \cot^2\theta = \csc^2\theta$
4. $\sin 2\theta = 2\sin\theta\cos\theta$
5. $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$
6. $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$

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If the integral involves

$a > 0$

1) $\sqrt{a^2 - x^2}$ assume $x = a\sin\theta \rightarrow dx = a\cos\theta\, d\theta$

$$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

2) $\sqrt{a^2 + x^2}$ assume $x = a\tan\theta \rightarrow dx = a\sec^2\theta\, d\theta$

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

3) $\sqrt{x^2 - a^2}$ assume $x = a\sec\theta \rightarrow dx = a\sec\theta\tan\theta\, d\theta$

$0 \leq \theta < \frac{\pi}{2}$ if $x \geq a$

$\frac{\pi}{2} < \theta \leq \pi$ if $x \leq a$

$$\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2\sin^2\theta}$$

$$= a\sqrt{1 - \sin^2\theta}$$

$$= a\sqrt{\cos^2\theta}$$

$$= a\,|\cos\theta|$$

$$= a\cos\theta$$

$(a^2 + x^2)^n \longrightarrow x = a\tan\theta$

$(a^2 - x^2)^n \longrightarrow x = a\sin\theta$

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e.g: Evaluate .

$$\int \frac{dx}{\sqrt{4+x^2}}$$

$a^2 = 4$

$a = 2$

Let $x = 2\tan\theta$

$dx = 2\sec^2\theta\, d\theta$

$$\int \frac{dx}{\sqrt{4+x^2}} = \int \frac{2\sec^2\theta\, d\theta}{\sqrt{4+4\tan^2\theta}}$$

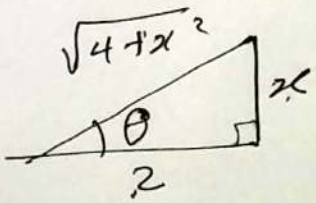
$$= \int \frac{2\sec^2\theta}{2\sqrt{1+\tan^2\theta}}\, d\theta$$

$$= \int \frac{\sec^2\theta}{\sec\theta}\, d\theta$$

$$= \int \sec\theta\, d\theta$$

$$= \ln|\sec\theta + \tan\theta| + C$$

but $x = 2\tan\theta \rightarrow \tan\theta = \frac{x}{2}$



$$\therefore \int \frac{dx}{\sqrt{4+x^2}} = \ln\left|\frac{\sqrt{4+x^2}}{2} + \frac{x}{2}\right| + C_1$$

$$= \ln\left|\frac{\sqrt{x^2+4}+x}{2}\right| + C_1$$

$$= \ln|\sqrt{x^2+4}+x| + C \quad .$$

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e.g ! Evaluate

$$\int \frac{\sqrt{16-x^2}}{x^2}\,dx$$

Let $x = 4\sin\theta \rightarrow dx = 4\cos\theta\,d\theta$

$$\therefore \int \frac{\sqrt{16-x^2}}{x^2}\,dx = \int \frac{\sqrt{16-16\sin^2\theta}}{16\sin^2\theta}\cdot 4\cos\theta\,d\theta$$

$$= \int \frac{4\sqrt{1-\sin^2\theta}}{4\sin^2\theta}\cos\theta\,d\theta$$

$$= \int \frac{4\cos\theta\cdot\cos\theta}{4\sin^2\theta}\,d\theta$$

$$= \int \frac{\cos^2\theta}{\sin^2\theta}\,d\theta$$

$$= \int \cot^2\theta\,d\theta \qquad \boxed{1+\cot^2\theta = \csc^2\theta}$$

$$= \int (\csc^2\theta - 1)\,d\theta$$

$$= -\cot\theta - \theta + C$$

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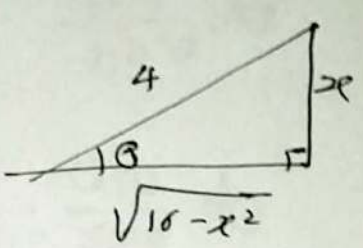
$x = 4\sin\theta \rightarrow \sin\theta = \frac{x}{4}$

$\theta = \sin^{-1}\left(\frac{x}{4}$



$\therefore \int \frac{\sqrt{16-x^2}}{x^2}\,dx = -\frac{\sqrt{16-x^2}}{x\theta} - \sin^{-1}\left(\frac{x}{4}\right) + C$

e.g: Evaluate $\int \frac{dx}{(x^2-4)^{3/2}}$

let $x = 2\sec\theta \rightarrow dx = 2\sec\theta\tan\theta\,d\theta$

$\therefore \int \frac{dx}{(x^2-4)^{3/2}} = \int \frac{2\sec\theta\tan\theta\,d\theta}{(4\sec^2\theta - 4)^{3/2}}$

$= \int \frac{2\sec\theta\tan\theta\,d\theta}{4^{3/2}(\sec^2\theta - 1)^{3/2}}$

$= \int \frac{2\sec\theta\tan\theta\,d\theta}{8(\tan^2\theta)^{3/2}}$

$= \frac{1}{4}\int \frac{\sec\theta\tan\theta}{\tan^3\theta}\,d\theta$ $\tan^2\theta$

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$$\therefore \int \frac{dx}{(x^2-4)^{3/2}} = \frac{1}{4}\int \frac{\sec\theta}{\tan^2\theta}\, d\theta$$

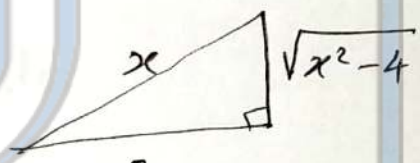
$$= \frac{1}{4}\int \frac{1}{\cos\theta} \cdot \frac{\cos^2\theta}{\sin^2\theta}\, d\theta$$

$$= \frac{1}{4}\int \frac{1}{\sin\theta} \cdot \frac{\cos\theta}{\sin\theta}\, d\theta$$

$$= \frac{1}{4}\int \csc\theta \cot\theta\, d\theta$$

$$= -\frac{1}{4}\csc\theta + C$$

$$x = 2\sec\theta \rightarrow \sec\theta = \frac{x}{2}$$



$$\therefore \int \frac{dx}{(x^2-4)^{3/2}} = -\frac{1}{4} \cdot \frac{x}{\sqrt{x^2-4}} + C$$

$$= \frac{-x}{4\sqrt{x^2-4}} + C$$

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e.g: Evaluate $\int_0^1 \sqrt{2-x^2}\,dx$

Let $x = \sqrt{2}\sin\theta \longrightarrow dx = \sqrt{2}\cos\theta\,d\theta$

when $x = 0 \longrightarrow \sin\theta = 0 \longrightarrow \theta = 0$

when $x = 1 \longrightarrow \sin\theta = \frac{1}{\sqrt{2}} \longrightarrow \theta = \frac{\pi}{4}$

$$\therefore \int_0^1 \sqrt{2-x^2}\,dx = \int_0^{\pi/4} \sqrt{2-2\sin^2\theta} \cdot \sqrt{2}\cos\theta\,d\theta$$

$$= \int_0^{\pi/4} \sqrt{2}\sqrt{1-\sin^2\theta} \cdot \sqrt{2}\cos\theta\,d\theta$$

$$= 2\int_0^{\pi/4} \cos^2\theta\,d\theta$$

$$= \frac{2}{2}\int_0^{\pi/4} (1+\cos 2\theta)\,d\theta$$

$$= \left[\theta + \frac{1}{2}\sin 2\theta\right]_0^{\pi/4}$$

$$= \frac{\pi}{4} + \frac{1}{2}\sin\frac{\pi}{2} - (0+0)$$

$$= \frac{\pi}{4} + \frac{1}{2}.$$

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e.g : Evaluate $\int \frac{x}{\sqrt{x^2-6x+25}}\,dx$

$$x^2-6x+25 = \underbrace{x^2-6x+9}-9+25$$

$$= (x-3)^2+16$$

$$\therefore \int \frac{x}{\sqrt{x^2-6x+25}}\,dx = \int \frac{x}{\sqrt{(x-3)^2+16}}\,dx$$

$$x-3 = 4\tan\theta \longrightarrow dx = 4\sec^2\theta\,d\theta$$

$$\longrightarrow x = 3+4\tan\theta$$

$$\therefore \int \frac{x}{\sqrt{x^2-6x+25}}\,dx = \int \frac{3+4\tan\theta}{\sqrt{16\tan^2\theta+16}} \cdot 4\sec^2\theta\,d\theta$$

$$= \int \frac{3+4\tan\theta}{4\sqrt{\tan^2\theta+1}} \cdot 4\sec^2\theta\,d\theta$$

$$= \int \frac{(3+4\tan\theta)\sec^2\theta}{\sec\theta}\,d\theta$$

$$= \int (3\sec\theta + 4\sec\theta\tan\theta)\,d\theta$$

$$= 3\ln|\sec\theta+\tan\theta| + 4\sec\theta + C$$

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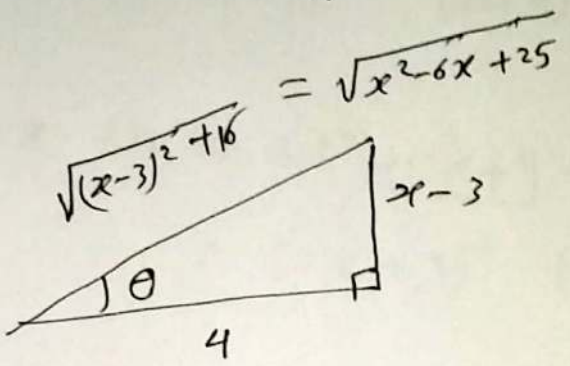
$x-3 = 4\tan\theta \rightarrow \tan\theta = \frac{x-3}{4}$



$\therefore \int \frac{x}{\sqrt{x^2+6x+25}} = 3\ln\left|\frac{\sqrt{x^2-6x+25}}{4} + \frac{x-3}{4}\right| +$

$\frac{\sqrt{x^2-6x+25}}{4} + C$

---

1) $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$

2) $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$

3) $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{x}{a}\right| + C$

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e.g: $\int \frac{dx}{4x^2+12x+11}$

$(a+b)^2 = a^2 + 2ab + b^2$

$4x^2 + 12x + 11 = \underline{4x^2+12x+9} - 9 + 11$

$= (2x+3)^2 + 2$

$\therefore \int \frac{dx}{4x^2+12x+11} = \int \frac{dx}{(2x+3)^2+2}$

$= \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{2x+3}{\sqrt{2}}\right) + C$

$= \frac{1}{2\sqrt{2}} \tan^{-1}\left(\frac{2x+3}{\sqrt{2}}\right) + C$

e.g: $\int \frac{\sin 2x}{5+\sin^4 x} dx = \int \frac{\sin 2x}{5+(\sin^2 x)^2} dx$

Let $t = \sin^2 x \rightarrow dt = 2\sin x \cos x \, dx$

$\leftrightarrow dx = \frac{dt}{\sin 2x}$

$\therefore \int \frac{\sin 2x}{5+\sin^4 x} dx = \int \frac{\sin 2x}{5+t^2} \cdot \frac{dt}{\sin 2x}$

$= \int \frac{dt}{5+t^2} = \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{t}{\sqrt{5}}\right) + C$

$= \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{\sin^2 x}{\sqrt{5}}\right) + C$.

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7.5 Integration by Partial Fractions:

(Integration of rational functions)

Introduction:

Prime (irreducible) Polynomials

- Linear $(ax+b)$
  like $2x+3$,
  x,
  $5x+10$,
  ...
- Quadratic (ax^2+bx+c) with negative discriminant
  i.e $\boxed{b^2-4ac<0}$
  like, x^2+x+1,
  x^2+4, ...

* Any polynomial of degree ≥ 3 is not prime (reducible)

e.g: Factorize the given polynomial as a product of prime factors

1) $f(x) = x^2-4$
   $= (x-2)(x+2)$

2) $f(x) = x^2+4$ prime (irreducible)

3) $f(x) = x^3-27$
   $= (x-3)(x^2+3x+9)$

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4) $f(x) = x^3 + 3x^2 + 6x + 18$

$= x^2(x+3) + 6(x+3)$

$= (x+3)(x^2+6)$

5) $f(x) = (x^2-1)(x^2-x-2)$

$= (x-1)(x+1)(x-2)(x+1)$

$= (x-1)(x-2)(x+1)^2$

6) $f(x) = x^4 + 2x^2 + 1$

$= (x^2+1)(x^2+1)$

$= (x^2+1)^2$

7) $f(x) = x^4 - 13x^2 + 36$

$= (x^2-4)(x^2-9)$

$= (x-2)(x+2)(x-3)(x+3)$

8) $f(x) = x^4 - 2x^2 - 3$

$= (x^2-3)(x^2+1)$

$= (x-\sqrt{3})(x+\sqrt{3})(x^2+1)$

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9) $f(x) = x^4 + 4$

$(a^2 \pm b)^2 = a^2 \pm 2ab + b^2$

$= x^4 + 4x^2 + 4 - 4x^2$

$= (x^2+2)^2 - 4x^2$

$= (x^2 + 2 - 2x)(x^2 + 2 + 2x)$

$= (x^2 - 2x + 2)(x^2 + 2x + 2)$

10) $f(x) = x^4 + 13x^2 + 36$

$= (x^2 + 4)(x^2 + 9)$

How to write the partial fraction decomposition

e.g: Write the partial fraction decomposition of:

1) $f(x) = \dfrac{3x+1}{(x-1)(x^2+1)}$

$= \dfrac{A}{x-1} + \dfrac{Bx+C}{x^2+1}$

2) $f(x) = \dfrac{5x+1}{x^3-8}$

$= \dfrac{5x+1}{(x-2)(x^2+2x+4)}$

$= \dfrac{A}{x-2} + \dfrac{Bx+C}{x^2+2x+4}$

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3| $f(x) = \dfrac{x}{(x-1)^3 (x^2+1)^2}$

$$= \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3} + \frac{Dx+E}{x^2+1} + \frac{Fx+G}{(x^2+1)^2}$$

---

* We use the integration by partial fractions to evaluate integrals of the form $\int \dfrac{\text{Polynomial}}{\text{Polynomial (reducible)}}$

such that the degree of the numerator must be less than the degree of the donominator, otherise use the long division first.

e.g.: Evaluate

1) $\int \dfrac{dx}{x^2-4}$

$$\frac{1}{x^2-4} = \frac{1}{(x-2)(x+2)}$$

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$$\left[\frac{1}{(x-2)(x+2)} = \frac{A}{x-2} + \frac{B}{x+2}\right] * (x-2)(x+2)$$

$$\boxed{1 = A(x+2) + B(x-2)} \quad \text{Identity.}$$

$x = 2 \longrightarrow 1 = 4A \longrightarrow \boxed{A = \frac{1}{4}}$

$x = -2 \longrightarrow 1 = -4B \longrightarrow \boxed{B = -\frac{1}{4}}$

$$\therefore \int \frac{dx}{x^2-4} = \int \frac{\frac{1}{4}}{x-2}\,dx + \int \frac{-\frac{1}{4}}{x+2}\,dx$$

$$= \frac{1}{4}\int \frac{1}{x-2}\,dx - \frac{1}{4}\int \frac{1}{x+2}\,dx$$

$$= \frac{1}{4}\ln|x-2| - \frac{1}{4}\ln|x+2| + C$$

$$= \frac{1}{4}\left[\ln|x-2| - \ln|x+2|\right] + C$$

$$= \frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C \quad .$$

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2) $\int \frac{3x^2+5x+1}{x^3+x}\,dx$

$$\left[\frac{3x^2+5x+1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}\right] * x(x^2+1)$$

$$3x^2+5x+1 = A(x^2+1) + (Bx+C)x$$

$x=0 \longrightarrow 1 = A \longrightarrow \boxed{A=1}$

$x=1 \longrightarrow 9 = 2A + B + C$

$9 = 2 + B + C \longrightarrow \boxed{B+C=7}$ — ①

$x=2 \longrightarrow 23 = 5A + 4B + 2C$

$23 = 5 + 4B + 2C \Rightarrow$

$4B + 2C = 18$

$\div 2 \longrightarrow \boxed{2B + C = 9}$ — ②

② − ① $\longrightarrow \boxed{B=2}$

Sub. in ① $\longrightarrow 2 + C = 7 \longrightarrow \boxed{C=5}$

$$\therefore \int \frac{3x^2+5x+1}{x^3+x}dx = \int \frac{1}{x}dx + \int \frac{2x+5}{x^2+1}dx$$

$$= \int \frac{1}{x}dx + \int \frac{2x}{x^2+1}dx + 5\int \frac{1}{x^2+1}dx$$

$$= \ln|x| + \ln(x^2+1) + 5\tan^{-1}x + C.$$

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$$3x^2+5x+1 = A(x^2+1)+(Bx+C)x$$

$$= Ax^2+A+Bx^2+Cx$$

$$3x^2+5x+1 = (A+B)x^2+Cx+A$$

$$\rightarrow \boxed{A=1}, \boxed{C=5}$$

$$A+B=3 \rightarrow 1+B=3$$

$$\rightarrow \boxed{B=2}$$

3) $\int \frac{3x^2+2x+1}{x^2-1}dx$

deg(num.) = deg(denom.) → long division

Quotient ← Q(x)

$$\begin{array}{r|l} & 3 \\ \hline x^2-1 & 3x^2+2x+1 \\ & \mp 3x^2 \pm 3 \\ \hline & 2x+4 \end{array}$$

g(x) | f(x)

r(x)

remainder

deg(r) < deg(g)

Stop.

$$\boxed{\int \frac{f(x)}{g(x)}dx = \int Q(x)dx + \int \frac{r(x)}{g(x)}dx}$$

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$$\int \frac{3x^2+2x+1}{x^2-1}dx = \int 3\,dx + \underbrace{\int \frac{2x+4}{x^2-1}dx} \quad \cdots (*)$$

partial fractions

$$\left[\frac{2x+4}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}\right] * (x-1)(x+1)$$

$$\rightarrow \boxed{2x+4 = A(x+1) + B(x-1)}$$

$x=1 \rightarrow 6 = 2A \rightarrow \boxed{A=3}$

$x=-1 \rightarrow 2 = -2B \rightarrow \boxed{B=-1}$

$$\therefore \int \frac{3x^2+2x+1}{x^2-1}dx = \int 3\,dx + 3\int \frac{1}{x-1}dx + (-1)\int \frac{1}{x+1}dx$$

$$= 3x + 3\ln|x-1| - \ln|x+1| + C$$

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4) $\int \frac{x}{x^3-x^2-x+1}\,dx$

$x^3-x^2-x+1 = x^2(x-1)-(x-1)$

$= (x-1)(x^2-1)$

$= (x-1)(x-1)(x+1)$

$= (x-1)^2(x+1)$

$$\left[\frac{x}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}\right] * (x-1)^2(x+1)$$

$$x = A(x-1)(x+1) + B(x+1) + C(x-1)^2$$

$x=1 \rightarrow 1 = 2B \rightarrow B = \frac{1}{2}$

$x=-1 \rightarrow -1 = 4C \rightarrow C = -\frac{1}{4}$

$x=0 \rightarrow 0 = -A + B + C$

$A = \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \rightarrow A = \frac{1}{4}$

$$\therefore \int \frac{x}{x^3-x^2-x+1}\,dx = \frac{1}{4}\int \frac{1}{x-1}\,dx + \frac{1}{2}\int \frac{1}{(x-1)^2}\,dx + \left(-\frac{1}{4}\right)\int \frac{1}{x+1}\,dx$$

$$= \frac{1}{4}\ln|x-1| + \frac{1}{2}\left(\frac{-1}{x-1}\right) - \frac{1}{4}\ln|x+1| + C$$

$$= \frac{1}{4}\ln|x-1| - \frac{1}{2(x-1)} - \frac{1}{4}\ln|x+1| + C$$

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e.g : Make a suitable substitution to convert the given integral into an integral of rational function.

1) $\int \frac{\cos x}{\sin^2 x - \sin x - 2}\, dx$

Let $u = \sin x \longrightarrow \frac{du}{dx} = \cos x \rightarrow dx = \frac{du}{\cos x}$

$\therefore \int \frac{\cos x}{\sin^2 x - \sin x - 2}\, dx = \int \frac{\cos x}{u^2 - u - 2} \cdot \frac{du}{\cos x}$

$= \int \frac{du}{(u-2)(u+1)}$

2) $\int \frac{dx}{x[(\ln x)^2 - \ln x^2 - 3]}\, dx$

Let $u = \ln x \longrightarrow \frac{du}{dx} = \frac{1}{x} \longrightarrow dx = x\,du$

$\int \frac{dx}{x[(\ln x)^2 - 2\ln x - 3]} = \int \frac{x\,du}{x[u^2 - 2u - 3]}$

$= \int \frac{du}{(u-3)(u+1)}$

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3) $\int \frac{dx}{e^{2x} - e^{x} - 2}$

$(e^x)^2 = e^{2x}$

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let $u = e^x \rightarrow \frac{du}{dx} = e^x \rightarrow dx = \frac{du}{e^x}$

$\therefore \int \frac{dx}{e^{2x} - e^{x} - 2} = \int \frac{1}{u^2 - u - 2} \cdot \frac{du}{e^x}$

$= \int \frac{du}{u(u^2 - u - 2)}$

$= \int \frac{du}{u(u-2)(u+1)}$

:

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7.8 Improper Integral:

Def:

1) Integral over $[a, \infty)$

$$\int_a^{\infty} f(x)\,dx = \lim_{l \to \infty} \int_a^{l} f(x)\,dx$$

- $A \in \mathbb{R}$ → the integral is convergent (converges to A)
- d.n.e → the integral is divergent. (diverges)

2) Integral over $(-\infty, b]$

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{l \to -\infty} \int_{l}^{b} f(x)\,dx$$

- $A \in \mathbb{R}$ → convergent (converges to A)
- d.n.e → is divergent (diverges)

3) Integral over $(-\infty, \infty)$

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_{c}^{\infty} f(x)\,dx$$

- converges to their sums ← [Conv. Conv.]
- diverges ← [else]

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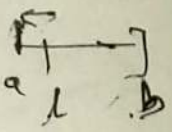
4) over $[a,b]$ such that f is continuous on $(a,b]$, f has infinite (essential) discontinuity at $x=a$

$$\int_a^b f(x)\,dx = \lim_{t\to a^+} \int_t^b f(x)\,dx$$

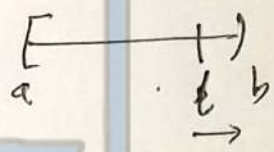
- $A \in \mathbb{R}$ → converges to A
- d.n.e → diverges



5) over $[a,b]$ such that f is continuous on $[a,b)$, f has infinite discontinuity (essential) at $x=b$

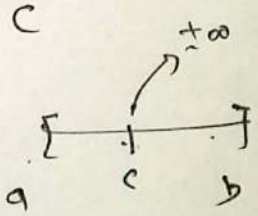
$$\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx$$

- $A \in \mathbb{R}$ → converges to A
- d.n.e → diverges



6) over $[a,b]$, $c \in (a,b)$

f is cont. on $[a,b]\setminus\{c\}$ and has infinite discontinuity (essential) at $x=c$



$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

- $\int_a^b$ → Convergent ← [Conv. Conv.]
- divergent ← [else]

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e.g: Evaluate the integrals that converge

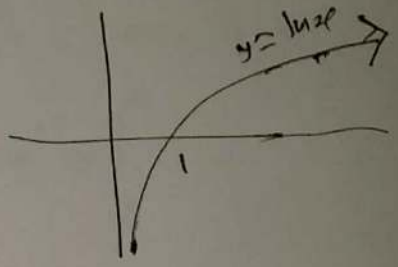
1) $\int_1^\infty \frac{1}{x}\,dx$

$$\int_1^\infty \frac{1}{x}\,dx = \lim_{l\to\infty} \int_1^l \frac{1}{x}\,dx$$

$$= \lim_{l\to\infty} \ln x \Big|_1^l$$

$$= \lim_{l\to\infty} (\ln l - \ln 1)$$

$$= \infty$$



$\ln 1 = 0$

$\ln \infty = \infty$

$\ln 0^+ = -\infty$

$\int_1^\infty \frac{1}{x}\,dx$ is divergent.

2) $\int_{-\infty}^4 \frac{dx}{(x-5)^2}$

$$\int_{-\infty}^4 \frac{dx}{(x-5)^2} = \lim_{l\to-\infty} \int_l^4 (x-5)^{-2}\,dx$$

$$= \lim_{l\to-\infty} \frac{(x-5)^{-1}}{(-1)(1)} \Big|_l^4$$

$$= \lim_{l\to-\infty} \left(\frac{-1}{x-5}\right)\Big|_l^4$$

$$= \lim_{l\to-\infty} \left(1 + \frac{1}{l-5}\right) = 1$$

$\therefore \int_{-\infty}^4 \frac{dx}{(x-5)^2}$ converges to 1

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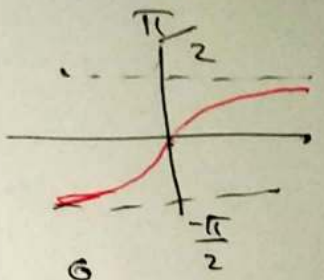
3) $\int_{-\infty}^{\infty} \frac{dx}{x^2+1}$

$$\int_{-\infty}^{\infty} \frac{dx}{x^2+1} = \int_{-\infty}^{0} \frac{dx}{x^2+1} + \int_{0}^{\infty} \frac{dx}{x^2+1}$$

$y = \tan^{-1} x$



$$\int_{-\infty}^{0} \frac{dx}{x^2+1} = \lim_{t \to -\infty} \int_{t}^{0} \frac{dx}{x^2+1} = \lim_{t \to -\infty} \tan^{-1} x \Big|_{t}^{0}$$

$$= \lim_{t \to -\infty} [0 - \tan^{-1} t]$$

$$= -\left(-\frac{\pi}{2}\right) = \frac{\pi}{2}$$

$\tan^{-1} \infty = \frac{\pi}{2}$

$\tan^{-1}(-\infty) = \frac{-\pi}{2}$

$$\therefore \int_{-\infty}^{0} \frac{dx}{x^2-1} = \frac{\pi}{2}$$

$$\int_{0}^{\infty} \frac{dx}{x^2+1} = \lim_{t \to \infty} \int_{0}^{t} \frac{dx}{x^2+1} = \lim_{t \to \infty} \tan^{-1} x \Big|_{0}^{t}$$

$$= \lim_{t \to \infty} [\tan^{-1} t - 0] = \frac{\pi}{2}.$$

$$\therefore \int_{-\infty}^{\infty} \frac{dx}{x^2+1} = \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

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(4) $\int_1^2 \frac{dx}{2-x}$ $\quad f(x) = \frac{1}{2-x}$ $\quad [1,2]$

$$\int_1^2 \frac{dx}{2-x} = \lim_{t\to 2^-} \int_1^t \frac{dx}{2-x}$$

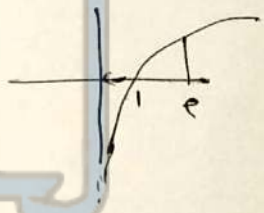
$$= \lim_{t\to 2^-} \left(-\ln(2-x) \Big|_1^t \right)$$

$$= \lim_{t\to 2^-} \left[-\ln(2-t) + \ln 1 \right] \quad (\ln 1 \to 0)$$

$$= \infty$$

$\therefore \int_1^2 \frac{dx}{2-x}$ is divergent.

(5) $\int_0^e \ln x\, dx$



$$\int_0^e \ln x\, dx = \lim_{t\to 0^+} \int_t^e \ln x\, dx \quad \text{(by parts)}$$

$$= \lim_{t\to 0^+} \left[x\ln x - x \right]_t^e$$

$$= \lim_{t\to 0^+} \left[e - e - (t\ln t - t) \right]$$

$$= \lim_{t\to 0^+} (-t\ln t) + \lim_{t\to 0^+} t \quad (t \to 0)$$

$$= \lim_{t\to 0^+} \frac{\ln t}{-\frac{1}{t}} = \lim_{t\to 0^+} \frac{\frac{1}{t}}{\frac{1}{t^2}} = \lim_{t\to 0^+} t = 0$$

$\therefore$ The integral converges to 0.

$u = \ln x \to du = \frac{d}{x}dx$

$dv = dx \to v = x$

$\int \ln x\, dx = x\ln x - \int dx$

$= x\ln x - x + C$

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6) $\int_1^{\infty} \frac{dx}{x\sqrt{x^2-1}}$

$f(x) = \frac{1}{x\sqrt{x^2-1}}$

$D_f = (-\infty,-1) \cup (1,\infty)$

$$\int_1^{\infty} \frac{dx}{x\sqrt{x^2-1}} = \int_1^{2} \frac{dx}{x\sqrt{x^2-1}} + \int_2^{\infty} \frac{dx}{x\sqrt{x^2-1}}$$

$$= \lim_{t_1 \to 1^+} \int_{t_1}^{2} \frac{dx}{x\sqrt{x^2-1}} + \lim_{t_2 \to \infty} \int_2^{t_2} \frac{dx}{x\sqrt{x^2-1}}$$

$$= \lim_{t_1 \to 1^+} \sec^{-1}x \Big|_{t_1}^{2} + \lim_{t_2 \to \infty} \sec^{-1}x \Big|_2^{t_2}$$

$$= \lim_{t_1 \to 1^+} \left[\cos^{-1}\left(\frac{1}{2}\right) - \cos^{-1}\left(\frac{1}{t_1}\right)\right] + \lim_{t_2 \to \infty} \left[\cos^{-1}\left(\frac{1}{t_2}\right) - \cos^{-1}\left(\frac{1}{2}\right)\right]$$

$$= \underbrace{\frac{\pi}{3} - 0}_{\text{Conv.}} + \underbrace{\frac{\pi}{2} - \frac{\pi}{3}}_{\text{Conv.}}$$

$$\therefore \int_1^{\infty} \frac{dx}{x\sqrt{x^2-1}} = \frac{\pi}{3} + \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{3}$$

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2.3: Def: The transform of the function $f(t)$ is defined by

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t) e^{-st} dt$$

use this definition to find the Laplace transform of

Ex

1) $f(t) = e^{2t}$ 2) $f(t) = t$ $\left(\frac{1}{s^2}\right)$

Sol:

1) $$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty e^{2t} e^{-st} dt$$

$$= \int_0^\infty e^{(2-s)t} dt$$

$$= \lim_{l \to \infty} \int_0^l e^{(2-s)t} dt$$

$$= \lim_{l \to \infty} \frac{e^{(2-s)t}}{2-s} \Big|_0^l$$

$$= \lim_{l \to \infty} \left[\frac{e^{(2-s)l}}{2-s} - \frac{1}{2-s} \right]$$

$$= 0 - \frac{1}{2-s} \quad , s > 2$$

$$= \frac{1}{s-2}$$

$$\boxed{\mathcal{L}\{e^{2t}\} = \frac{1}{s-2}}$$

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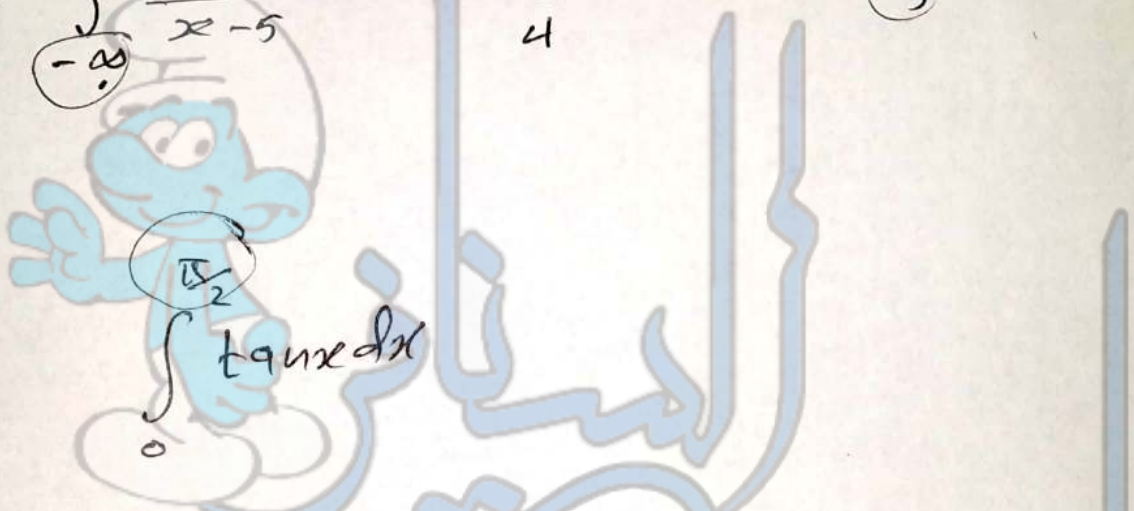
Mo fes

* $\int_{-\infty}^{\infty} \frac{x}{x-5}\,dx$

$= \int_{-\infty}^{4} \frac{x}{x-5}\,dx + \int_{4}^{5} \frac{x}{x-5}\,dx + \int_{5}^{7} \frac{x}{x-5}\,dx + \int_{7}^{\infty} \frac{x}{x-5}\,dx$

* $\int_{0}^{\frac{\pi}{2}} \tan x\,dx$



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Ch.9

Infinite Series

9.1 sequence :

Def: A sequence is a function whose domain is $\mathbb{N} = \{1, 2, 3, \dots\}$ and range is a subset of $\mathbb{R}$.

$$a : \mathbb{N} \longrightarrow \mathbb{R}$$

$$a(1) = a_1$$
$$a(2) = a_2$$
$$\vdots$$
$$a(n) = a_n \leftarrow n^{th} \text{ term (general term)}$$
$$\vdots$$

Notations

$$\{a_1, a_2, a_3, \dots, a_n, \dots\} = \{a_n\}_{n=1}^{\infty} = \{a_n\}$$

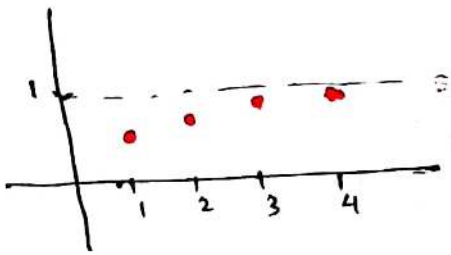
e.g: write the first four terms of the given sequence

a_n

1) $\left\{\frac{n}{n+1}\right\}_{n=1}^{\infty} = \{\underset{a_1}{\frac{1}{2}}, \underset{a_2}{\frac{2}{3}}, \frac{3}{4}, \frac{4}{5}, \dots\}$

as $n \to \infty \Rightarrow \frac{n}{n+1} \to 1$

Convergent seq. converges to 1



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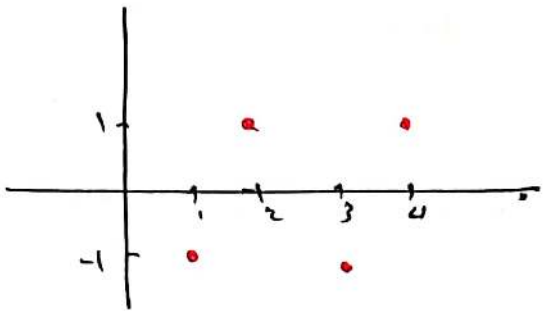
2) $\{(-1)^n\}_{n=1}^{\infty} = \{-1, 1, -1, 1, \dots\}$

there exists no $L \in \mathbb{R}$, such that $(-1)^n \to L$

divergent sequence (diverges)



3) $\{n^2\}_{n=1}^{\infty} = \{1, 4, 9, 16, \dots\}$

as $n \to \infty$, $n^2 \to \infty$

the sequence is divergent.

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Def: A sequence $\{a_n\}_{n=1}^{\infty}$ is convergent, if

$\lim_{n\to\infty} a_n = L$ for some $L \in \mathbb{R}$,

and L is called the limit of the sequence,

otherwise, $\{a_n\}_{n=1}^{\infty}$ is called divergent

(has no limit)

Thm: If $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are two convergent sequences, such that $\lim_{n\to\infty} a_n = L$ and $\lim_{n\to\infty} b_n = M$, $L, M \in \mathbb{R}$.

And if C is a constant, then

1) $\lim_{n\to\infty} C = C$

2) $\lim_{n\to\infty} (a_n + b_n) = L + M$

3) $\lim_{n\to\infty} (a_n - b_n) = L - M$

4) $\lim_{n\to\infty} c\,a_n = cL$

5) $\lim_{n\to\infty} (a_n b_n) = LM$

6) $\lim_{n\to\infty} \left(\frac{a_n}{b_n}\right) = \frac{L}{M}$, $M \neq 0$

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e.g: Determine whether the given sequence converges or diverges. Find its limit if it converges.

1) $\left\{ \frac{2n^2+1}{n^3+n+3} \right\}_{n=1}^{\infty}$

$$\lim_{n\to\infty} \frac{2n^2+1}{n^3+n+3} = 0$$

∴ The sequence is convergent and its limit = 0

2) $\left\{ \frac{5n^2+3n+8}{-n^2+n+3} \right\}_{n=1}^{\infty}$

$$\lim_{n\to\infty} \frac{5n^2+3n+8}{-n^2+n+3} = \frac{5}{-1} = -5$$

∴ the sequence converges to (-5)

3) $\left\{ \frac{n^4+n+1}{-n^3+7n^2+3} \right\}_{n=1}^{\infty}$

$$\lim_{n\to\infty} \frac{n^4+n+1}{-n^3+7n^2+3} = \lim_{n\to\infty} \frac{n^4}{-n^3} = \lim_{n\to\infty} (-n) = -\infty$$

∴ the sequence is divergent.

4) $\left\{ \sin\left(\frac{\pi n+1}{3n+2}\right) \right\}_{n=1}^{\infty}$

$$\lim_{n\to\infty} \sin\left(\frac{\pi n+1}{3n+2}\right) = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

∴ the sequence converges to $\frac{\sqrt{3}}{2}$.

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5) $\left\{ (-1)^n \frac{n+2}{5n+1} \right\}_{n=1}^{\infty}$

$(-1)^n$ (5)

$\cos(n\pi) = (-1)^n$ like.

$$\lim_{\substack{n\to\infty \\ (n:\text{even})}} (-1)^n \frac{n+2}{5n+1} = \lim_{\substack{n\to\infty \\ (n:\text{even})}} \frac{n+2}{5n+1} = \frac{1}{5}$$

$\neq$

$$\lim_{\substack{n\to\infty \\ (n:\text{odd})}} (-1)^n \frac{n+2}{5n+1} = \lim_{\substack{n\to\infty \\ (n:\text{odd})}} \left(-\frac{n+2}{5n+1}\right) = \frac{-1}{5}$$

$\therefore \lim_{n\to\infty} (-1)^n \frac{n+2}{5n+1}$ d.n.e

$\therefore \left\{ (-1)^n \frac{n+2}{5n+1} \right\}_{n=1}^{\infty}$ diverges.

6) $\left\{ (-1)^n + 7 \right\}_{n=1}^{\infty}$

$$\lim_{\substack{n\to\infty \\ (n:\text{even})}} \left((-1)^n + 7\right) = \lim_{\substack{n\to\infty \\ (n:\text{even})}} 8 = 8$$

$\neq$

$$\lim_{\substack{n\to\infty \\ (n:\text{odd})}} \left((-1)^n + 7\right) = \lim_{\substack{n\to\infty \\ (n:\text{odd})}} (-1+7) = \lim_{\substack{n\to\infty \\ n:\text{odd}}} (6) = 6$$

$\therefore$ The sequence diverges

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7/ $\left\{ 2 + \frac{n+1}{n+2} \right\}_{n=1}^{\infty}$

$$\lim_{n\to\infty}\left(2 + \frac{n+1}{n+2}\right) = \lim_{n\to\infty} 2 + \lim_{n\to\infty}\left(\frac{n+1}{n+2}\right)$$

$$= 2 + 1$$

$$= 3$$

8/ $\left\{ n e^{-2n} \right\}_{n=1}^{\infty}$

$$\lim_{n\to\infty} n e^{-2n} \qquad (\infty \cdot 0)$$

$$= \lim_{x\to\infty} x e^{-2x} \qquad (\infty \cdot 0) \qquad x \in [1, \infty)$$

$$= \lim_{x\to\infty} \frac{x}{e^{2x}} \qquad \frac{\infty}{\infty}$$

$$= \lim_{x\to\infty} \frac{1}{2e^{2x}} = 0$$

?? $e^{-\infty} = 0$

∴ The sequence converges to 0.

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(9) $\left\{\frac{\ln n}{n}\right\}_{n=1}^{\infty}$

$$\lim_{n\to\infty}\frac{\ln n}{n} = \lim_{x\to\infty}\frac{\ln x}{x}$$

$$= \lim_{x\to\infty}\frac{\frac{1}{x}}{1} = 0$$

$\therefore$ The sequence converges to 0.

(10) $\left\{\sqrt[n]{n}\right\}_{n=1}^{\infty}$

$$\lim_{n\to\infty}\sqrt[n]{n} = \lim_{n\to\infty} n^{\frac{1}{n}}$$

$e^{\ln \square} = \square$

$$= \lim_{n\to\infty} e^{\ln n^{\frac{1}{n}}}$$

$$= \lim_{n\to\infty} e^{\frac{\ln n}{n}}$$

$$= e^{0} = 1$$

(11) $\left\{\sqrt{n} - n\right\}_{n=1}^{\infty}$

$$\lim_{n\to\infty}(\sqrt{n} - n) = \lim_{n\to\infty}(-n) = -\infty$$

$\therefore$ The seq. is divergent.

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12) $\left\{ \dfrac{\sqrt{4n^2+3n+2} + 7n + 1}{5n+8} \right\}_{n=1}^{\infty}$

$$\lim_{n\to\infty} \frac{\sqrt{4n^2+3n+2} + 7n + 1}{5n+8}$$

$$= \lim_{n\to\infty} \frac{\sqrt{4n^2} + 7n}{5n}$$

$$= \lim_{n\to\infty} \frac{2n + 7n}{5n}$$

$$= \lim_{n\to\infty} \frac{9n}{5n} = \frac{9}{5}$$

∴ the sequence converges to $\frac{9}{5}$

13) $\left\{ \sqrt{n^2+7n} - n \right\}_{n=1}^{\infty}$

$$\lim_{n\to\infty} (\sqrt{n^2+7n} - n) \qquad \infty - \infty$$

$$= \lim_{n\to\infty} \left(\frac{\sqrt{n^2+7n} - n}{1} * \frac{\sqrt{n^2+7n} + n}{\sqrt{n^2+7n} + n} \right)$$

$$= \lim_{n\to\infty} \frac{n^2 + 7n - n^2}{\sqrt{n^2+7n} + n} = \lim_{n\to\infty} \frac{7n}{\sqrt{n^2+7n} + n}$$

$$= \lim_{n\to\infty} \frac{7n}{n+n} = \lim_{n\to\infty} \frac{7n}{2n} = \frac{7}{2}$$ convergent

↑ its limit.

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Rules:

① $\lim_{n\to\infty} \left(1+\frac{a}{n}\right)^n = e^a$

② $a_n \le b_n \le c_n$ $\forall n$ (for all) and squeezing theorem

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, $L \in \mathbb{R}$

$\Longrightarrow \lim_{n\to\infty} b_n = L$

3) $\lim_{n\to\infty} x^n = 0$ when $|x| < 1 \iff -1 < x < 1$

4) $\lim_{n\to\infty} |a_n| = 0 \Longrightarrow \lim_{n\to\infty} a_n = 0$

e.g.: Determine whether the given sequence is converges or diverges:

1) $\left\{ \frac{\sin n}{n} \right\}_{n=1}^{\infty}$

$-1 \le \sin n \le 1$

$\div n \Longrightarrow -\frac{1}{n} \le \frac{\sin n}{n} \le \frac{1}{n}$

and $\lim_{n\to\infty} \left(\frac{-1}{n}\right) = \lim_{n\to\infty} \left(\frac{1}{n}\right) = 0$

$\therefore \lim_{n\to\infty} \frac{\sin n}{n} = 0$

$\therefore$ The sequence converges to 0

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2) $\left\{ n \sin\left(\frac{1}{n}\right) \right\}_{n=1}^{\infty}$

$\lim_{n\to\infty} n \sin\left(\frac{1}{n}\right)$ $\quad \infty \cdot 0$

$\lim_{x\to 0} \frac{\sin x}{x} = 1$

$= \lim_{n\to\infty} \frac{\sin\left(\frac{1}{n}\right)}{\left(\frac{1}{n}\right)} = 1$

$\therefore$ the sequence converges to 1

3) $\left\{ \frac{\cos n}{n+1} + \left(\frac{2}{7}\right)^n \right\}_{n=1}^{\infty}$ $\quad a_n$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \left(\frac{\cos n}{n+1} \right) + \lim_{n\to\infty} \left(\frac{2}{7}\right)^n$

$-1 \le \cos n \le 1$

$\frac{-1}{n+1} \le \frac{\cos n}{n+1} \le \frac{1}{n+1}$

$\to 0 \qquad \to 0$

$= 0 + 0$

$= 0$

$\therefore$ The seq. converges to 0.

4) $\left\{ \left(1 - \frac{1}{n}\right)^n \right\}_{n=1}^{\infty}$

$\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n = e^{-1}$ conv.

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5/ $\left\{ \left(\frac{n+1}{n+4} \right)^{3n} \right\}_{n=1}^{\infty}$

$$\lim_{n\to\infty} \left(1 + \frac{a}{n}\right)^n = e^a$$

$$\lim_{n\to\infty} \left(\frac{n+1}{n+4} \right)^{3n}$$

$$= \lim_{n\to\infty} \left(\frac{1 + \frac{1}{n}}{1 + \frac{4}{n}} \right)^{3n}$$

$$= \lim_{n\to\infty} \left[\frac{\left(1 + \frac{1}{n}\right)^n}{\left(1 + \frac{4}{n}\right)^n} \right]^3$$

$$= \left(\frac{e}{e^4} \right)^3 = e^{-9}$$

$\therefore$ the sequence converges to e^{-9}

Rule :

If $\lim_{n\to\infty} a_n = L$, $L \in \mathbb{R}$ then

$$\lim_{n\to\infty} a_{n \pm n_0} = L$$

n_0 : fixed number

$$\lim_{n\to\infty} a_n = 7 \Longrightarrow \lim_{n\to\infty} a_{n+3} = 7$$

$$\lim_{n\to\infty} a_{n-4} = 7$$

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e.g: Given that

$$a_n = \sqrt{2 + a_{n-1}} \qquad n = 2, 3, 4, \cdots$$

$$a_1 = \sqrt{2}$$

1) find the first 4 terms of the sequence $\{a_n\}_{n=1}^{\infty}$

2) Given that $\{a_n\}_{n=1}^{\infty}$ is convergent, find its limit.

Sol:

1) $a_1 = \sqrt{2}$ $\qquad a_n = \sqrt{2 + a_{n-1}}$

$$a_2 = \sqrt{2 + a_1} = \sqrt{2 + \sqrt{2}}$$

$$a_3 = \sqrt{2 + a_2} = \sqrt{2 + \sqrt{2 + \sqrt{2}}}$$

$$a_4 = \sqrt{2 + a_3} = \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2}}}}$$

$$\vdots$$

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2) $a_n = \sqrt{2 + a_{n-1}}$

since $\{a_n\}_{n=1}^{\infty}$ is convergent.

$\therefore$ it has a limit

assume $\lim_{n\to\infty} a_n = L$

$a_n = \sqrt{2 + a_{n-1}} \Rightarrow a_n^2 = 2 + a_{n-1}$

$\therefore \lim_{n\to\infty} a_n^2 = \lim_{n\to\infty} (2 + a_{n-1})$

$L^2 = 2 + L$

$L^2 - L - 2 = 0$

$(L-2)(L+1) = 0$

$\rightarrow L = 2$ or $L = -1$ $\quad a_n \geq 0$

rejected

$\therefore \lim_{n\to\infty} a_n = 2$

END

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9.2 Monotone sequences:

Def: A sequence $\{a_n\}_{n=1}^{\infty}$ is called

1) strictly increasing if $a_1 < a_2 < a_3 < \cdots$

$(a_n < a_{n+1} \quad \forall n = 1, 2, 3, \ldots)$

2) increasing if $a_1 \leq a_2 \leq a_3 \leq \cdots$

$(a_n \leq a_{n+1} \quad \forall n = 1, 2, \ldots)$

3) strictly decreasing if $a_1 > a_2 > a_3 > \cdots$

$(a_n > a_{n+1} \quad \forall n = 1, 2, 3, \ldots)$

4) decreasing if $a_1 \geq a_2 \geq a_3 \geq \cdots$

$(a_n \geq a_{n+1} \quad \forall n = 1, 2, 3, \ldots)$

* if $\{a_n\}_{n=1}^{\infty}$ is strictly increasing or strictly decreasing then it is called strictly monotone.

* if $\{a_n\}_{n=1}^{\infty}$ is increasing or decreasing, it is called monotone.

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Examples:

* $\{n\}_{n=1}^{\infty} = \{1, 2, 3, \dots\}$ is strictly increasing (strictly monotone)

* $\{2, 2, 4, 4, 6, 6, \dots\}$ is increasing (monotone but not strictly monotone)

* $\{-n\}_{n=1}^{\infty} = \{-1, -2, -3, \dots\}$ is strictly decreasing (strictly monotone)

* $\{-2, -2, -4, -4, -6, -6, \dots\}$ is decreasing (monotone but not strictly monotone)

To show that the sequence $\{a_n\}$ is

1) strictly increasing, it is enough to show that

$(a_{n+1} - a_n > 0 \ \forall n)$, $\left(\frac{a_{n+1}}{a_n} > 1 \ \forall n \in \mathbb{N}\right)$ or

$(f'(x) > 0 \ \forall x \geq 1$ given that $f(n) = a_n)$

2) increasing, if

$(a_{n+1} - a_n \geq 0 \ \forall n)$, $\left(\frac{a_{n+1}}{a_n} \geq 1 \ \forall n \in \mathbb{N}\right)$

or $(f'(x) \geq 0 \ \forall x \geq 1$ given that $f(n) = a_n)$

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3) Strictly decreasing if

$(a_{n+1} - a_n < 0 \ \forall n)$, $\because \left(\frac{a_{n+1}}{a_n} < 1 \ \forall n\right)$

or $(f'(x) < 0 \ \forall x \geq 1$, given that $f(n) = a_n)$

4) decreasing , if

$(a_{n+1} - a_n \leq 0 \ \forall n)$, $\left(\frac{a_{n+1}}{a_n} \leq 1 \ \forall n\right)$

or $(f'(x) \leq 0 \ \forall x \geq 1$, given that $f(n) = a_n)$

e.g: Use three different methods, to show that $\{a_n\}_{n=1}^{\infty} = \left\{\frac{n+1}{n+3}\right\}_{n=1}^{\infty}$ is strictly increasing

method 1 : $a_{n+1} = \frac{(n+1)+1}{(n+1)+3} = \frac{n+2}{n+4}$, $a_n = \frac{n+1}{n+3}$

$$a_{n+1} - a_n = \frac{n+2}{n+4} - \frac{n+1}{n+3}$$

$$= \frac{(n+2)(n+3) - (n+4)(n+1)}{(n+4)(n+3)}$$

$$= \frac{n^2+5n+6 - (n^2+5n+4)}{n^2+7n+12}$$

$$\therefore a_{n+1} - a_n = \frac{2}{n^2+7n+12} > 0 \quad \forall n = 1, 2, 3, \ldots$$

$\therefore \left\{\frac{n+1}{n+3}\right\}_{n=1}^{\infty}$ is strictly increasing

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method 2: $\left\{\frac{n+1}{n+3}\right\}_{n=1}^{\infty}$

$$a_{n+1} = \frac{n+2}{n+4} \quad , \quad a_n = \frac{n+1}{n+3}$$

$$\frac{a_{n+1}}{a_n} = \frac{\frac{n+2}{n+4}}{\frac{n+1}{n+3}} = \frac{n+2}{n+4} \cdot \frac{n+3}{n+1}$$

$$\therefore \frac{a_{n+1}}{a_n} = \frac{n^2+5n+6}{n^2+5n+4} = \frac{(n^2+5n+4)+2}{n^2+5n+4} > 1$$

for $n = 1, 2, 3, \dots$

$\therefore \left\{\frac{n+1}{n+3}\right\}_{n=1}^{\infty}$ is strictly increasing.

method 3 $\left\{\frac{n+1}{n+3}\right\}_{n=1}^{\infty}$

$a_n = \frac{n+1}{n+3}$, Let $f(x) = \frac{x+1}{x+3}$, $x \in [1, \infty)$

$$f'(x) = \frac{x+3-(x+1)}{(x+3)^2} = \frac{+2}{(x+3)^2} > 0 \quad \forall x \geq 1$$

$\therefore$ f is strictly increasing

$\therefore$ $\left\{\frac{n+1}{n+3}\right\}_{n=1}^{\infty}$ is strictly increasing.

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Factorial Concept (!)

Def: 1) If $n \in \mathbb{N}$ then

$$n! = n(n-1)(n-2)\cdots(2)(1)$$

2) $0! = 1$

Examples

* $1! = 1$, $2! = 2(1) = 2$, $3! = 3(2)(1) = 6$

$4! = 4(3)(2)(1) = 24$, $5! = 5(4)(3)(2)(1) = 120$

* $\frac{7!}{5!} = \frac{(7)(6)\cancel{5!}}{\cancel{5!}} = 42$

* $\frac{(n+1)!}{n!} = \frac{(n+1)\cancel{n!}}{\cancel{n!}} = n+1$

* $\frac{(n-1)!}{(n+1)!} = \frac{\cancel{(n-1)!}\ 1}{(n+1)n\cancel{(n-1)!}} = \frac{1}{n^2+n}$

*Notes:

* $2n! \neq (2n)!$

* $(2+3)! \neq 2! + 3!$

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e.g. Show that the sequence, $\left\{ \frac{(n!)^2}{(2n)!} \right\}_{n=1}^{\infty}$ is strictly decreasing.

Sol: want to show $a_n = \frac{n!\,n!}{(2n)!}$

$\frac{a_{n+1}}{a_n} < 1 \quad \forall n \in \mathbb{N}$

$$a_{n+1} = \frac{(n+1)!\,(n+1)!}{(2(n+1))!} = \frac{(n+1)!\,(n+1)!}{(2n+2)!}$$

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)!\,(n+1)!}{(2n+2)!} \cdot \frac{(2n)!}{n!\,n!}$$

$$= \frac{(n+1)\,n!\,(n+1)n!}{(2n+2)(2n+1)(2n)!} \cdot \frac{(2n)!}{n!\,n!}$$

$$= \frac{(n+1)(n+1)}{(2n+2)(2n+1)}$$

$$= \frac{(n+1)(n+1)}{2(n+1)(2n+1)}$$

$$= \frac{n+1}{4n+2}$$

$$\frac{a_{n+1}}{a_n} = \frac{(n+1)}{(n+1)+(3n+1)} < 1$$

$\therefore \frac{a_{n+1}}{a_n} < 1 \quad \forall n.$

$\therefore \left\{ \frac{(n!)^2}{(2n)!} \right\}_{n=1}^{\infty}$ is strictly increasing

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Def:

A sequence $\{a_n\}_{n=1}^{\infty}$ is

eventually satisfying a property $\equiv$

the sequence satifying the property from any term and on.

For example:

The sequence $\{1, 5, 2, 10, 3, 9, 27, 81, 243, \dots\}$

eventualy strictly increasing.

e.g: Show that $\left\{\frac{50^n}{n!}\right\}_{n=1}^{\infty}$ is eventually strictly increasing.

$$\frac{a_{n+1}}{a_n} = \frac{50^{n+1}}{(n+1)!} \cdot \frac{n!}{50^n} = \frac{(50)\,n!}{(n+1)\,n!}$$

$$\frac{a_{n+1}}{a_n} = \frac{50}{n+1} < 1 \qquad \forall n \geq 50$$

$\searrow$

$a_{50}, \dots$

$\therefore$ The sequence is eventually strictly decreasing.

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Def: A sequence $\{a_n\}_{n=1}^{\infty}$ is called

1) bounded above, if there exists $M \in \mathbb{R}$ such that $a_n \leq M$ for any $n \in \mathbb{N}$

and such M is called an upper bound for the sequence.

2) bounded below, if there exists $m \in \mathbb{R}$ such that $a_n \geq m$ for any $n \in \mathbb{N}$

and such m is called a lower bound for the sequence.

3) bounded if it is bounded above and below.

Examples

* the sequence $\{-1, -2, -3, \dots\}$ is bounded above by

0 | or 2 | or 5 $\longleftrightarrow$ upper bounds

* The sequence $\{2, 4, 6, 8, \dots\}$ is bounded below by

2 / 0 / -3 | lower bounds

* The sequence $\{\frac{1}{n}\}_{n=1}^{\infty}$ is bounded

$0 \leq \frac{1}{n} \leq 1$

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Thm:

If $\{a_n\}_{n=1}^{\infty}$ is eventually strictly increasing and bounded above by M, then it is convergent and $\lim_{n\to\infty} a_n \leq M$

but if it is eventually strictly increasing and not bounded above, then

$\lim_{n\to\infty} a_n = \infty$ (divergent)

Thm: If $\{a_n\}_{n=1}^{\infty}$ is eventually strictly decreasing and bounded below by m, then it is convergent and $\lim_{n\to\infty} a_n \geq m$.

but if it is eventually strictly decreasing and not bounded below, then

$\lim_{n\to\infty} a_n = -\infty$ (divergent)

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e.g: Show that $\left\{\frac{50^n}{n!}\right\}_{n=1}^{\infty}$ is convergent and find its limit.

Sol:

$\left\{\frac{50^n}{n!}\right\}_{n=1}^{\infty}$ is eventually strictly decreasing

(by a previous example)

Now $\frac{50^n}{n!} \geq 0 \quad \forall n \in \mathbb{N}$.

$\therefore \left\{\frac{50^n}{n!}\right\}_{n=1}^{\infty}$ is bounded below. $\rightarrow$ Conv.

Assume $a_n = \frac{50^n}{n!}$ and $\lim_{n\to\infty} a_n = L$, $L \in \mathbb{R}$

$$\frac{a_{n+1}}{a_n} = \frac{50^{n+1}}{(n+1)!} \cdot \frac{n!}{50^n} = \frac{50}{n+1}$$

$$\frac{a_{n+1}}{a_n} = \frac{50}{n+1} \rightarrow a_{n+1} = \frac{50}{n+1} a_n$$

$$\Rightarrow \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \left(\frac{50}{n+1} \cdot a_n\right)$$

$$L = 0 \cdot L$$

$$L = 0$$

$$\therefore \lim_{n\to\infty} \frac{50^n}{n!} = 0$$

END.

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9.3 : Infinite series:

Introduction (Sigma 'notation) Σ

$$\sum_{K=1}^{m} f(K) = f(1) + f(2) + f(3) + \cdots + f(m)$$

* $\sum_{K=1}^{5} K^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$

Properties:

1) $\sum_{K=1}^{n} C = nC$, C: constant

* $\sum_{K=1}^{3} 4 = 4 + 4 + 4 = 12$

2) $\sum_{K=m}^{n} f(K) = \sum_{K=m-K_0}^{n-K_0} f(K+K_0) = \sum_{K=m+K_0}^{n+K_0} f(K-K_0)$ $n < m$

* $\sum_{K=3}^{5} K^3 = 3^3 + 4^3 + 5^3$

* $\sum_{K=0}^{2} (K+3)^3 = (0+3)^3 + (1+3)^3 + (2+3)^3$
$= 3^3 + 4^3 + 5^3$

* $\sum_{K=5}^{7} (K-2)^3 = 3^3 + 4^3 + 5^3$

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3) $\sum_{k=m}^{n} (f(k) \mp g(k)) = \sum_{k=m}^{n} f(k) \mp \sum_{k=m}^{n} g(k)$

$m < n$

* $\sum_{k=1}^{2} (k^2 + 2^k) = (1^2 + 2^1) + (2^2 + 2^2)$

$= (1+2) + (4+4)$

$= 11$

* $\sum_{k=1}^{2} k^2 + \sum_{k=1}^{\infty} 2^k = (1^2 + 2^1) + (2^1 + 2^2)$

$= 5 + 6$

$= 11$

4) $\sum_{k=m}^{n} C f(k) = C \sum_{k=m}^{n} f(k)$ C: Constant

$m < n$

* $\sum_{k=1}^{3} 2k^3 = 2(1^3) + 2(2^3) + 2(3^3)$

$= 2 + 16 + 54$

$= 72$

* $2 \sum_{k=1}^{3} k^3 = 2[1^3 + 2^3 + 3^3]$

$= 2[1 + 8 + 27]$

$= 2(36) = 72$

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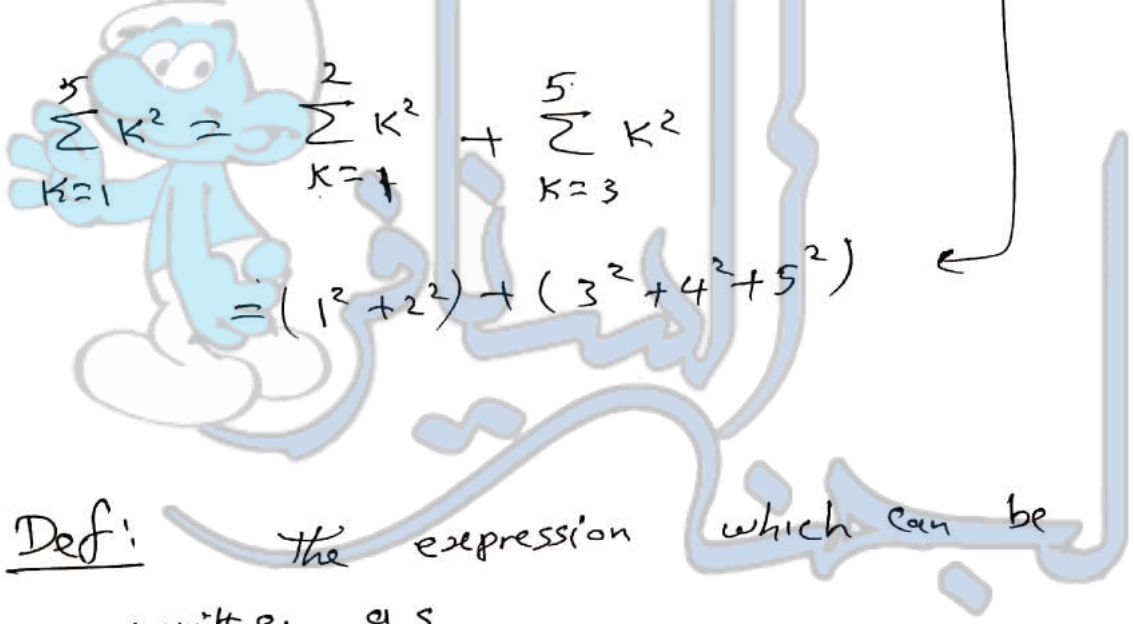
5) $$\sum_{K=m}^{n} f(K) = \sum_{K=m}^{K_0} f(K) + \sum_{K=K_0+1}^{n} f(K)$$

$$m < K_0 < n$$

* $$\sum_{K=1}^{5} K^2 = 1^2+2^2+3^2+4^2+5^2$$

$$\sum_{K=1}^{5} K^2 = \sum_{K=1}^{2} K^2 + \sum_{K=3}^{5} K^2$$

$$= (1^2+2^2) + (3^2+4^2+5^2)$$



Def: the expression which can be written as

$$\sum_{K=1}^{\infty} u_K = u_1 + u_2 + u_3 + \cdots + u_{15} + \cdots$$

is called an infinite series

u_K : the K^{th} term (or the general) term of the series.

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Note : $\sum_{k=1}^{\infty} u_k = \sum_{k=2}^{\infty} u_{(k-1)} = \sum_{k=5}^{\infty} u_{(k-5)}$

$= u_1 + u_2 + u_3 + \cdots$

i.e. the starting point of the k need not be 1.

If $\sum_{k=1}^{\infty} u_k$ is any series then

$S_1 = u_1$ the first partial sum

$S_2 = u_1 + u_2$ " 2nd " "

$S_3 = u_1 + u_2 + u_3$ " 3rd " "

$\vdots$

$S_n = u_1 + u_2 + \cdots + u_n$ the n^{th} partial sum.

$\{S_1, S_2, S_3, \cdots, S_n, \cdots\} = \{S_n\}_{n=1}^{\infty}$ the sequence of n^{th} partial sums.

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e.g: Given that

$$\sum_{k=1}^{\infty} u_k = \sum_{k=1}^{\infty} k^2$$

find 1) u_1 and u_3

2) S_1 and S_3

3) $S_5 - S_4$

sol:

$$u_k = k^2$$

1) $u_1 = 1^2 = 1$, $u_3 = 3^2 = 9$

2) $S_1 = u_1 = 1$

$S_3 = u_1 + u_2 + u_3 = 1^2 + 2^2 + 3^2 = 14$

3) $S_5 - S_4$

$S_5 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2$

$S_4 = 1^2 + 2^2 + 3^2 + 4^2$

$\therefore S_5 - S_4 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 - (1^2 + 2^2 + 3^2 + 4^2)$

$= 5^2 = u_5$

Notice that

$$u_n = S_n - S_{n-1}$$

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e.g: If $\sum_{k=1}^{\infty} u_k$ is a series,

such that its nth partial sum

$S_n = 2n^2$, find u_k ?

sol.

$u_k = S_k - S_{k-1}$.

$\therefore u_k = 2k^2 - 2(k-1)^2$

$= 2k^2 - 2(k^2 - 2k + 1)$

$= 2k^2 - 2k^2 + 4k - 2$

$u_k = 4k - 2$

$\therefore \sum_{k=1}^{\infty} u_k = \sum_{k=1}^{\infty} (4k-2)$

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Def: Let $\sum_{k=1}^{\infty} a_k$ be any series,

$\{s_n\}_{n=1}^{\infty}$ is the sequence of the n^{th} partial sums of this series.

then the series is called convergent

if $\lim_{n\to\infty} s_n = s$ for some $s \in \mathbb{R}$

otherwise, the series is called divergent.

and s is called the sum of the series.

e.g: Determine whether the given series converges, if so, find its sum.

1) $\sum_{k=1}^{\infty} 2$

$a_k = 2$

$s_n = a_1 + a_2 + a_3 + \cdots + a_n$

$s_n = \underbrace{2 + 2 + 2 + \cdots + 2}_{n\text{-terms}}$

$s_n = 2n$ $\qquad \{s_n\}_{n=1}^{\infty} = \{2, 4, 6, \cdots\}$

$\lim_{n\to\infty} s_n = \lim_{n\to\infty} (2n) = \infty$

$\therefore \sum_{k=1}^{\infty} 2$ is divergent.

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2) $\sum_{k=0}^{\infty} 0$

$S_n = 0$

$\lim_{n\to\infty} S_n = 0$

∴ This seris is Convergent and its sum = 0

3) $\sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1}\right)$ ← Telescopic seris.

$f(k) - f(k+1)$

$u_k = \frac{1}{k} - \frac{1}{k+1}$

$S_n = u_1 + u_2 + \cdots + u_n$

$= \underbrace{\left(\frac{1}{1} - \frac{1}{2}\right)}_{u_1} + \underbrace{\left(\frac{1}{2} - \frac{1}{3}\right)}_{u_2} + \underbrace{\left(\frac{1}{3} - \frac{1}{4}\right)}_{u_3} + \cdots + \underbrace{\left(\frac{1}{n-1} - \frac{1}{n}\right)}_{u_{n-1}} + \underbrace{\left(\frac{1}{n} - \frac{1}{n+1}\right)}_{u_n}$

$S_n = 1 - \frac{1}{n+1}$

$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \left(1 - \frac{1}{n+1}\right) = 1$

∴ The seris is Convergent and its sum = 1.

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4) $\sum_{k=1}^{\infty} \frac{1}{k^2+k}$

$$\frac{1}{k(k+1)} = \frac{A}{k} + \frac{B}{k+1}$$

$\times k(k+1)$ →

$$1 = A(k+1) + Bk$$

$k = 0 \rightarrow A = 1$

$k = -1 \rightarrow 1 = -B \rightarrow B = -1$

$$\sum_{k=1}^{\infty} \frac{1}{k^2+k} = \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1} \right)$$ telescopic series

$$\sum \left(f(k+\dots) - f(k+\dots) \right)$$

5) $\sum_{k=3}^{\infty} \ln\left(\frac{k-1}{k+1}\right)$

$$= \sum_{k=3}^{\infty} \left(\ln(k-1) - \ln(k+1) \right)$$

$$S_n = u_3 + u_4 + u_5 + \dots + u_n$$

$$= (\ln 2 - \ln 4) + (\ln 3 - \ln 5) + (\ln 4 - \ln 6) +$$

$$(\ln 5 - \ln 7) + \dots + (\ln(n-2) - \ln n) + (\ln(n-1) - \ln(n+1))$$

$$S_n = \ln 2 + \ln 3 - \ln n - \ln(n+1)$$

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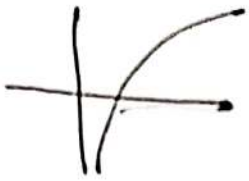
$\therefore S_n = \ln 2 + \ln 3 - \ln n - \ln(n+1)$



$\therefore \lim_{n\to\infty} S_n = \lim_{n\to\infty} (\ln 2 + \ln 3 - \ln n - \ln(n+1))$

$= -\infty$

$\therefore$ the seris is divergent.

6) $\sum_{k=0}^{\infty} (\tan^{-1} k - \tan^{-1}(k+1))$

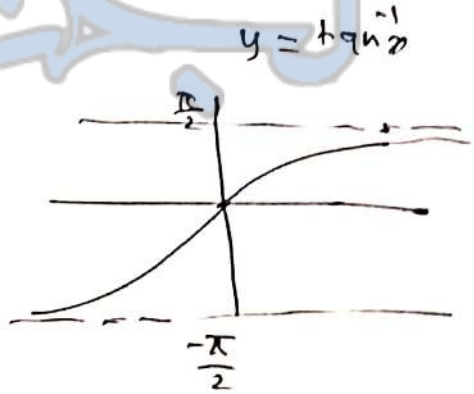
$S_n = u_0 + u_1 + u_2 + \cdots + u_n$

$S_n = (\tan^{-1}0 - \tan^{-1}1) + (\tan^{-1}1 - \tan^{-1}2) + \cdots$

$+ (\tan^{-1}n - \tan^{-1}(n+1))$

$S_n = 0 - \tan^{-1}(n+1)$

$S_n = -\tan^{-1}(n+1)$



$\lim_{n\to\infty} S_n = \lim_{n\to\infty} (-\tan^{-1}(n+1))$

$= -\frac{\pi}{2}$

$\therefore$ The seris is convergent

and its sum $= 1$.

END.

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Infinite Series - Part 2

Geometric series (G.S)

Def:

If $\sum_{k=1}^{\infty} a_k = a_1 + a_2 + a_3 + \cdots + a_k + \cdots$ a series with non-zero terms

then this series is called geometric (G.S) if for any k, $\frac{a_{k+1}}{a_k} = r$, r: non-zero fixed real number

r: common ratio (C.r)

Now If $\sum_{k=1}^{\infty} a_k$ is a G.S with C.r $= r$ and $a_1 = a \neq 0$

then $a_1 = a$

$a_2 = a.r$

$a_3 = ar^2$

$\vdots$

$a_k = ar^{k-1}$ → k^{th} term (general term)

$\vdots$

so $\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} a\, r^{k-1}$

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If $\sum_{k=1}^{\infty} a r^{k-1}$ is g.s then

$S_n = a_1 + a_2 + \cdots + a_n$

$S_n = a + ar + ar^2 + \cdots + ar^{n-1}$ ——— (1)

Multiply (1) by r $\Longrightarrow$

$rS_n = ar + ar^2 + ar^3 + \cdots + ar^n$ ——— (2)

(1) − (2) $\Rightarrow$ $S_n - rS_n = a - ar^n$

$(1-r)S_n = a(1-r^n)$

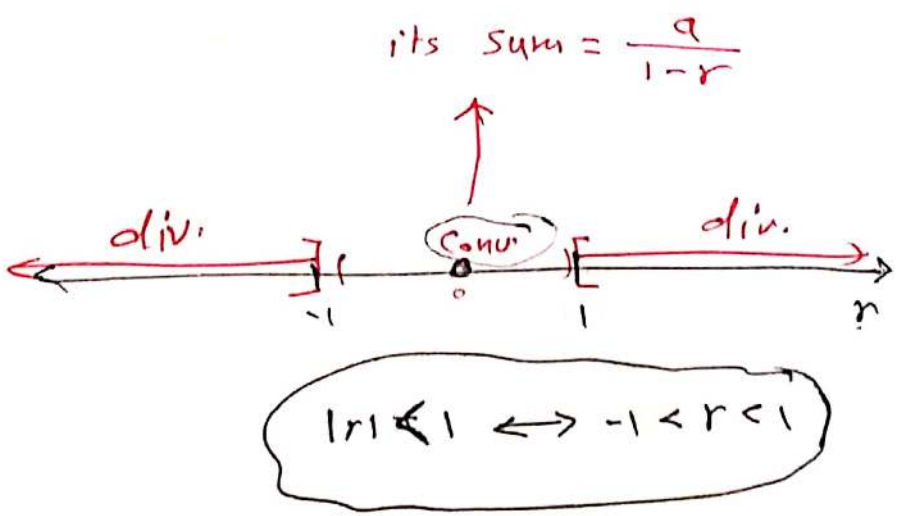
$\therefore S_n = \dfrac{a(1-r^n)}{1-r}$

$\lim_{n\to\infty} r^n = 0$ if $|r|<1$

then $\lim_{n\to\infty} S_n = \begin{cases} \dfrac{a}{1-r} & \text{if } |r|<1 \\ \text{d.n.e} & \text{if } |r|\geq 1 \end{cases}$

$\therefore$ The g.s converges if $|r|<1$ and its sum $= \dfrac{a}{1-r}$

and diverges if $|r|\geq 1$



$|r|<1 \iff -1<r<1$

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Notice that

$$\sum_{k=1}^{\infty} a r^{k-1} = \sum_{k=0}^{\infty} a r^{k} = \sum_{k=5}^{\infty} a r^{k-5}$$

OR

$$\sum_{k=k_1}^{\infty} (\text{constant})(\text{constant})^{k} \longrightarrow \text{g.s}$$

$k \pm k_0 \qquad k_0 \in \mathbb{Z}$

r

a = substitute k_1 in the general term

e.g: Determine whether the given seris is geometric, if it is geometric, find its sum if it is convergent

1) $0.3 + 0.03 + 0.003 + 0.00003 + \cdots + 0.3\overline{}$

$$\frac{0.03}{0.3} = 0.1, \quad \frac{0.003}{0.03} = 0.1, \quad \frac{0.0003}{0.003} = 0.1$$

g.s c.r $r = 0.1$ → the seris is convergent.

$$S = \frac{0.3}{1 - 0.1} = \frac{0.3}{0.9} = \frac{3}{9} = \frac{1}{3}$$

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2) $1 + 2 + 4 + 6 + 8 + \cdots$

$\frac{2}{1} = 2$, $\frac{4}{2} = 2$, $\frac{6}{4} = 1.5$ ✗

this seq is not geometric.

3) $1 - 3 + 9 - 27 + 81 - 243 + \cdots$

$\frac{-3}{1} = -3$, $\frac{9}{-3} = -3$, $\frac{-27}{9} = -3$, $\frac{81}{-27} = -3$

$\frac{-243}{81} = -3$, . . .

∴ g.s $r = -3$

∴ this seq diverges

4) $\sum_{k=0}^{\infty} (-1)^k (2^k)(3^{1-k})$

$= \sum_{k=0}^{\infty} (-1)^k (2^k)(\frac{1}{3})(3^{-k})$

$= \sum_{k=0}^{\infty} 3 \frac{(-1)^k (2^k)}{3^k} = \sum_{k=0}^{\infty} 3 \left(\frac{-2}{3}\right)^k$

g.s c.v $r = \frac{-2}{3}$

Conv.

$\text{sum} = \frac{3}{1 - (\frac{-2}{3})} = \frac{3}{1 + \frac{2}{3}} = \frac{9}{3+2} = \frac{9}{5}$

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5) $\sum_{k=0}^{\infty} \left(2^{3k+1} \cdot 5^{3-k}\right)$

$= \sum_{k=0}^{\infty} \left(2^{3k} \cdot 2^{1} \cdot 5^{3} \cdot 5^{-k}\right)$

$= \sum_{k=0}^{\infty} (250) \frac{(2^3)^k}{5^k}$

$= \sum_{k=0}^{\infty} 250 \left(\frac{8}{5}\right)^k$ g.s

$r = \frac{8}{5} \longrightarrow$ div.

6) $\sum_{k=1}^{\infty} e^k \pi^{2-k}$

$= \sum_{k=1}^{\infty} \left(e^k \pi^2 \cdot \pi^{-k}\right)$

$= \sum_{k=1}^{\infty} \pi^2 \frac{e^k}{\pi^k} = \sum_{k=1}^{\infty} \pi^2 \left(\frac{e}{\pi}\right)^k$

g.s c.r $r = \frac{e}{\pi} < 1$

∴ the series is convergent.

its sum $= \frac{\pi^2 \left(\frac{e}{\pi}\right)}{1 - \frac{e}{\pi}} = \frac{\pi^2 e}{\pi - e}$

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7) $\sum_{K=2}^{\infty} (-1)^K \frac{1}{4^K}$

$= \sum_{K=2}^{\infty} \left(\frac{-1}{4}\right)^K \longrightarrow$ g.s.

$r = \frac{-1}{4}$, $|r| < 1$

$\therefore$ The series is conv.

its sum $= \dfrac{\frac{1}{16}}{1 - \left(\frac{-1}{4}\right)}$

$= \dfrac{\frac{1}{16}}{1 + \frac{1}{4}}$

$= \dfrac{1}{16 + 4} = \dfrac{1}{20}$

8) $\sum_{K=3}^{\infty} \frac{1}{2^K}$

$= \sum_{K=3}^{\infty} \left(\frac{1}{2}\right)^K \longrightarrow$ g.s $r = \frac{1}{2}$

$\rightarrow$ conv.

sum $= \dfrac{\frac{1}{8}}{1 - \frac{1}{2}} = \dfrac{1}{8 - 4} = \dfrac{1}{4}$

— END —

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1 Convergence Tests

Thm: If $\sum_{k=1}^{\infty} a_k$ is a convergent series

then $\lim_{k\to\infty} a_k = 0$

Proof:

Let $S_n = a_1 + a_2 + \cdots + a_n$ n^{th} partial sums

since $\sum_{k=1}^{\infty} a_k$ is convergent $\Rightarrow$

$\lim_{n\to\infty} S_n = S$ for some $S \in \mathbb{R}$

Now $a_n = S_n - S_{n-1}$

$\Rightarrow \lim_{n\to\infty} a_n = \lim_{n\to\infty} (S_n - S_{n-1})$

$= \lim_{n\to\infty} S_n - \lim_{n\to\infty} S_{n-1}$

$= S - S$

$= 0$

The converse of this theorem is not true.

because the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ is

divergent (as will see) but $\lim_{k\to\infty} \frac{1}{k} = 0$

i.e $\sum_{k=1}^{\infty} a_k$ is convergent $\Longrightarrow$ $\lim_{k\to\infty} a_k = 0$

$\Longleftarrow$ ✗

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Thm: If $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ are two convergent series, their sums S_1 and S_2 respectively, then

1) $\sum_{k=1}^{\infty}(a_k + b_k)$ is convergent and its sum $= S_1 + S_2$

2) $\sum_{k=1}^{\infty}(a_k - b_k)$ is convergent and its sum $= S_1 - S_2$

3) $\sum_{k=1}^{\infty} c\,a_k$, "c any real constant" is convergent and its sum is $(c.S_1)$

Thm: If $c \neq 0$ then

$\sum_{k=1}^{\infty} a_k$ is convergent $\iff \sum_{k=1}^{\infty} c\,a_k$ is convergent

i.e ($\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} c\,a_k$ are both convergent or both divergent given that $c \neq 0$)

Thm: $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=k_0}^{\infty} a_k$ are both convergent or both divergent.

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e.g: Determine, if the given series is convergent, if so, find its sum.

$$\sum_{k=0}^{\infty} \left(\frac{2^k + 3^k}{5^k} \right)$$

$$\sum_{k=0}^{\infty} \left(\frac{2^k + 3^k}{5^k} \right) = \sum_{k=0}^{\infty} \left(\frac{2^k}{5^k} + \frac{3^k}{5^k} \right)$$

$$= \sum_{k=0}^{\infty} \left(\left(\frac{2}{5}\right)^k + \left(\frac{3}{5}\right)^k \right)$$

Now $\sum_{k=0}^{\infty} \left(\frac{2}{5}\right)^k$ is g.s, $r = \frac{2}{5}$ → this series is convergent and its sum $= \frac{1}{1 - \frac{2}{5}} = \frac{1}{\frac{3}{5}} = \frac{5}{3}$

And $\sum_{k=0}^{\infty} \left(\frac{3}{5}\right)^k$ is g.s, $r = \frac{3}{5}$ → this series is convergent and its sum $= \frac{1}{1 - \frac{3}{5}}$

$$= \frac{1}{\frac{2}{5}} = \frac{5}{2}$$

$\therefore \sum_{k=0}^{\infty} \left(\frac{2^k + 3^k}{5^k} \right)$ is convergent and

its sum $= \frac{5}{3} + \frac{5}{2} = \frac{10 + 15}{6} = \frac{25}{6}$

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1 Divergence Test (D.T)

If $\sum_{k=1}^{\infty} a_k$ is any series such that

$\lim_{k\to\infty} a_k \neq 0$, then the series is divergent

Notice that: $\sum_{k=1}^{\infty} a_k$ is divergent $\not\Rightarrow \lim_{k\to\infty} a_k \neq 0$

e.g: Determine, whether the given series is convergent or divergent ? (if possible)

1) $\sum_{k=1}^{\infty} \frac{2k+1}{5k+2}$

$\lim_{k\to\infty} \left(\frac{2k+1}{5k+2}\right) = \frac{2}{5} \neq 0$

$\therefore$ The series is divergent. (D.T)

2) $\sum_{k=1}^{\infty} \left(1+\frac{1}{k}\right)^k$

$\lim_{k\to\infty} \left(1+\frac{1}{k}\right)^k = e \neq 0$

$\therefore$ The series is divergent (D.T)

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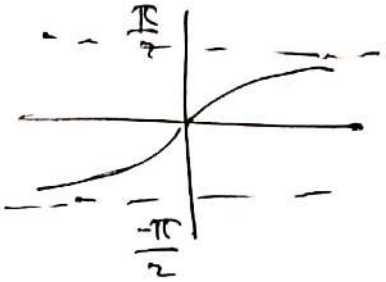
3) $\sum_{k=1}^{\infty} \tan^{-1} k$

$\lim_{k\to\infty} \tan^{-1} k = \frac{\pi}{2} \neq 0$

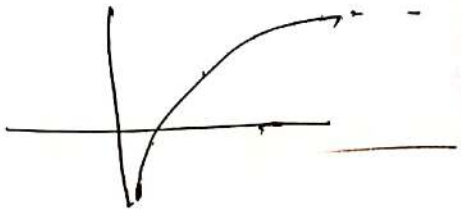
$\therefore$ The series is div. (D.T)



4) $\sum_{k=5}^{\infty} \ln k$

$\lim_{k\to\infty} (\ln k) = \infty \neq 0$

$\therefore$ The series is div. (D.T)



5) $\sum_{k=1}^{\infty} \cos\left(\frac{k}{k^2+1}\right)$

$\lim_{k\to\infty} \cos\left(\frac{k}{k^2+1}\right) = \cos 0 = 1 \neq 0$

$\Rightarrow$ The series is div (D.T)

6) $\sum_{k=1}^{\infty} \frac{1}{2^k}$

$\lim_{k\to\infty} \frac{1}{2^k} = \lim_{k\to\infty} \left(\frac{1}{2}\right)^k = 0$ D.T fails

$\sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k$ g.s $r = \frac{1}{2} \longrightarrow$ Conv.

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7) $\sum_{k=1}^{\infty} \frac{1}{k}$ (Harmonic series)

$\lim_{k\to\infty} \frac{1}{k} = 0$ D.T fails

but this series is divergent as will see.

2 Integral Test (I.T)

If f is continuous, positive and decreasing on $[a, \infty)$ then

$\sum_{k=1}^{\infty} f(k)$ is convergent $\iff$ $\int_a^{\infty} f(x)dx$ is convergent

i.e $\sum_{k=1}^{\infty} f(k)$ and $\int_a^{\infty} f(x)dx$ are both convergent or both divergent

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e.g: Determine whether the given series is convergent or divergent.

1) $\sum_{k=1}^{\infty} \frac{1}{k}$ (Harmonic Series)

Let $f(x) = \frac{1}{x}$ is cont., +ve and ↓ on $[1, \infty)$

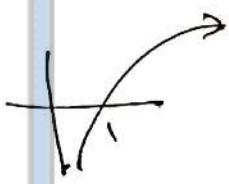
$$\int_1^{\infty} f(x)\,dx = \lim_{L\to\infty} \int_1^{L} \frac{1}{x}\,dx$$

$$= \lim_{L\to\infty} \ln x \Big|_1^{L}$$

$$= \lim_{L\to\infty} (\ln L - \ln 1) \quad (\ln L \to \infty,\ \ln 1 \to 0)$$

$$= \infty$$

∴ the integral is div.



∴ by I.T, the series

$\sum_{k=1}^{\infty} \frac{1}{k}$ is divergent.

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2) $\sum_{K=1}^{\infty} \frac{1}{K^2+4}$

$f(x) = \frac{1}{x^2+4}$

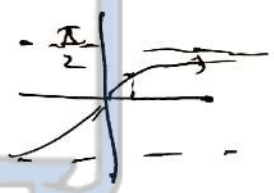
f is cont., ↓ and positive on $[1,\infty)$

$$\int_1^{\infty} \frac{1}{x^2+4}\,dx = \lim_{L\to\infty} \int_1^{L} \frac{1}{x^2+4}\,dx$$

$$= \lim_{L\to\infty} \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)\Big|_1^L$$

$$= \lim_{L\to\infty} \frac{1}{2}\left[\tan^{-1}\left(\frac{L}{2}\right) - \tan^{-1}\left(\frac{1}{2}\right)\right]$$

$$= \frac{1}{2}\left[\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{2}\right)\right]$$



∴ The integral is conv.

∴ $\sum_{K=1}^{\infty} \frac{1}{K^2+4}$ is conv. (I.T)

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3) $\sum_{k=2}^{\infty} \frac{\ln k}{k}$

Let $f(x) = \frac{\ln x}{x}$

$$f'(x) = \frac{x \cdot \frac{1}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2} \leftarrow +ve$$

$1 - \ln x = 0 \rightarrow \ln x = 1 \rightarrow x = e$

$\therefore$ f is cont., +ve and decreasing on $[e, \infty)$

$$\int_2^{\infty} \frac{\ln x}{x} dx = \lim_{L \to \infty} \int_2^{L} \frac{\ln x}{x} dx$$

$$= \lim_{L \to \infty} \frac{(\ln x)^2}{2} \Big|_2^L$$

$$= \lim_{L \to \infty} \left[\frac{(\ln L)^2}{2} - \frac{(\ln 2)^2}{2} \right] = \infty$$

$\therefore$ the integral is divergent.

$\therefore \sum_{k=1}^{\infty} \frac{\ln k}{k}$ is divergent

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A) $\sum_{k=1}^{\infty} k e^{-k}$

$f(x) = x e^{-x} = \frac{x}{e^x}$

$f'(x) = -x e^{-x} + e^{-x}$

$= e^{-x}(-x+1) \leq 0 \quad \forall x \geq 1$

$\therefore$ f is +ve, cont. and $\downarrow$ on $[1, \infty)$

$$\int_1^{\infty} x e^{-x} dx = \lim_{L \to \infty} \int_1^{L} x e^{-x} dx$$

x	e^{-x}
1	$-e^{-x}$
0	e^{-x}

$$= \lim_{L \to \infty} \left[-x e^{-x} - e^{-x} \right]_1^{L}$$

$$= \lim_{L \to \infty} \left[-L e^{-L} - e^{-L} - (-e^{-1} - e^{-1}) \right]$$

$$= \lim_{L \to \infty} \frac{-L}{e^L} - \lim_{L \to \infty} \frac{1}{eL} + 2e^{-1}$$

($\frac{-\infty}{\infty}$; $\frac{1}{eL} \to 0$; e^{-1})

$$= \lim_{L \to \infty} \frac{-1}{e^L} - 0 + 2e^{-1}$$

($\frac{-1}{e^L} \to 0$)

$= 2e^{-1} \Rightarrow$ the integral is convergent.

$\therefore$ by I.T

$\sum_{k=1}^{\infty} k e^{-k}$ is conv.

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5) $\sum_{k=1}^{\infty} \frac{1}{k^p}$, $p > 1$

Let $f(x) = \frac{1}{x^p}$.

f is +ve, ↓ and Cont. on $[1, \infty)$

$$\int_1^{\infty} \frac{1}{x^p}\,dx = \lim_{L\to\infty} \int_1^{L} x^{-p}\,dx$$

$$= \lim_{L\to\infty} \left(\frac{x^{1-p}}{1-p} \Big|_1^{L} \right)$$

$$= \lim_{L\to\infty} \left[\frac{L^{1-p}}{1-p} - \frac{1}{1-p} \right]$$

$$= 0 - \frac{1}{1-p} = \frac{1}{p-1}$$ Conv.

$\therefore \sum_{k=1}^{\infty} \frac{1}{k^p}$, $p > 1$ is Conv. (I.T)

6) $\sum_{k=1}^{\infty} \frac{1}{k^p}$ $0 < p < 1$

$f(x) = \frac{1}{x^p}$ is Cont., +ve and ↓ on $[1, \infty)$

Show that $\int_1^{\infty} \frac{1}{x^p}\,dx$ is div.

$\therefore \sum_{k=1}^{\infty} \frac{1}{k^p}$ is div. (I.T).

- END -

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Convegence Tests

P- series and Comparison Tests:

Recall

[1] D.T $\sum_{k=1}^{\infty} a_k$, $\lim_{k\to\infty} a_k \neq 0 \Rightarrow \sum_{k=1}^{\infty} a_k$ is divergent.

[2] I.T, if f is cont, +ve and $\downarrow$ (eventually)

$\sum_{k=1}^{\infty} f(k)$ is conv. $\iff \int_{a}^{\infty} f(x)\,dx$ is conv.

[3] P- series

$\sum_{k=1}^{\infty} \frac{1}{k^p}$

$p>1 \rightarrow$ The series is conv.

$p\leq 1 \rightarrow$ the series is div.

e.g: Test for convergence

1) $\sum_{k=1}^{\infty} \frac{1}{k}$ (Harmonic series)

$p=1 \Rightarrow$ the series is div.

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2) $\sum_{k=1}^{\infty} \frac{10}{\sqrt{k^3}}$

$= \sum_{k=1}^{\infty} \frac{10}{k^{3/2}}$ $\quad p = \frac{3}{2} > 1 \rightarrow$ the seris is Convergent.

3) $\sum_{k=5}^{\infty} \frac{1}{\sqrt[3]{k^2}}$

$= \sum_{k=5}^{\infty} \frac{1}{k^{\frac{2}{3}}}$ $\quad p = \frac{2}{3} \rightarrow$ the seris is divergent.

4) $\sum_{k=1}^{\infty} \frac{1}{k^{1.0001}}$

$p = 1.0001 > 1 \rightarrow$ the seris is Convergent.

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4 Comparison Test: (C.T)

If $\sum a_k$ and $\sum b_k$ are two series with non-negative terms, such that

$a_k \leq b_k$ for any. k (or eventually)

then

1) If $\sum b_k$ is convergent then $\sum a_k$ is conv.

2) If $\sum a_k$ is divergent then $\sum b_k$ is div.

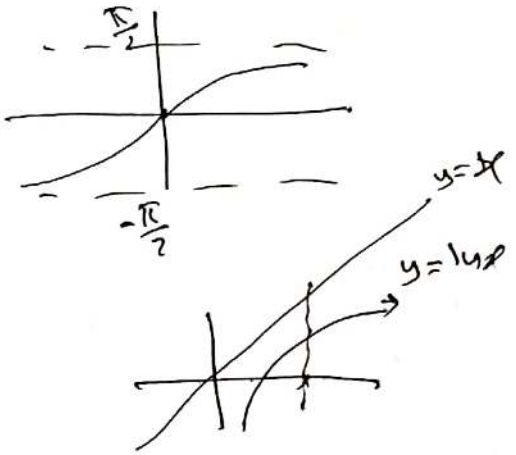
Important inequalities

1) $-1 \leq \sin f(k) \leq 1$

2) $-1 \leq \cos f(k) \leq 1$

3) $\frac{-\pi}{2} \leq \tan^{-1} f(k) \leq \frac{\pi}{2}$

4) $\ln k \leq k \quad \forall k \geq 1$



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e.g: Test for Convergence

1) $$\sum_{k=1}^{\infty} \frac{2+\sin k}{k^2} = \sum_{k=1}^{\infty} a_k$$

$\sin k \le 1 \quad \forall k$

$\xrightarrow{+2}$ $2+\sin k \le 3$

$\xrightarrow{\div k^2}$ $\frac{2+\sin k}{k^2} \le \frac{3}{k^2} \quad \forall k$

$\sum_{k=1}^{\infty} \frac{3}{k^2}$ $\quad p=2 \rightarrow$ Conv.

$\therefore$ by C.T, $\sum_{k=1}^{\infty} \frac{2+\sin k}{k^2}$ is Conv.

2) $$\sum_{k=1}^{\infty} \frac{5+\cos k^3}{\sqrt{k}}$$

$-1 \le \cos k^3 \quad \forall k$

$\xrightarrow{+5}$ $4 \le 5+\cos k^3$

$\xrightarrow{\div \sqrt{k}}$ $\frac{4}{\sqrt{k}} \le \frac{5+\cos k^3}{\sqrt{k}} \quad \forall k$

$\sum_{k=1}^{\infty} \frac{4}{\sqrt{k}} = \sum_{k=1}^{\infty} \frac{4}{k^{\frac{1}{2}}}$ $\quad p=\frac{1}{2} \rightarrow$ div.

$\therefore \sum_{k=1}^{\infty} \frac{5+\cos k^3}{\sqrt{k}}$ is div. (by C.T).

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3) $\sum_{k=1}^{\infty} \frac{\tan^{-1} k}{k^4}$

$\tan^{-1} k \leq \pi/2 \quad \forall k$

$\therefore k^4 \rightarrow \frac{\tan^{-1} k}{k^4} \leq \frac{\pi/2}{k^4} \quad \forall k$

$\sum_{k=1}^{\infty} \frac{\pi/2}{k^4}$ is Convergent $p = 4 > 1$

$\therefore \sum_{k=1}^{\infty} \frac{\tan^{-1} k}{k^4}$ is Conv. (by C.T)

4) $\sum_{k=2}^{\infty} \frac{1}{\ln k}$

$\ln k \leq k \quad k = 2, 3, \dots$

$\rightarrow \frac{1}{\ln k} \geq \frac{1}{k} \quad k = 2, 3, \dots$

$\sum_{k=1}^{\infty} \frac{1}{k} \quad p = 1 \rightarrow$ div.

$\therefore \sum_{k=2}^{\infty} \frac{1}{\ln k}$ is div. (C.T)

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5) $\sum_{k=1}^{\infty} \frac{1}{k + 2^k}$

$\frac{1}{2^k} \geq \frac{1}{k+2^k} \quad \forall k$

$\sum_{k=1}^{\infty} \frac{1}{2^k} = \sum_{k=1}^{\infty} (\frac{1}{2})^k$ g.s, $r = \frac{1}{2} \rightarrow$ Conv.

$\therefore \sum_{k=1}^{\infty} \frac{1}{k+2^k}$ is conv.

C.T

5) Limit Comparison Test (L.C.T)

If $\sum a_k$ and $\sum b_k$ are series with positive terms such that

$\rho = \lim_{k \to \infty} \frac{a_k}{b_k}$ where ρ is a positive real number $\boxed{\text{i.e } \rho \in (0, \infty)}$ then

both series are convergent or both of them are divergent.

e.g: Test for Convergence

1) $\sum_{k=1}^{\infty} \frac{k+5}{2k^2+1}$

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Let $\sum_{k=1}^{\infty} \frac{k+5}{2k^2+1} = \sum_{k=1}^{\infty} a_k$

Let $\sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} \frac{1}{k^{2-1}} = \sum_{k=1}^{\infty} \frac{1}{k}$ $\quad p=1 \rightarrow$ div.

$$\lim_{k\to\infty} \frac{a_k}{b_k} = \lim_{k\to\infty} \left(\frac{k+5}{2k^2+1} \div \frac{1}{k}\right) = \lim_{k\to\infty} \left(\frac{k^2+5k}{2k^2+1}\right) = \frac{1}{2} = \rho$$

$\therefore \sum_{k=1}^{\infty} \frac{k+5}{2k^2+1}$ is divergent (L.C.T)

2) $\sum_{k=1}^{\infty} \frac{k^3+k^2+\sqrt{k}+1}{5k^7+k^4+2} = \sum_{k=1}^{\infty} a_k$

Let $\sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} \frac{1}{k^4}$ $\quad p=4 \rightarrow$ conv.

$$\lim_{k\to\infty} \frac{a_k}{b_k} = \lim_{k\to\infty} \left(\frac{k^3+k^2+\sqrt{k}+1}{5k^7+k^4+2} \cdot \frac{k^4}{1}\right)$$

$$= \lim_{k\to\infty} \left(\frac{k^7+k^6+k^{4.5}+k^4}{5k^7+k^4+2}\right) = \frac{1}{5}$$

$\therefore \sum_{k=1}^{\infty} \frac{k^3+k^2+\sqrt{k}+1}{5k^7+k^4+2}$ is convergent (L.C.T).

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31) $\sum_{k=1}^{\infty} \frac{\sqrt{k+1}}{k+5} = \sum_{k=1}^{\infty} a_k$

let $\sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} \frac{1}{k^{\frac{1}{2}}}$ $\quad p = \frac{1}{2} \longrightarrow$ div

$$\lim_{k\to\infty} \frac{a_k}{b_k} = \lim_{k\to\infty} \left(\frac{\sqrt{k+1}}{k+5} \cdot \frac{\sqrt{k}}{1} \right) = \lim_{k\to\infty} \frac{\sqrt{k^2+k}}{k+1}$$

$$= \lim_{k\to\infty} \frac{\sqrt{k^2}}{k} = \lim_{k\to\infty} \frac{k}{k} = 1$$

$\therefore \sum_{k=1}^{\infty} \frac{\sqrt{k+1}}{k+5}$ is div. (L.C.T)

4) $\sum_{k=1}^{\infty} a_k$, $a_k > 0 \ \forall k$ and $\lim_{k\to\infty} k^2 a_k = 10$

$\sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} \frac{1}{k^2}$ $\longrightarrow$ $p = 2 \longrightarrow$ Conv.

$$\lim_{k\to\infty} \frac{a_k}{b_k} = \lim_{k\to\infty} \left(a_k \cdot \frac{k^2}{1} \right) = \lim_{k\to\infty} (k^2 a_k) = 10$$

$\therefore \sum_{k=1}^{\infty} a_k$ is Conv. (L.C.T).

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5) $\sum_{k=1}^{\infty} \sin^2\left(\frac{1}{k}\right) = \sum_{k=1}^{\infty} a_k$

Let $\sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} \frac{1}{k^2}$ $\quad p = 2 \rightarrow$ Conv!

$$\lim_{k\to\infty} \frac{a_k}{b_k} = \lim_{k\to\infty} \frac{\sin^2\left(\frac{1}{k}\right)}{\frac{1}{k^2}} = \lim_{k\to\infty} \left(\frac{\sin\left(\frac{1}{k}\right)}{\frac{1}{k}}\right)^2$$

$$= 1^2 = 1$$

$\therefore \sum_{k=1}^{\infty} \sin^2\left(\frac{1}{k}\right)$ is Conv. (L.C.T).

6) $\sum_{k=1}^{\infty} \ln k$

$\lim_{k\to\infty} \ln k = \infty \neq 0$

$\therefore$ by D.T, $\sum_{k=1}^{\infty} \ln k$ is div.

- END -

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Comparison Tests --- Ratio and Root Tests

· Recall

1) Divergence Test (D.T)

2) Integral Test (I.T)

3) p-series $\left(\sum \frac{1}{k^p}\right)$ $p>1 \rightarrow$ conv. $\quad p \leq 1 \rightarrow$ div.

4) comparison Test (C.T)

5) Limit Comparison Test (L.C.T)

---

6 **Ratio Test**

If $\sum_{k=1}^{\infty} a_k$ is a series with positive terms,

$$\rho = \lim_{k \to \infty} \frac{a_{k+1}}{a_k}$$ then

1) if $\rho < 1$ the series is convergent

2) if $\rho > 1$ (or ∞) the series is divergent.

No conclusion, if $\rho = 1$

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Root Test

$\sum (\dots k)^k$ ←

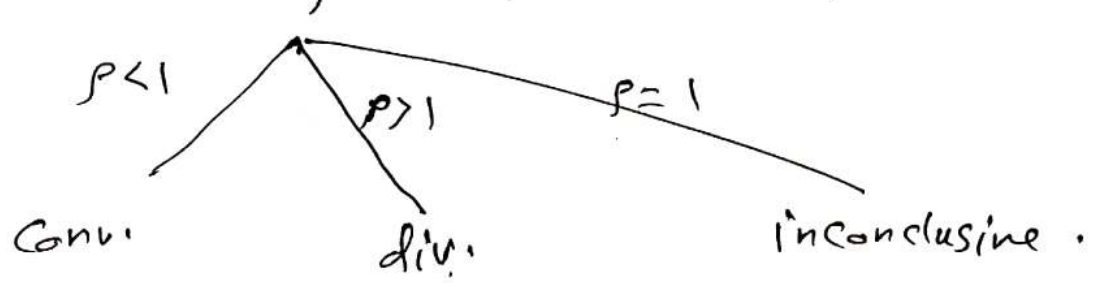
If $\sum a_k$ is a series with positive terms, $\rho = \lim_{k\to\infty} \sqrt[k]{a_k} = \lim_{k\to\infty} (a_k)^{\frac{1}{k}}$

then

1) if $\rho < 1$ then the series is convergent

2) if $\rho > 1$ (or ∞) the series is divergent

No conclusion if $\rho = 1$ → Try D.T



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e.g Test for convergence.

1) $\sum_{k=0}^{\infty} \frac{1}{k!}$

$a_k = \frac{1}{k!}$

$$\rho = \lim_{k\to\infty} \frac{a_{k+1}}{a_k} = \lim_{k\to\infty} \left(\frac{1}{(k+1)!} \cdot \frac{k!}{1}\right)$$

$$= \lim_{k\to\infty} \frac{k! \cdot 1}{(k+1)\, k!}$$

$$= \lim_{k\to\infty} \frac{1}{k+1} = 0$$

$\rho = 0 < 1$

$\therefore \sum_{k=0}^{\infty} \frac{1}{k!}$ is conv. (ratio test)

2) $\sum_{k=1}^{\infty} \left(\frac{k+1}{2k+1}\right)^k$

$a_k = \left(\frac{k+1}{2k+1}\right)^k$

$$\rho = \lim_{k\to\infty} \sqrt[k]{\left(\frac{k+1}{2k+1}\right)^k} = \lim_{k\to\infty} \frac{k+1}{2k+1} = \frac{1}{2}$$

$\rho = \frac{1}{2} < 1 \longrightarrow$ the series is conv. (root test).

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31) $\sum_{k=1}^{\infty} \frac{k!}{k^k}$

$$\rho = \lim_{k\to\infty} \frac{(k+1)!}{(k+1)^{k+1}} \cdot \frac{k^k}{k!}$$

$$= \lim_{k\to\infty} \frac{(k+1)\,k! \cdot k^k}{(k+1)^{k+1}\,k!}$$

$$= \lim_{k\to\infty} \frac{(k+1)^1 k^k}{(k+1)^{k+1}}$$

$$= \lim_{k\to\infty} \frac{k^k}{(k+1)^k}$$

$$= \lim_{k\to\infty} \left(\frac{k}{k+1}\right)^k$$

$$= \lim_{k\to\infty} \left(\frac{k+1}{k}\right)^{-k}$$

$$= \left[\lim_{k\to\infty} \left(1+\frac{1}{k}\right)^k\right]^{-1}$$

$$= (e^1)^{-1} = \frac{1}{e} < 1$$

$$\rho = e^{-1} = \frac{1}{e} < 1$$

$\therefore \sum_{k=1}^{\infty} \frac{k!}{k^k}$ is conv. (ratio test).

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4) $\sum_{k=1}^{\infty} \left(1+\frac{1}{k}\right)^{k^2}$

$$\rho = \lim_{k\to\infty} \left[\left(1+\frac{1}{k}\right)^{k^2}\right]^{\frac{1}{k}}$$

$$\rho = \lim_{k\to\infty} \left(1+\frac{1}{k}\right)^{k} = e > 1$$

$\therefore$ The series is div. (root test).

5) $\sum_{k=1}^{\infty} \left(1+\frac{1}{k}\right)^{k}$

$$\rho = \lim_{k\to\infty} \left[\left(1+\frac{1}{k}\right)^{k}\right]^{\frac{1}{k}}$$

$$= \lim_{k\to\infty} \left(1+\frac{1}{k}\right) = 1$$

$\rho = 1 \rightarrow$ No result.

D.T $\lim_{k\to\infty} \left(1+\frac{1}{k}\right)^{k} = e \neq 0$

$\therefore$ the seris is divergent (D.T).

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6) $\sum_{k=1}^{\infty} \frac{1}{k^2 5^k}$

6

$\lim_{k\to\infty} \sqrt[k]{k} = 1$

$$\rho = \lim_{k\to\infty} \sqrt[k]{\frac{1}{k^2 5^k}}$$

$$= \lim_{k\to\infty} \frac{1}{\sqrt[k]{k^2}\,(5)}$$

$$\rho = \lim_{k\to\infty} \frac{1}{5\,(\sqrt[k]{k})^2} = \frac{1}{5(1)} = \frac{1}{5} < 1$$

$\therefore$ The series is conv. (root test).

Sol 2: $\sum_{k=1}^{\infty} \frac{1}{k^2 5^k}$

$$\rho = \lim_{k\to\infty} \left(\frac{1}{(k+1)^2 \frac{5^{k+1}}{5}} \cdot \frac{k^2 5^k}{1} \right)$$

$$= \lim_{k\to\infty} \frac{k^2}{5(k+1)^2} = \frac{1}{5}$$

$$\rho = \frac{1}{5} < 1$$

$\therefore \sum_{k=1}^{\infty} \frac{1}{k^2 5^k}$ is conv. (ratio test)

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(7)

7) $\sum_{k=0}^{\infty} \frac{(k!)^2}{(2k)!}$

$$\rho = \lim_{k \to \infty} \frac{((k+1)!)^2}{(2(k+1))!} \cdot \frac{(2k)!}{(k!)^2}$$

$$= \lim_{k \to \infty} \frac{(k+1)\,k!\,(k+1)\,k!\,(2k)!}{(2k+2)!\,k!\,k!}$$

$$= \lim_{k \to \infty} \frac{(k+1)^2\,(2k)!}{(2k+2)(2k+1)\,(2k)!}$$

$$= \lim_{k \to \infty} \frac{(k+1)^2}{(2k+2)(2k+1)} = \frac{1}{4}$$

$$\rho = \frac{1}{4} < 1$$

$\therefore \sum_{k=0}^{\infty} \frac{(k!)^2}{(2k)!}$ is convergent (ratio test)

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8) $\sum_{k=0}^{\infty} \frac{3^k}{k!}$

$$\rho = \lim_{k\to\infty} \frac{3^{k+1}}{(k+1)!} \cdot \frac{k!}{3^k}$$

$$= \lim_{k\to\infty} \frac{3\, k!}{(k+1)\, k!}$$

$$= \lim_{k\to\infty} \frac{3}{k+1} = 0$$

$\rho = 0 < 1 \rightarrow$ the seris is conv. (ratio test)

9) $\sum_{k=1}^{\infty} \left(\frac{4k+1}{3k+2}\right)^k$

$$\rho = \lim_{k\to\infty} \left[\left(\frac{4k+1}{3k+2}\right)^k\right]^{\frac{1}{k}}$$

$$= \lim_{k\to\infty} \left(\frac{4k+1}{3k+2}\right) = \frac{4}{3}$$

$\rho = \frac{4}{3} > 1$.

$\therefore$ The seris is divergent (by root test).

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10) $\sum_{k=0}^{\infty} \frac{5 + (-1)^k}{7^k}$

$5 + (-1)^k \leq 6 \quad \forall k$

$\rightarrow \frac{5 + (-1)^k}{7^k} \leq \frac{6}{7^k} \quad \forall k$

$\sum_{k=0}^{\infty} \frac{6}{7^k} = \sum_{k=0}^{\infty} 6\left(\frac{1}{7}\right)^k$ g.s

$r = \frac{1}{7} \rightarrow$ conv.

$\therefore \sum_{k=0}^{\infty} \frac{5 + (-1)^k}{7^k}$ is conv. (C.T)

END

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Alternating series, conditional and absolute convergence:

Def: Alternating Series (A.S):

If $b_k > 0$ for any $k = 1, 2, 3, \dots$ then

$\sum_{k=1}^{\infty} (-1)^k b_k$ (or $\sum_{k=1}^{\infty} (-1)^{k+1} b_k$) is called an alternating series.

$\cos(k\pi) = (-1)^k$

$k = 1, 2, 3, \dots$

* $2 - 4 + 6 - 8 + 10 - \dots$ A.S

* $-1 + \frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \frac{1}{16} + \dots$ A.S

* $-1, -1, 2, 2, -3, -3, 4, 4, \dots$ not A.S

* $\sum_{k=1}^{\infty} \frac{k\cos(k\pi)}{k^2+1} = \sum_{k=1}^{\infty} (-1)^k \frac{k}{k^2+1}$ A.S.

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Alternating Series Test (A.S.T)

The A.S $\sum_{k=1}^{\infty} (-1)^k b_k$ (or $\sum_{k=1}^{\infty} (-1)^{k+1} b_k$), $b_k > 0$

is convergent, if

1) $\{b_k\}_{k=1}^{\infty}$ is strictly decreasing $(b_1 > b_2 > b_3 > \dots)$
or (eventually strictly decreasing)

and 2) $\lim_{k \to \infty} b_k = 0$

---

e.g.: Determine whether the given seris converges or diverges.

1) $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = \sum_{k=1}^{\infty} (-1)^k \left(\frac{1}{k}\right)$

$\{b_k\} = \left\{\frac{1}{k}\right\} \searrow$ ✓

$\lim_{k \to \infty} \frac{1}{k} = 0$ ✓

$\therefore \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$ is conv. (A.S.T)

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2) $$\sum_{k=1}^{\infty} \frac{(-1)^k}{\ln(k+1)} = \sum_{k=1}^{\infty} (-1)^k \frac{1}{\ln(k+1)}$$

$$\{b_k\}_{k=1}^{\infty} = \left\{\frac{1}{\ln(k+1)}\right\}_{k=1}^{\infty} \searrow \checkmark$$

$$\lim_{k\to\infty} \frac{1}{\ln(k+1)} = 0 \checkmark$$

$\therefore$ The seris is conv. (A.S.T)

Absolute and conditional convergence:

Thm: If $\sum |a_k|$ is convergent then $\sum a_k$ is convergent.

Notes

1) $\sum |a_k|$ converges $\Rightarrow$ $\sum a_k$ converges.

2) $\sum a_k$ converges $\not\Rightarrow$ $\sum |a_k|$ converges.

3) $\sum a_k$ diverges $\Rightarrow$ $\sum |a_k|$ diverges

4) $\sum a_k$ converges $\not\Rightarrow$ $\sum |a_k|$ converges.

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Def: A series $\sum a_k$ is called

1) absolutely convergent if $\sum |a_k|$ is convergent

2) Conditionally convergent if

$\sum a_k$ is convergent but $\sum |a_k|$ divergent

---

e.g: Determine whether the given series is absolutely convergent, conditionally convergent or divergent?

1) $\sum_{k=1}^{\infty} \frac{(-1)^k}{k^2}$

$\sum_{k=1}^{\infty} \left| \frac{(-1)^k}{k^2} \right| = \sum_{k=1}^{\infty} \frac{1}{k^2}$ $\quad p=2 \rightarrow$ Conv.

$\therefore \sum_{k=1}^{\infty} \frac{(-1)^k}{k^2}$ is abs. conv.

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2) $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{\sqrt{k}}$

$\sum_{k=1}^{\infty} \left| \frac{(-1)^{k+1}}{\sqrt{k}} \right| = \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}} = \sum_{k=1}^{\infty} \frac{1}{k^{\frac{1}{2}}}$, $p = \frac{1}{2} \rightarrow$ div.

$\therefore$

$\sum_{k=1}^{\infty} (-1)^k \frac{1}{\sqrt{k}}$ A.S

$\frac{1}{\sqrt{k}} \searrow$, $\lim_{k \to \infty} \frac{1}{\sqrt{k}} = 0$

$\therefore \sum_{k=1}^{\infty} (-1)^k \frac{1}{\sqrt{k}}$ is conv. (A.S.T)

$\therefore \sum_{k=1}^{\infty} (-1)^k \frac{1}{\sqrt{k}}$ is conditionally convergent.

3) $\sum_{k=1}^{\infty} (-1)^k \frac{k}{2k+1}$

$b_k = \frac{k}{2k+1}$

$\lim_{k \to \infty} b_k = \lim_{k \to \infty} \frac{k}{2k+1} = \frac{1}{2} \neq 0$

$\therefore \sum_{k=1}^{\infty} (-1)^k \frac{k}{2k+1}$ is divergent.

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4) $\sum_{k=1}^{\infty} \frac{\sin k}{k^2}$.

$$\sum_{k=1}^{\infty} \left| \frac{\sin k}{k^2} \right| = \sum_{k=1}^{\infty} \frac{|\sin k|}{k^2}$$

$$|\sin k| \le 1 \longrightarrow \frac{|\sin k|}{k^2} \le \frac{1}{k^2} \quad \forall k$$

$\sum_{k=1}^{\infty} \frac{1}{k^2}$ is Conv. , $p=2$

$\therefore \sum_{k=1}^{\infty} \left| \frac{\sin k}{k^2} \right|$ is Conv. (C.T)

$\therefore \sum_{k=1}^{\infty} \frac{\sin k}{k^2}$ is abs. Conv.

5) $\sum_{k=3}^{\infty} (-1)^k \frac{\ln k}{k}$

$$\sum_{k=3}^{\infty} \left| (-1)^k \frac{\ln k}{k} \right| = \sum_{k=3}^{\infty} \frac{\ln k}{k}$$

$$f(x) = \frac{\ln x}{x} \longrightarrow f'(x) = \frac{x \cdot \frac{1}{x} - \ln x}{x^2}$$

$$= \frac{1 - \ln x}{x^2} < 0 \quad \forall x \ge 3$$

$\therefore$ f is ~~Laeere~~ strictly decreasing

f is +ve , Cont. on $[3, \infty)$

$$\int_3^{\infty} \frac{\ln x}{x} dx = \lim_{L \to \infty} \int_3^{L} \frac{\ln x}{x} = \lim_{L \to \infty} \left[\frac{(\ln x)^2}{2} \right]_3^{L}$$

$$= \lim_{L \to \infty} \left[\frac{(\ln L)^2}{2} - \frac{(\ln 3)^2}{2} \right] = \infty$$

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$\therefore \int_3^{\infty} \frac{\ln x}{x}\,dx$ is divergent.

$\therefore \sum_{k=3}^{\infty} \frac{\ln k}{k}$ is div. (I.T)

$\sum_{k=3}^{\infty} (-1)^k \frac{\ln k}{k}$ A.S

$b_k = \frac{\ln k}{k}$ ↓ ✓

$\lim_{k\to\infty} \frac{\ln k}{k} = \lim_{k\to\infty} \frac{\frac{1}{k}}{1} = 0$ ✓

$\therefore \sum_{k=3}^{\infty} (-1)^k \frac{\ln k}{k}$ is conv.

$\therefore \sum_{k=3}^{\infty} (-1)^k \frac{\ln k}{k}$ is conditionally convergent

6) $\sum_{k=2}^{\infty} \left(-\frac{1}{\ln k}\right)^k = \sum_{k=1}^{\infty} (-1)^k \left(\frac{1}{\ln k}\right)^k$

$\sum_{k=1}^{\infty} \left|(-1)^k \left(\frac{1}{\ln k}\right)^k\right| = \sum_{k=1}^{\infty} \left(\frac{1}{\ln k}\right)^k$

$\lim_{k\to\infty} \sqrt[k]{\left(\frac{1}{\ln k}\right)^k} = \lim_{k\to\infty} \frac{1}{\ln k} = 0 < 1$

$\therefore$ By root test $\sum_{k=2}^{\infty} \left|\left(-\frac{1}{\ln k}\right)^k\right|$ conv.

$\therefore \sum_{k=2}^{\infty} \left(-\frac{1}{\ln k}\right)^k$ is abs. conv.

7) $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k^2+3k}}$

$\sum_{k=1}^{\infty} \left|\frac{(-1)^k}{\sqrt{k^2+3k}}\right| = \sum_{k=1}^{\infty} \frac{1}{\sqrt{k^2+3k}} = \sum_{k=1}^{\infty} a_k$

Let $\sum_{k=1}^{\infty} b_k = \sum_{k=1}^{\infty} \frac{1}{k^1}$ $\quad p=1 \to$ div.

$\sum_{k=1}^{\infty} (-1)^k \frac{1}{\sqrt{k^2+3k}}$ A.S

$\frac{1}{\sqrt{k^2+3k}} \searrow$ ✓ $\quad \because \lim_{k\to\infty} \frac{1}{\sqrt{k^2+3k}} = 0$ ✓

$\therefore \sum_{k=1}^{\infty} (-1)^k \frac{1}{\sqrt{k^2+3k}}$ is conv. (A.S.T)

$\therefore \sum_{k=1}^{\infty} (-1)^k \frac{1}{\sqrt{k^2+3k}}$ is Conditionally conv.

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8) $\sum_{k=1}^{\infty} (-1)^k \frac{e^k}{k^2+1}$ A.S

$$\lim_{k\to\infty} \frac{e^k}{k^2+1} \quad \frac{\infty}{\infty}$$

$$= \lim_{k\to\infty} \frac{e^k}{2k} \quad \frac{\infty}{\infty}$$

$$= \lim_{k\to\infty} \frac{e^k}{2} = \infty \neq 0$$

$\therefore \sum_{k=1}^{\infty} (-1)^k \frac{e^k}{k^2+1}$ is div.

9) $\sum_{k=1}^{\infty} (-1)^k \frac{k!}{(2k+1)!}$ A.S

$$\sum_{k=1}^{\infty} \left| (-1)^k \frac{k!}{(2k+1)!} \right| = \sum_{k=1}^{\infty} \frac{k!}{(2k+1)!}$$

$$\rho = \lim_{k\to\infty} \frac{(k+1)!}{(2(k+1)+1)!} \cdot \frac{(2k+1)!}{k!}$$

$$= \lim_{k\to\infty} \frac{(k+1)\,k!\,(2k+1)!}{(2k+3)!\,k!} = \lim_{k\to\infty} \frac{(2k+1)!\,(k+1)}{(2k+3)(2k+2)(2k+1)!}$$

$$\rho = \lim_{k\to\infty} \frac{k+1}{(2k+3)(2k+2)} = 0 < 1$$

$\therefore \sum_{k=1}^{\infty} \left| (-1)^k \frac{k!}{(2k+1)!} \right|$ is conv.

$\therefore \sum_{k=1}^{\infty} (-1)^k \frac{k!}{(2k+1)!}$ is abs. conv.

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Power series:

Recall

* $\sum |a_k|$ converges $\Rightarrow$ $\sum a_k$ converges

* $\sum |a_k|$ diverges $\not\Rightarrow$ $\sum a_k$ diverges

Ratio Test for absolute convergence

Let $\sum a_k$ be a series with non zero terms, $\rho = \lim_{k\to\infty} \left|\frac{a_{k+1}}{a_k}\right|$ then

1) $\rho < 1 \Rightarrow \sum a_k$ converges

2) $\rho > 1$ (or ∞) $\Rightarrow \sum a_k$ diverges

3) No conclusion if $\rho = 1$

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Any series of the form

$$\sum_{k=0}^{\infty} C_k (x-x_0)^k$$ is called a power series in $(x-x_0)$

(or a power series about $x=x_0$)

* $\sum_{k=0}^{\infty} \frac{2^k}{5+k^2}(x-3)^k$ → power series in $x-3$

(power series about $x=3$)

$C_k = \frac{2^k}{5+k^2}$

---

* $\sum_{k=0}^{\infty} \frac{(-1)^k}{k!}(x+2)^k$ → power series in $x+2$

(power series about $x=-2$

$C_k = \frac{(-1)^k}{k!}$

---

* $\sum_{k=0}^{\infty} \frac{k^2}{2^k} x^k$ → power series in x

(power series about $x=0$)

$C_k = \frac{k^2}{2^k}$

---

* $\sum_{k=0}^{\infty} (x-5)^k$ → power series in $x-5$

(power series about $x=5$)

$C_k = 1 \quad \forall k$

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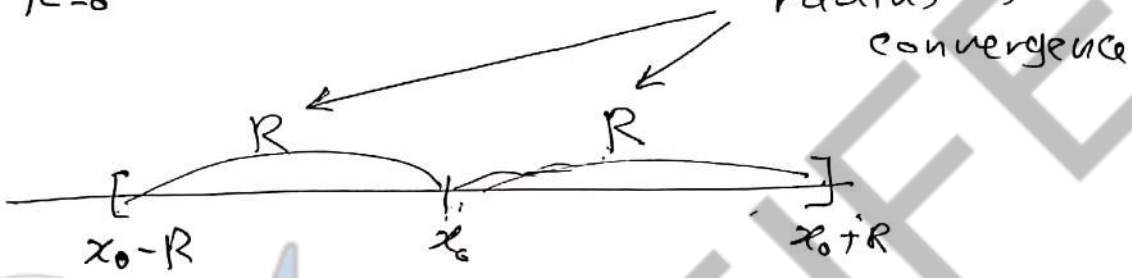
Interval and Radius of Convergence of the Power Series.

$$\sum_{k=0}^{\infty} c_k (x - x_0)^k$$



Interval of Convergence

i.e the interval convergence will be

1) $[x_0 - R, x_0 + R]$ $\quad R \in \mathbb{R}^+$

2) $(x_0 - R, x_0 + R]$ $\quad R \in \mathbb{R}^+$

3) $[x_0 - R, x_0 + R)$ $\quad R \in \mathbb{R}^+$

4) $(x_0 - R, x_0 + R)$ $\quad R \in \mathbb{R}^+$

5) $(-\infty, \infty) \longrightarrow R = +\infty$

6) $[x_0, x_0] = \{x_0\} \longrightarrow R = 0$

If $f(x) = \sum_{k=0}^{\infty} c_k (x - x_0)^k$ then

Domain f = the interval of convergence

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e.g: Find the interval and radius of convergence of the given power series.

1) $\sum_{k=0}^{\infty} \frac{(x-3)^k}{k+2}$

$$\rho = \lim_{k\to\infty} \left| \frac{(x-3)^{k+1}}{k+3} \cdot \frac{k+1}{(x-3)^k} \right|$$

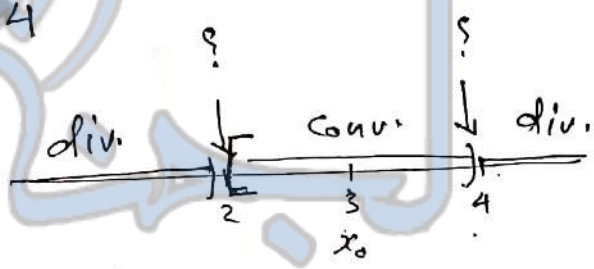
$$= \lim_{k\to\infty} |x-3| \frac{k+1}{k+3} = |x-3| \lim_{k\to\infty} \frac{k+1}{k+3} = |x-3|$$

$$\rho = |x-3|$$

$$\rho < 1 \iff |x-3| < 1$$

$$\iff -1 < x-3 < 1$$

$$\iff 2 < x < 4$$



$x = 2 \rightarrow \sum_{k=0}^{\infty} \frac{(-1)^k}{k+2}$ A.S

$\left\{\frac{1}{k+2}\right\} \searrow$, $\lim_{k\to\infty} \frac{1}{k+2} = 0 \rightarrow$ conv. (A.S.T)

$x = 4 \rightarrow \sum_{k=1}^{\infty} \frac{1}{k+2}$

$\sum_{k=1}^{\infty} \frac{1}{k}$ div. $\rho = 1$

$$\lim_{k\to\infty} \left(\frac{1}{k+2} \cdot \frac{k}{1}\right) = \lim_{k\to\infty} \frac{k}{k+2} = 1 \ (+ve)$$

$\therefore \sum_{k=1}^{\infty} \frac{1}{k+2}$ is divergent

Interval of conv. $= [2,4)$, radius of conv. $= 1$

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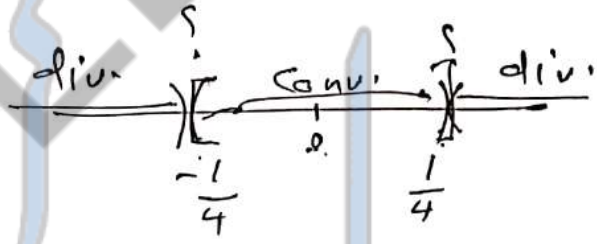
2) $\sum_{k=1}^{\infty} \frac{4^k}{k^3} x^k$

$$\rho = \lim_{k\to\infty} \left| \frac{4^{k+1}}{(k+1)^3} \cdot \frac{x^{k+1}}{x^k} \cdot \frac{k^3}{4^k} \right|$$

$$= \lim_{k\to\infty} |x| \left(\frac{4k^3}{(k+1)^3}\right) = 4|x|$$

$$\rho < 1 \iff 4|x| < 1 \iff |x| < \frac{1}{4}$$

$$\iff -\frac{1}{4} < x < \frac{1}{4}$$



$$x = -\frac{1}{4} \to \sum_{k=1}^{\infty} \frac{4^k}{k^3} \cdot \left(\frac{-1}{4}\right)^k$$

$$= \sum_{k=1}^{\infty} \frac{(-1)^k 4^k}{k^3 4^k} = \sum_{k=1}^{\infty} \frac{(-1)^k}{k^3} \to \text{conv.}$$

$$x = \frac{1}{4} \to \sum_{k=1}^{\infty} \frac{4^k}{k^3} \left(\frac{1}{4}\right)^k = \sum_{k=1}^{\infty} \frac{1}{k^3} \quad \text{conv.} \quad p=3$$

$\therefore$ the interval of conv. is $[-\frac{1}{4}, \frac{1}{4}]$

Radius of conv. $= \frac{1}{4}$

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3) $\sum_{k=0}^{\infty} (x-5)^{2k+1} = (x-5)^3 + (x-5)^5 + (x-5)^7 + \cdots$

$$\rho = \lim_{k\to\infty} \left| \frac{(x-5)^{2(k+1)+1}}{(x-5)^{2k+1}} \right|$$

$$= \lim_{k\to\infty} \left| \frac{(x-5)^{2k+3}}{(x-5)^{2k}} \right| = \lim_{k\to\infty} \left| (x-5)^2 \right| = (x-5)^2$$

$\rho = (x-5)^2$

$\rho < 1 \longrightarrow (x-5)^2 < 1$

$\longrightarrow \sqrt{(x-5)^2} < \sqrt{1}$

$\longrightarrow |x-5| < 1$

$\longrightarrow -1 < x-5 < 1$

$\longrightarrow 4 < x < 6$

div conv div

4 5 6

$x = 4 \longrightarrow \sum_{k=0}^{\infty} (4-5)^{2k+1} = \sum_{k=0}^{\infty} (-1)^{2k+1} = \sum_{k=0}^{\infty} -1$

div., $\lim_{k\to\infty}(-1) = -1 \neq 0$ (D.T)

$x = 6 \longrightarrow \sum_{k=0}^{\infty} 1$ div, $\lim_{k\to\infty} 1 = 1$ (D.T)

∴ the interval of convergence $= (4,6)$

Radius of conv. $= \frac{2}{2} = 1$

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4) $\sum_{k=0}^{\infty} \frac{(-1)^k x^k}{7^k}$

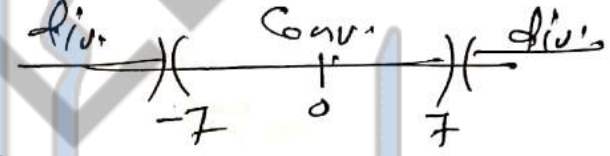
$$\rho = \lim_{k\to\infty} \left| \frac{(-1)^{k+1} x^{k+1}}{7^{k+1}} \cdot \frac{7^k}{(-1)^k x^k} \right|$$

$$= \lim_{k\to\infty} \left| \frac{x}{7} \right| = \frac{1}{7}|x|$$

$\rho < 1 \longrightarrow \frac{1}{7}|x| < 1$

$\longrightarrow |x| < 7$

$\longrightarrow -7 < x < 7$



$x = -7 \longrightarrow \sum_{k=0}^{\infty} \frac{(-1)^k (-7)^k}{7^k} = \sum_{k=0}^{\infty} \left(\frac{(-1)(-7)}{7} \right)^k = \sum_{k=0}^{\infty} 1$

$\lim_{k\to\infty} 1 = 1 \neq 0$

$\longrightarrow$ div. D.T

$x = 7 \longrightarrow \sum_{k=0}^{\infty} \frac{(-1)^k 7^k}{7^k} = \sum_{k=0}^{\infty} (-1)^k \longrightarrow$ div.

Interval of Conv. $= (-7, 7)$

Radius of Conv. $= 7$

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5) $\sum_{k=0}^{\infty} \frac{(-1)^k (x+1)^k}{k!}$

$$\rho = \lim_{k\to\infty} \left| \frac{(x+1)^{k+1}}{(k+1)!} \cdot \frac{k!}{(x+1)^k} \right| = \lim_{k\to\infty} \left(|x+1| \frac{1}{k+1} \right)$$

$$= 0 < 1 \quad \forall x$$

$\rho < 0$ for any x

$\therefore$ the interval of convergence $= (-\infty, \infty)$
$= \mathbb{R}$

radius of ~~Conv. = ∞~~
Convergence $= +\infty$

6) $\sum_{k=0}^{\infty} k! (x-6)^k$

$$\rho = \lim_{k\to\infty} \left| \frac{(k+1)! (x-6)^{k+1}}{k! (x-6)^k} \right| = \lim_{k\to\infty} \left(|x-6| \cdot (k+1) \right)$$

$= \infty$ given $x \neq 6$

div. $\forall x \neq 6$

$\therefore$ The interval of convergence $= [6,6]$
$= \{6\}$

Radius of Conv. $= 0$

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e.g: Given that $f(x) = \sum_{k=0}^{\infty} \frac{(x-1)^k}{4^k}$

a) find the domain of f ?

b) find $f(0)$

---

a) D_f = Interval of convergence.

$$\rho = \lim_{k\to\infty} \left| \frac{(x-1)^{k+1}}{4^{k+1}} \cdot \frac{4^k}{(x-1)^k} \right|$$

$$= \lim_{k\to\infty} \frac{|x-1|}{4} = \frac{|x-1|}{4}$$

$\rho < 1 \longrightarrow \frac{|x-1|}{4} < 1 \longrightarrow |x-1| < 4$

$\longrightarrow -4 < x-1 < 4$

$\longrightarrow -3 < x < 5$

div (-3) conv. (5) div

$x = -3 \quad \sum_{k=0}^{\infty} \frac{(-4)^k}{4^k} = \sum_{k=0}^{\infty} (-1)^k$ div.

$x = 5 \longrightarrow \sum_{k=0}^{\infty} \frac{4^k}{4^k} = \sum_{k=0}^{\infty} 1$ div.

interval of convergence $= (-3, 5)$

$\therefore$ Domain $(f) = (-3, 5)$

b) $f(0) = \sum_{k=0}^{\infty} \frac{(-1)^k}{4^k} = \sum_{k=0}^{\infty} \left(\frac{-1}{4}\right)^k$ g.s, $r = -\frac{1}{4} \to$ Conv

$$f(0) = \text{sum} = \frac{1}{1-(-\frac{1}{4})} = \frac{1}{1+\frac{1}{4}}$$

$$= \frac{4}{4+1} = \frac{4}{5}$$

END.

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Taylor and Maclaurin series (Expansions)

Taylor series about $x = a$ (in $(x-a)$)

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!} + \cdots$$

$$= \sum_{k=0}^{\infty} \frac{f^{(k)}(a)(x-a)^k}{k!}$$

If $a = 0$ the Taylor series is called Maclaurin series.

Maclaurin series

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \cdots$$

$$= \sum_{k=0}^{\infty} \frac{f^{(k)}(0)x^k}{k!}$$

$$f^{(0)}(x) = f(x)$$

$$f^{(1)}(x) = f'(x)$$

$$f^{(2)}(x) = f''(x)$$

$$\vdots$$

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e.g: Find Taylor expansion of $f(x) = e^x$ about $x = 1$.

Sol:

$$f(x) = f(1) + f'(1)(x-1) + \frac{f''(1)(x-1)^2}{2!} + \cdots$$

$f(x) = e^x \Rightarrow f(1) = e^1 = e$

$f'(x) = e^x \Rightarrow f'(1) = e$

$f''(x) = e^x \Rightarrow f''(1) = e$

$$\therefore f(x) = e + e(x-1) + \frac{e(x-1)^2}{2!} + \cdots$$

$$\Rightarrow f(x) = e \sum_{k=0}^{\infty} \frac{(x-1)^k}{k!}$$

e.g: Find Maclaurin series corresponding to the given function.

1) $f(x) = e^x$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \cdots$$

$f(x) = e^x \Rightarrow f(0) = 1$

$f'(x) = e^x \Rightarrow f'(0) = 1$

$\vdots$

$$\therefore f(x) = 1 + 1x + \frac{1x^2}{2!} + \frac{1x^3}{3!} + \cdots$$

$$= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \qquad -\infty < x < \infty$$

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2) $f(x) = \sin x$

Sol:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \cdots$$

$f(x) = \sin x \Rightarrow f(0) = 0$

$f'(x) = \cos x \Rightarrow f'(0) = 1$

$f''(x) = -\sin x \Rightarrow f''(0) = 0$

$f'''(x) = -\cos x \Rightarrow f'''(0) = -1$

$f^{(4)}(x) = \sin x \Rightarrow f^{(4)}(0) = 0$

$\vdots$

$$\therefore f(x) = 0 + 1x + \frac{0x^2}{2!} + \frac{(-1)x^3}{3!} + \frac{0x^4}{4!} + \frac{1x^5}{5!} + \frac{0x^6}{6!} + \cdots$$

$$\Rightarrow \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

$$\therefore \sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} \qquad -\infty < x < \infty$$

3) $f(x) = \cos x$ EX.

Ans:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

$$= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} \qquad -\infty < x < \infty$$

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4) $f(x) = \cosh x$

Sol.:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \frac{f^{(4)}(0)x^4}{4!} + \cdots$$

$f(x) = \cosh x \Rightarrow f(0) = 1$

$f'(x) = \sinh x \Rightarrow f'(0) = 0$

$f''(x) = \cosh x \Rightarrow f''(0) = 1$

$\cosh 0 = 1$
$\sinh 0 = 0$

$\vdots$

$$\therefore f(x) = 1 + 0 + \frac{x^2}{2!} + 0 + \frac{x^4}{4!} + 0 + \cdots$$

$$\Rightarrow \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots$$

$$= \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!} \qquad -\infty < x < \infty$$

5) $f(x) = \sinh x$ EX

Sol:

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots$$

$$\therefore \sinh x = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$$

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6) $f(x) = \frac{1}{1-x}$

Sol: $f(x) = (1-x)^{-1}$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \frac{f^{(4)}(0)x^4}{4!} + \cdots$$

$f(x) = (1-x)^{-1} \Rightarrow f(0) = 1$

$f'(x) = +1(1-x)^{-2} \Rightarrow f'(0) = 1$

$f''(x) = +2(1)(1-x)^{-3} \Rightarrow f''(0) = 2(1)$

$f'''(x) = +3(2)(1)(1-x)^{-4} \Rightarrow f'''(0) = 3!$

$$\therefore f(x) = 1 + 1x + \frac{2(1)x^2}{2!} + \frac{3!x^3}{3!} + \frac{4!x^4}{4!} + \cdots$$

$$\Rightarrow \frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k \quad , -1 < x < 1$$

Note

you can get the same result by using geometric series

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7) $f(x) = (1+x)^m$, $m \in \mathbb{R} \setminus \{0, 1, 2, 3, \dots\}$

(Binomial series)

Sol:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \dots$$

$f(x) = (1+x)^m \Rightarrow f(0) = 1$

$f'(x) = m(1+x)^{m-1} \Rightarrow f'(0) = m$

$f''(x) = m(m-1)(1+x)^{m-2} \Rightarrow f''(0) = m(m-1)$

$f'''(x) = m(m-1)(m-2)(1+x)^{m-3} \Rightarrow f'''(0) = m(m-1)(m-2)$

$\Rightarrow f^{(k)}(0) = m(m-1)(m-2)\cdots(m-k+1)$

$$\therefore f(x) = 1 + mx + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-2)}{2!}x^3 + \dots$$

$$\Rightarrow (1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-2)}{2!}x^3 + \dots$$

$$= 1 + \sum_{k=1}^{\infty} \frac{m(m-1)(m-2)\cdots(m-k+1)}{k!}x^k \quad , -1 < x < 1$$

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Important Maclaurin Series:

1) $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \qquad -\infty < x < \infty$

2) $\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \qquad -\infty < x < \infty$

3) $\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \qquad -\infty < x < \infty$

4) $\sinh x = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots \qquad -\infty < x < \infty$

5) $\cosh x = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \frac{x^8}{8!} + \cdots \qquad -\infty < x < \infty$

6) $\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + x^3 + \cdots \qquad -1 < x < 1$ (or $|x| < 1$)

7) $(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-2)}{3!}x^3 + \cdots$

$-1 < x < 1$ (or $|x| < 1$)

E.g.: Find Maclaurin series of the given function

1) $f(x) = x^3 e^{2x}$

Sol: $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \Rightarrow e^{2x} = \sum_{k=0}^{\infty} \frac{(2x)^k}{k!}$

$\therefore f(x) = x^3 \sum_{k=0}^{\infty} \frac{2^k x^k}{k!} \Rightarrow f(x) = \sum_{k=0}^{\infty} \frac{2^k x^{k+3}}{k!}$

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2) $f(x) = x^3 \cos x^4$

Sol:

$$\cos x^4 = \sum_{K=0}^{\infty} (-1)^K \frac{(x^4)^{2K}}{(2K)!} = \sum_{K=0}^{\infty} (-1)^K \frac{x^{8K}}{(2K)!}$$

$$\therefore f(x) = x^3 \cos x^4 = \sum_{K=0}^{\infty} (-1)^K \frac{x^{8K+3}}{(2K)!}$$

e.g: Find Maclaurin expansion corresponding to the given function and state its interval of convergence.

1) $f(x) = \frac{x^2}{2+3x}$

$$f(x) = \frac{x^2}{2}\left(\frac{1}{1-\left(\frac{-3x}{2}\right)}\right)$$

$$= \frac{x^2}{2} \sum_{K=0}^{\infty} \left(\frac{-3x}{2}\right)^K \quad , \left|\frac{-3x}{2}\right| < 1$$

$$\frac{1}{1-x} = \sum_{K=0}^{\infty} x^K \quad , |x| < 1$$

$$\longleftrightarrow |x| < \frac{2}{3} \longleftrightarrow \frac{-2}{3} < x < \frac{2}{3}$$

$$= \frac{x^2}{2} \sum_{K=0}^{\infty} (-1)^K \frac{3^K x^K}{2^K}$$

$$= \sum_{K=0}^{\infty} (-1)^K \frac{3^K x^{K+2}}{2^{K+1}} \quad , \frac{-2}{3} < x < \frac{2}{3}$$

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2) $f(x) = \tan^{-1} x$

Sol:

$$f'(x) = \frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{k=0}^{\infty} (-x^2)^k \quad , |-x^2| < 1$$

$$\Longleftrightarrow |x| < 1$$

$$\Longleftrightarrow -1 < x < 1$$

$$\Rightarrow \frac{1}{1+x^2} = \sum_{k=0}^{\infty} (-1)^k x^{2k} \qquad -1 < x < 1$$

integrate w.r.t x (You can integrate and differentiate any power serie in the interval of convergence)

$$\Rightarrow \int \frac{1}{1+x^2}\,dx = \int \sum_{k=0}^{\infty} (-1)^k x^{2k}\,dx$$

$$\Rightarrow \tan^{-1} x = \sum_{k=0}^{\infty} (-1)^k \int x^{2k}\,dx$$

$$\Rightarrow \tan^{-1} x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{2k+1} + C$$

Take $x = 0 \Longrightarrow C = 0$

$$\therefore f(x) = \tan^{-1} x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{2k+1}$$

for sure the series converges $-1 < x < 1$

Now check at $x = 1$ and $x = -1$

(use alternating series test, you'll find the series converges at $x = \pm 1$)

$\therefore$ the interval of convergence $= [-1, 1]$

or $|x| \leq 1$

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e.g: Find the first three non-zero terms of Maclaurin series corresponding to the given function and state the interval of convergence

1) $f(x) = x^5 \sin x^2$

$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$

Sol.

$\sin x^2 = x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \cdots$

$f(x) = x^5 \sin x^2 = x^5 \left(x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \cdots \right)$

$= x^7 - \frac{x^{11}}{3!} + \frac{x^{15}}{5!} + \cdots \qquad -\infty < x < \infty$

2) $f(x) = \sqrt[3]{1+2x}$

$= (1+2x)^{\frac{1}{3}}$

use Binomial series, $m = \frac{1}{3}$

$= 1 + \frac{1}{3}(2x) + \frac{1}{3}\left(\frac{1}{3} - 1\right)\frac{(2x)^2}{2!} + \cdots$

$= 1 + \frac{2}{3}x - \frac{4}{9}x^2 + \cdots$

(Interval of convergence): since $|2x| < 1$, $|x| < \frac{1}{2}$

$\left(-\frac{1}{2}, \frac{1}{2}\right)$

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e.g: Find the sum of series

$$\sum_{K=2}^{\infty} \frac{x^{4K}}{(2K)!}$$

Sol:

$$\sum_{K=2}^{\infty} \frac{x^{4K}}{(2K)!} = \sum_{K=2}^{\infty} \frac{(x^2)^{2K}}{(2K)!}$$

But $$\sum_{K=0}^{\infty} \frac{(x^2)^{2K}}{(2K)!} = 1 + \frac{1}{2}x^4 + \sum_{K=2}^{\infty} \frac{x^{4K}}{(2K)!}$$

$$\Rightarrow \cosh x^2 = 1 + \frac{1}{2}x^4 + \sum_{K=2}^{\infty} \frac{x^{4K}}{(2K)!}$$

$$\therefore \sum_{K=2}^{\infty} \frac{x^{4K}}{(2K)!} = -1 - \frac{1}{2}x^4 + \cosh x^2$$

e.g: Find the coefficient of x^3 in Maclaurin expansion corresponding to $f(x) = e^x \sin x$

Sol:

$$f(x) = e^x \sin x = \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots\right)$$

Coefficient of x^3 is $\left(-\frac{1}{3!} + \frac{1}{2!}\right) = \frac{-1}{6} + \frac{1}{2} = \frac{2}{6}$

$$= \frac{1}{3}$$

END

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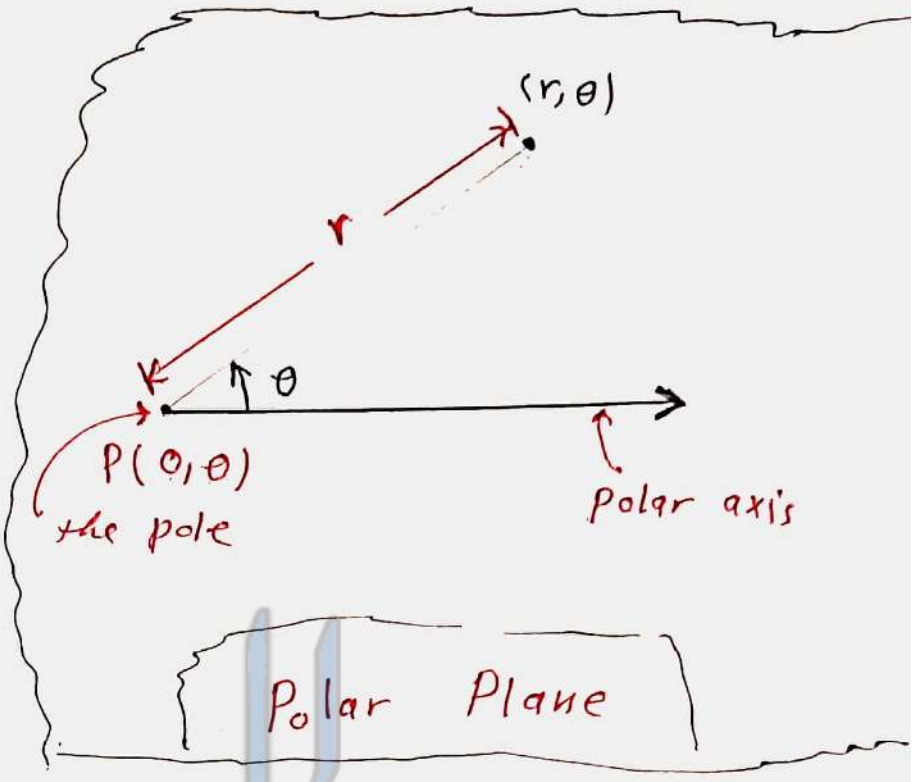
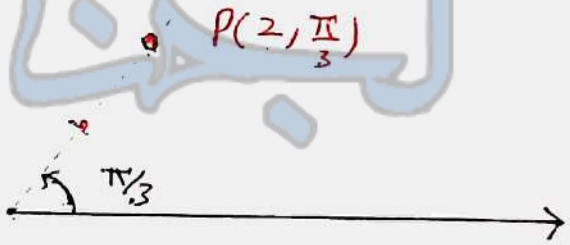
Polar Coordinates:

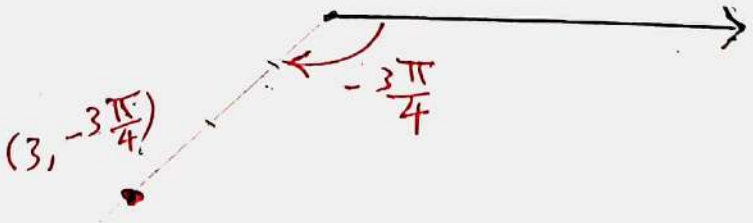


e.g: Plot the following points in the polar plane

1) $(2, \frac{\pi}{3})$



2) $(3, -\frac{3\pi}{4})$



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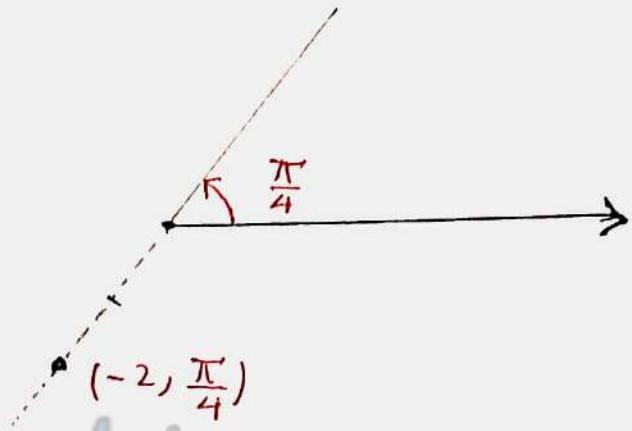
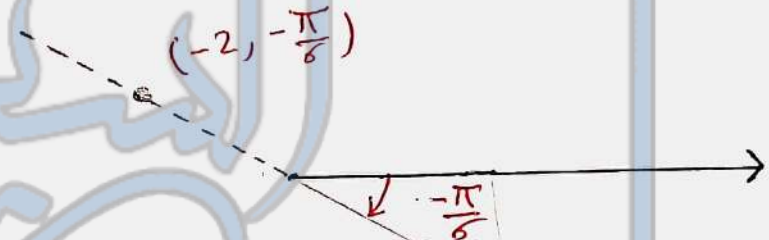
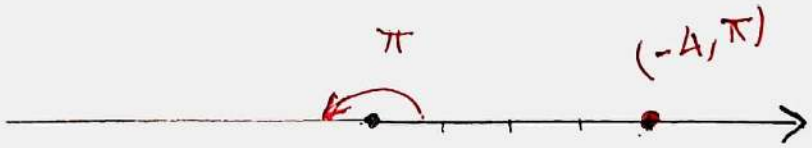
3) $(-2, \frac{\pi}{4})$



3) $(-2, -\frac{\pi}{6})$



4) $(-4, \pi)$



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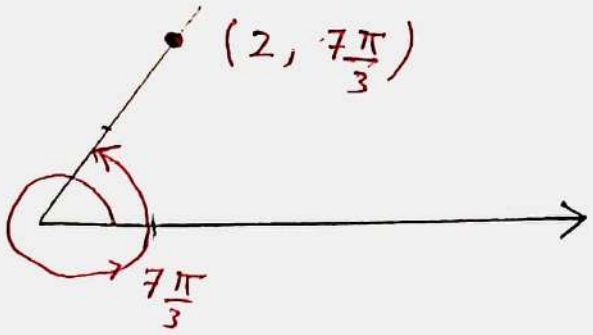
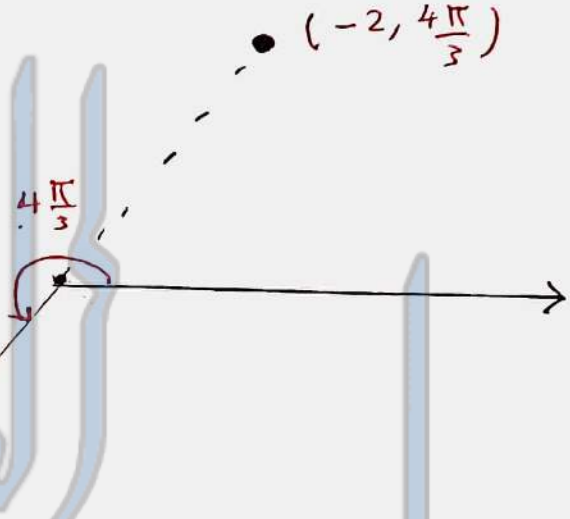
5) $(2, \frac{7\pi}{3})$



6) $(-2, \frac{4\pi}{3})$



Note: (r, θ) , $(r, \theta + 2n\pi)$, $n \in \mathbb{Z}$

$(-r, \theta + \pi)$ are polar coordinates of the same point.

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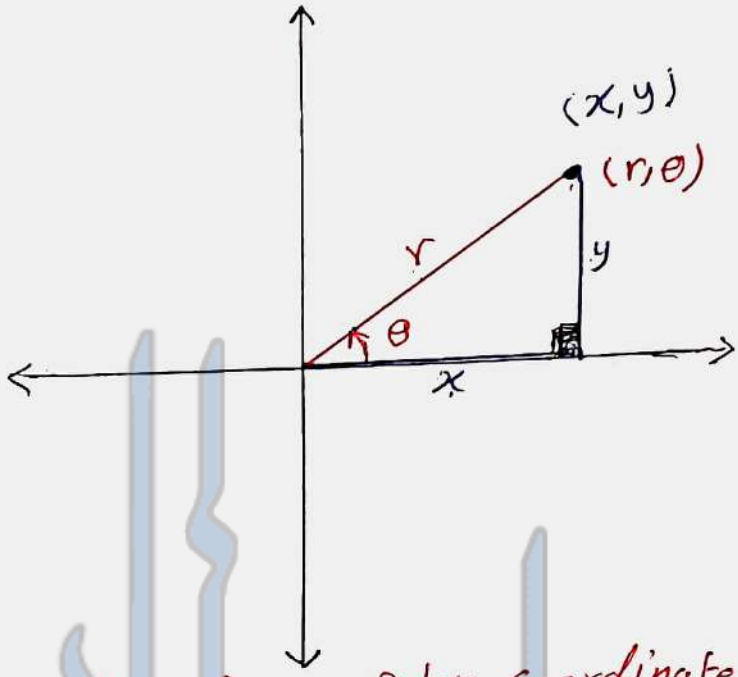
Relations between rectangular and polar coordinates

1) $x^2+y^2=r^2 \iff r=\pm\sqrt{x^2+y^2}$

2) $\cos\theta=\frac{x}{r} \iff x=r\cos\theta$

3) $\sin\theta=\frac{y}{r} \iff y=r\sin\theta$

4) $\tan\theta=\frac{y}{x}$



e.g: Convert the following points from polar coordinates into rectangular coordinates

$(r,\theta) \longrightarrow (x,y)$

$x=r\cos\theta$

$y=r\sin\theta$

1) $(4,\frac{\pi}{6})$

$x=4\cos\frac{\pi}{6}=4(\frac{\sqrt{3}}{2})=2\sqrt{3}$

$y=4\sin\frac{\pi}{6}=4(\frac{1}{2})=2$

$\therefore \underset{r\ \ \theta}{(4,\frac{\pi}{6})} \longrightarrow \underset{x\ \ \ y}{(2\sqrt{3},2)}$

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2) $(-6, \frac{2\pi}{3})$

$x = -6\cos(\frac{2\pi}{3}) = 3$

$y = -6\sin(\frac{2\pi}{3}) = -3\sqrt{3}$

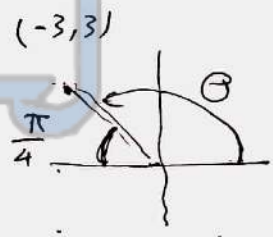
$\overset{r}{(-6}, \overset{\theta}{\frac{2}{3}\pi}) \longrightarrow \overset{x}{(3}, \overset{y}{-3\sqrt{3}})$

e.g: Find all polar coordinates of the given rectangular point.

Firstly we find the polar coordinates such that $r > 0$ and $0 \leq \theta < 2\pi$

$r = \sqrt{x^2+y^2}$ - and we find the principlal angle by $\tan^{-1}|\frac{y}{x}|$

1) $(-3,3)$



$r = \sqrt{x^2+y^2} = \sqrt{18} = 3\sqrt{2}$

$\tan^{-1}(|\frac{3}{-3}|) = \tan^{-1}(1) = \frac{\pi}{4}$

$\Rightarrow \theta = \frac{3\pi}{4} \Rightarrow \overset{x}{(-3}, \overset{y}{3}) \longrightarrow \overset{r}{(3\sqrt{2}}, \overset{\theta}{\frac{3\pi}{4}})$

all the other polar coordinates

$(3\sqrt{2}, \frac{3\pi}{4} + 2n\pi)$ or $(-3\sqrt{2}, \frac{7\pi}{4} + 2n\pi)$

$n \in \mathbb{Z}$

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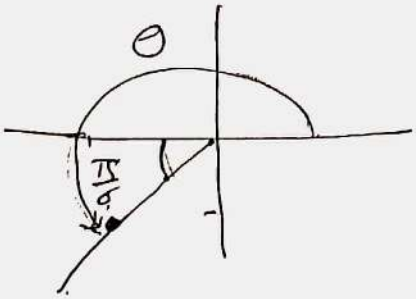
2) $(-\sqrt{3}, -1)$

$$r = \sqrt{(-\sqrt{3})^2 + (-1)^2} = 2$$

$$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$\theta = 7\frac{\pi}{6}$$



$$\therefore (\overset{x}{-\sqrt{3}}, \overset{y}{-1}) \longrightarrow (\overset{r}{2}, \overset{\theta}{7\frac{\pi}{6}})$$

all the other polar coordinates

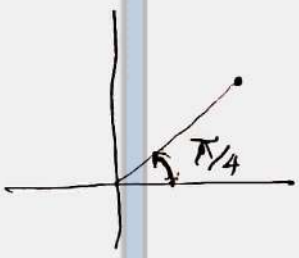
$$(2, 7\frac{\pi}{6} + 2n\pi) \text{ or } (-2, \frac{\pi}{6} + 2n\pi)$$

3) $(1, 1)$

$$r = \sqrt{1^2+1^2} = \sqrt{2}$$

$$\tan^{-1}\left|\frac{1}{1}\right| = \tan^{-1} 1 = \frac{\pi}{4}$$

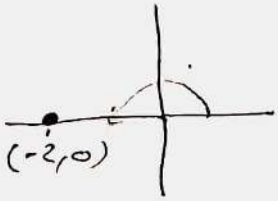
$$\Rightarrow \theta = \frac{\pi}{4}$$



$$(\overset{x}{1}, \overset{y}{1}) \longrightarrow (\overset{r}{\sqrt{2}}, \overset{\theta}{\frac{\pi}{4}})$$

$$\longrightarrow (\sqrt{2}, \frac{\pi}{4} + 2n\pi) \text{ or } (-\sqrt{2}, \frac{5}{4}\pi + 2n\pi) \quad n \in \mathbb{Z}$$

4) $(-2, 0)$



$r = 2, \quad \theta = \pi$

$(-2, 0) \longrightarrow (2, \pi)$

all other polar coordinates

$(2, \pi + 2n\pi)$ or $(-2, 2n\pi)$, $n \in \mathbb{Z}$

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Recall:

* If $a > 0$ then

$(x-h)^2 + (y-k)^2 = a^2$ is a circle with center (h, k) and radius (a)

* If $x^2 + y^2 + ax + by + c = 0$ represents a circle

then its center $(-\frac{a}{2}, -\frac{b}{2}) = (h, k)$

and radius $= \sqrt{h^2 + k^2 - c}$

For example $x^2 + y^2 - 2x + 6y - 1 = 0$

is a circle with center $(1, -3)$ and

radius $= \sqrt{(1)^2 + (-3)^2 - (-1)} = \sqrt{11}$

e.g: Convert the given equation from rectangular coordinates into polar coordinate

1) $x^2 + y^2 = 25$

circle with center $(0, 0)$ and radius $= 5$

$r^2 = 25 \rightarrow r = 5$

In polar coordinate $r =$ constant is a circle with center at the origin and radius $= |\text{constant}|$

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2) $x = a$, a constant (vertical line)

$r\cos\theta = a$ or $r = a\sec\theta$

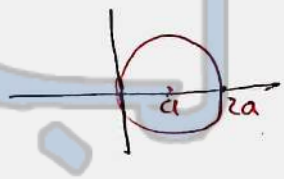
3) $y = b$, b constant (horizontal line)

$r\sin\theta = b$ or $r = b\csc\theta$

e.g: Convert the given equation from the rectangular system to Polar system

1) $x^2 + y^2 - 2ax = 0$, $a > 0$

(circle with center $(a, 0)$, radius $= a$)



$x^2 + y^2 - 2ax = 0$

$\downarrow$

$r^2 - 2ar\cos\theta = 0$

$\Rightarrow r^2 = 2ar\cos\theta$

$\Rightarrow r = 2a\cos\theta$

circle with center $(a, 0)$ and radius a

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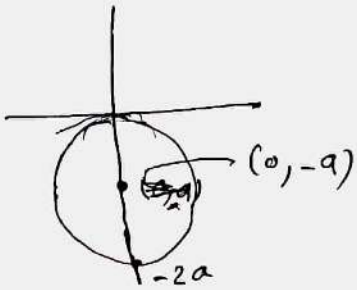
2) $x^2 + y^2 + 2ay = 0 \quad , a > 0$

(circle with center $(0, -a)$ and radius a)



$x^2 + y^2 + 2ay = 0$

$\downarrow$

$r^2 + 2ar\sin\theta = 0$

$\Rightarrow \boxed{r = -2a\sin\theta}$ circle with center $(0, -a)$ and radius a.

e.g.: Show that $\theta = \frac{\pi}{3}$ is a line that passes through the origin.

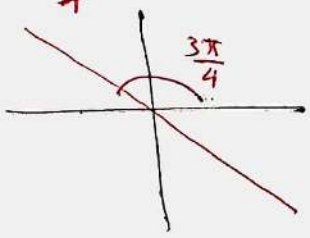
sol:

$\theta = \frac{\pi}{3} \Rightarrow \tan\theta = \tan\frac{\pi}{3} \Rightarrow \frac{y}{x} = \sqrt{3} \rightarrow y = \sqrt{3}x$

line passes through the origin

$\theta = \theta_0$, θ_0: constant line passes through the origin, its angle of inclination $= \theta_0$

$\theta = \frac{3\pi}{4}$



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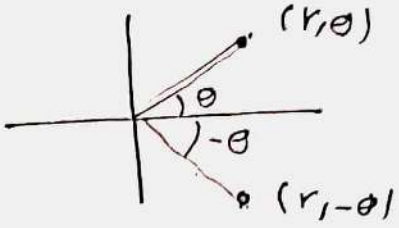
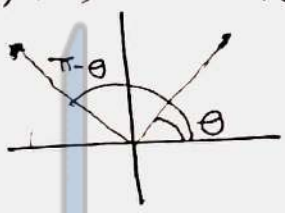
Symmetry Tests

A curve in polar coordinates is symmetric about

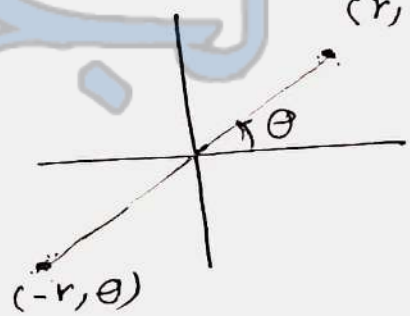
1) the x-axis if replacing (θ) by $(-\theta)$ gives an equivalent equation



2) the y-axis if replacing (θ) by $(\pi - \theta)$ gives an equivalent equation $(r, \pi-\theta)$ (r,θ)



3) the origin if replacing r by $(-r)$ [or θ by $\theta+\pi$] gives an equivalent equation.



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e.g: Test for symmetry the given polar curve

1) $r = a + a\sin\theta \quad , \quad a > 0$

* about x-axis:
replace θ by $(-\theta)$

$r = a + a\sin(-\theta)$

$r = a - a\sin\theta$ → Not symmetric about x-axis

* about y-axis
replace θ by $(\pi - \theta)$

$r = a + a\sin(\pi - \theta)$

$\Rightarrow r = a + a\sin\theta$ ⟹ symmetric about y-axis

* about the origin
replace r by $(-r)$

$-r = a + a\sin\theta$

$r = -a - a\sin\theta$ ⟹ Not symmetric about the origin.

2) $r^2 = 2a^2\cos 2\theta$

* about x-axis

$r^2 = 2a^2\cos(2(-\theta))$

$r^2 = 2a^2\cos 2\theta$ ⟹ sym. about the x-axis ✓

* about y-axis

$r^2 = 2a^2\cos(2(\pi - \theta))$

$r^2 = 2a^2\cos(2\pi - 2\theta)$

$\Rightarrow r^2 = 2a^2\cos(-2\theta)$

$\Rightarrow r^2 = 2a^2\cos(2\theta)$ ✓

* about the origin

$(-r)^2 = 2a^2\cos(2\theta)$

$\Rightarrow r^2 = 2a^2\cos 2\theta$ ✓

END

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Graphs, Arc length and Area in Polar Coordinates:

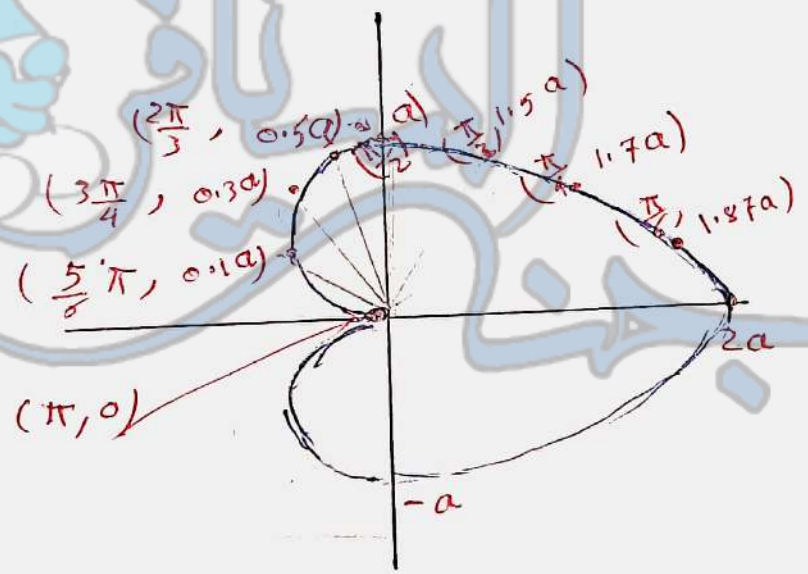
e.g: Sketch the graph of $r = a + a\cos\theta$, $a > 0$

Sol:

$r = a + a\cos\theta = a(1+\cos\theta)$

Show that this curve is symmetric about the x-axis

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
r	$2a$	$1.87a$	$1.7a$	$1.5a$	a	$0.5a$	$0.3a$	$0.1a$	0



Cardioid

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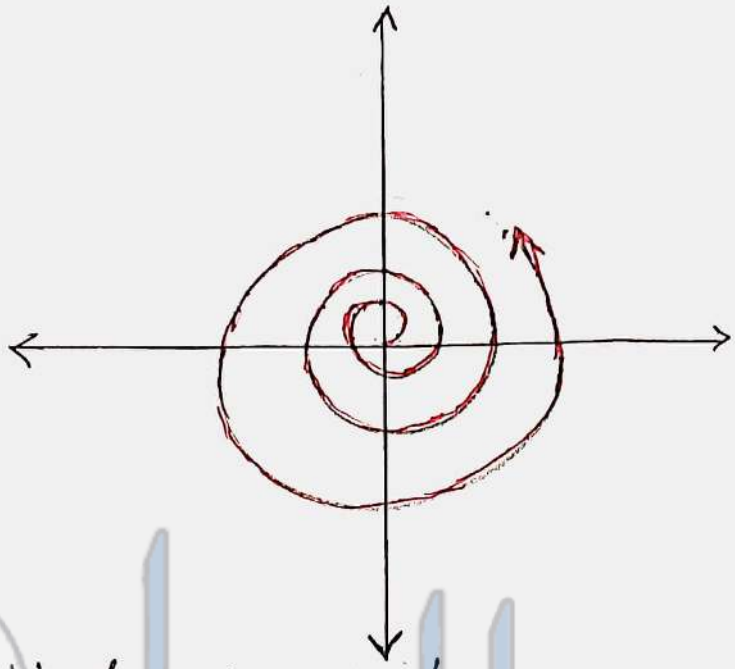
e.g : Sketch the graph of $r = \theta$, $\theta \geq 0$

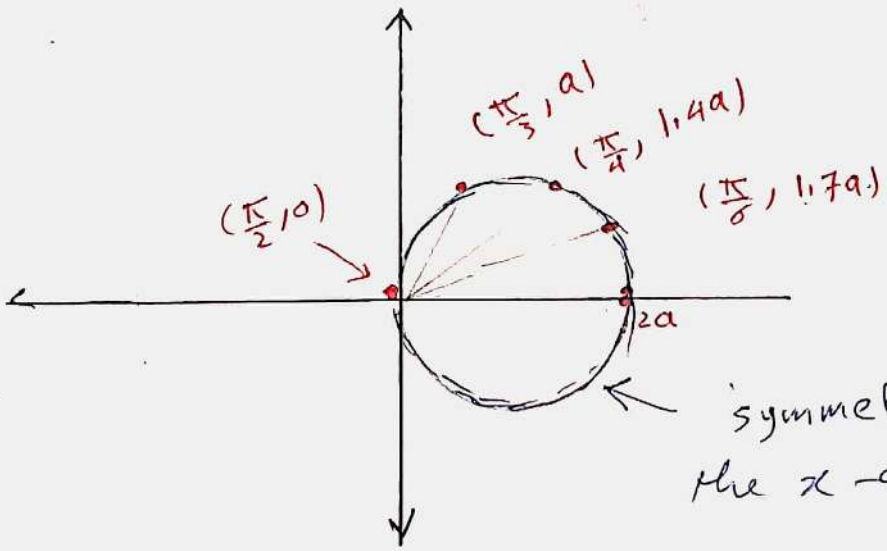


(Archimedean) spiral

e.g: Sketch $r = 2a\cos\theta$, $a > 0$ $\quad 0 \leq \theta \leq \pi$

Sol:

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
r	$2a$	$1.7a$	$1.4a$	a	0



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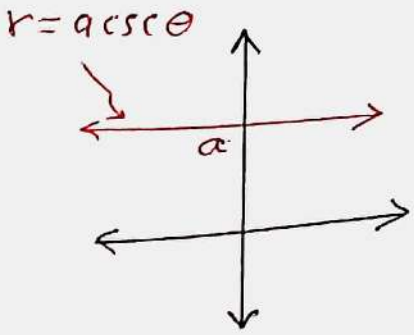
Basic Graphs in polar coordinates:

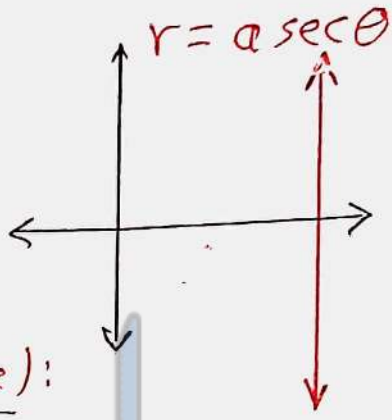
1 Lines:



* **Horizontal lines:**

$r\sin\theta = a$ or $r = a\csc\theta$

a: constant



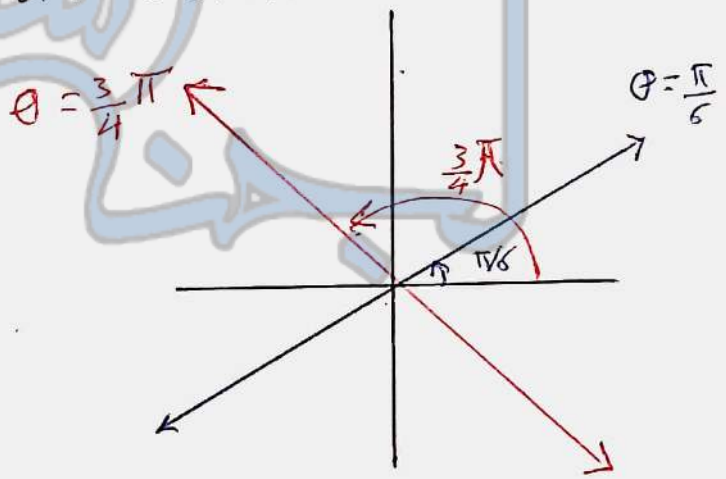
* **Vertical lines:**

$r\cos\theta = a$ or $r = a\sec\theta$

a: constant

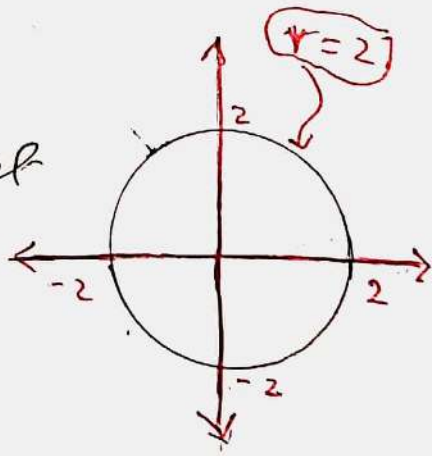
* **Lines pass-through the origin (the pole):**

$\theta = a$, a: constant



2 Circles: $a > 0$

* $r = a$ → circle with center at the origin and radius (a).



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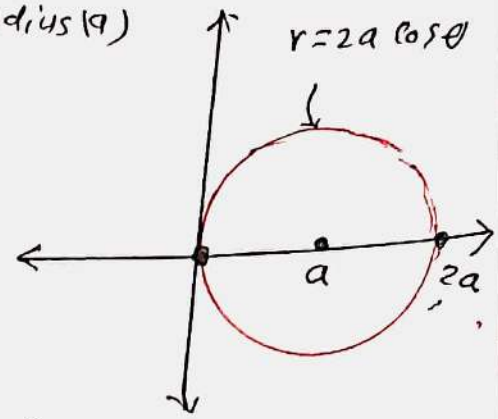
⇝ Circles touch the coordinate axes at the origin.

* $a > 0$

$r = 2a\cos\theta$, $0 \le \theta \le \pi$

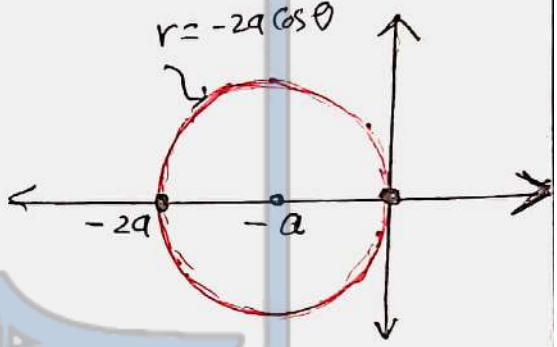
Circle with center $(a, 0)$ and radius (a)



* $a > 0$

$r = -2a\cos\theta$ $\quad 0 \le \theta \le \pi$ or $\frac{\pi}{2} \le \theta \le \frac{3\pi}{2}$

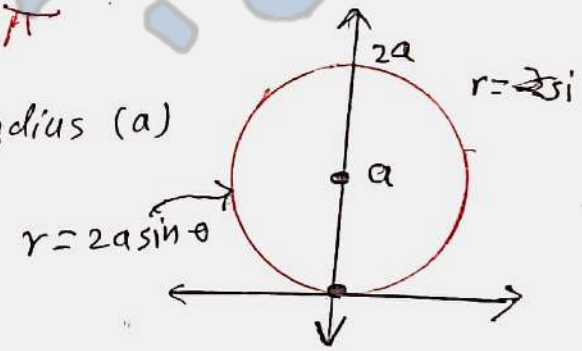
Circle with center $(-a, 0)$ and radius (a)



* $a > 0$

$r = 2a\sin\theta$ $\quad 0 \le \theta \le \pi$

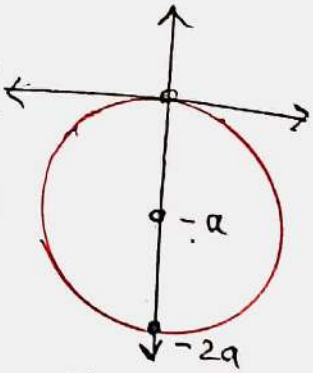
Circle with center $(0, a)$ and radius (a)



* $a > 0$

$r = -2a\sin\theta$ $\quad 0 \le \theta \le \pi$ or $\pi \le \theta \le 2\pi$

Circle with center $(0, -a)$ and radius (a)



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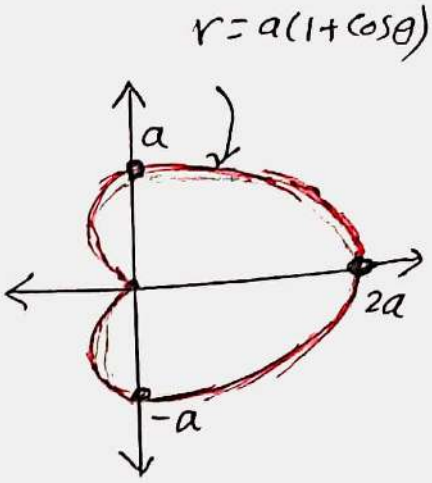
3 **Cardioids :**

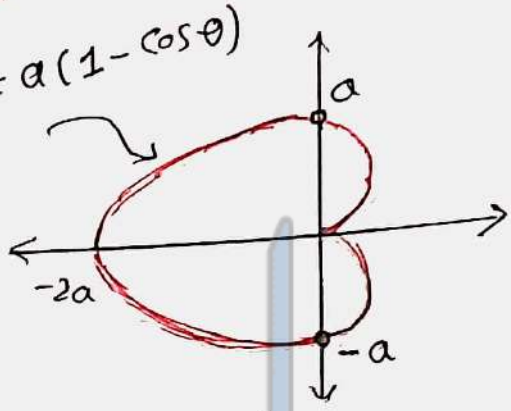
If $a > 0$

* $r = a + a\cos\theta$, $0 \le \theta \le 2\pi$

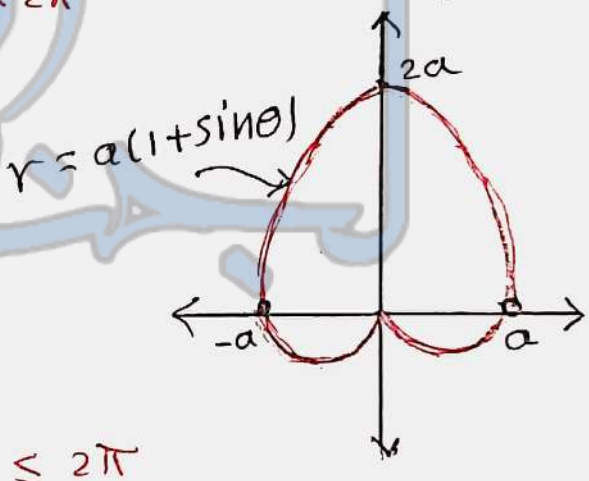
or $r = a(1 + \cos\theta)$.



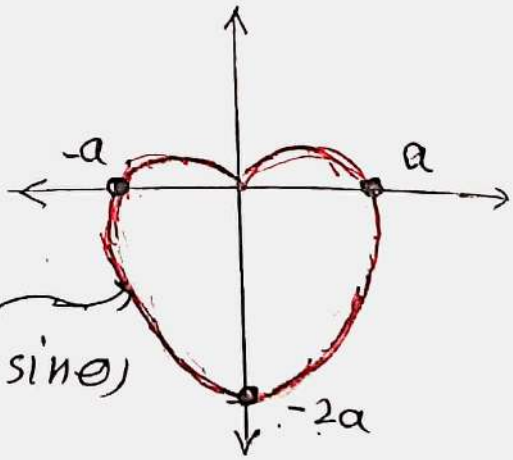
* $r = a - a\cos\theta$, $0 \le \theta \le 2\pi$



* $r = a + a\sin\theta$ $0 \le \theta \le 2\pi$



* $r = a - a\sin\theta$ $0 \le \theta \le 2\pi$



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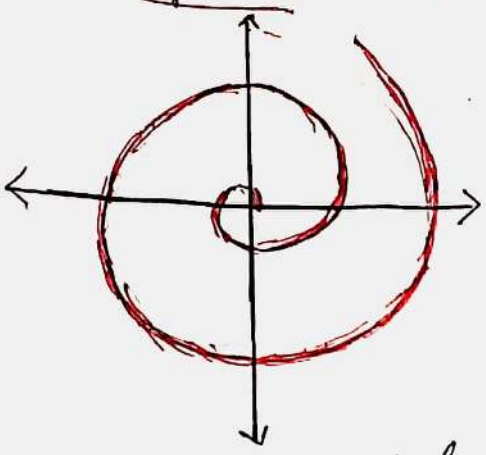
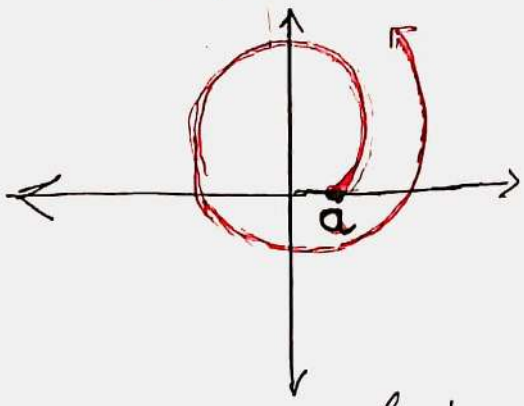
4 Spirals $(a>0, b>0, \theta \geq 0)$



Archimedean spiral

$r = a\theta$



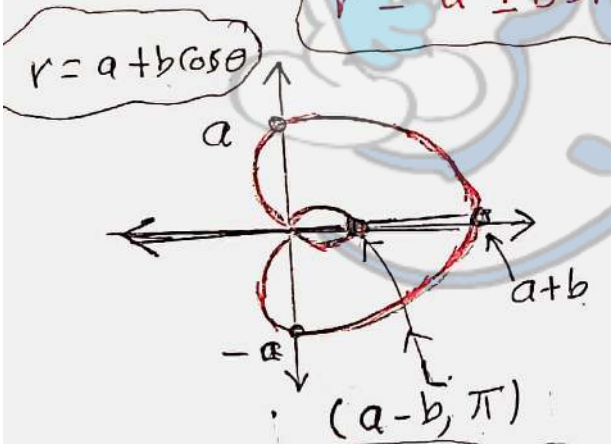
Logarithmic spiral

$r = a e^{b\theta}$

6

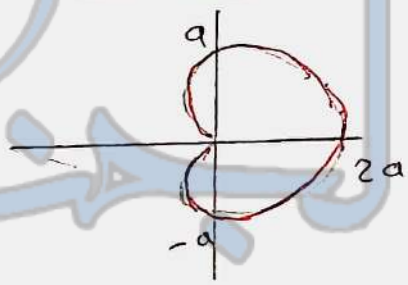
5 Cardioids and Limacon $(a>0, b>0)$

$r = a \pm b\sin\theta$, $r = a \pm b\cos\theta$



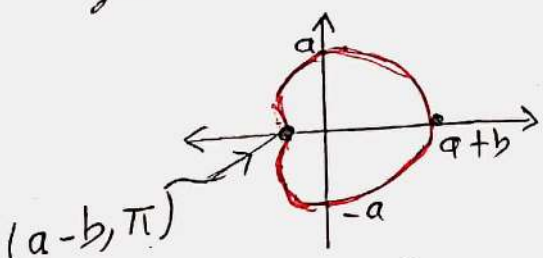
$\frac{a}{b} < 1$ (Loop)

Limacon with inner loop



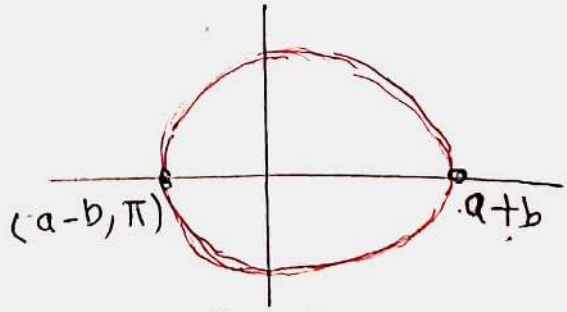
$\frac{a}{b} = 1$

Cardioid



$1 < \frac{a}{b} < 2$

Dimpled Limacon



$\frac{a}{b} \geq 2$

Convex Limacon

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Roses: $(a>0)$

$r = a\sin(n\theta)$ or $r = a\cos(n\theta)$, $n = 2,3,4,5,\dots$ $\Longrightarrow$ Rose

- with (n) petals if n is odd $(3,5,7,\dots)$
- with $(2n)$ petals if n is even $(2,4,6,\dots)$

The axes of the petals are equally spaced in the plane and the angle between any two adjacent axes equals $\frac{360^\circ}{\text{number of petals}}$

e.g. Sketch the graph of

1) $r = 3\cos(2\theta)$

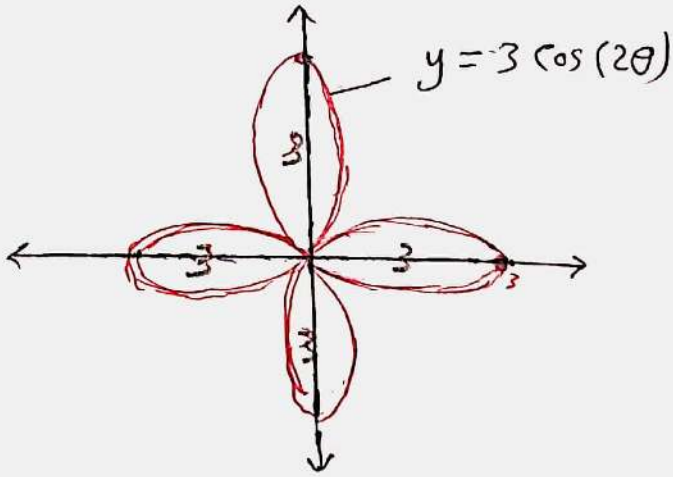
Sol:

$r = 3\cos(2\theta)$ is a rose with 4 petals

Each axis occurs when $\cos(2\theta) = 1$

$\Longrightarrow 2\theta = 0 \Longrightarrow \theta = 0$ one of the axes

the angle between any two adjacent axes is $\frac{360^\circ}{4} = 90^\circ$



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2) $r = 5\sin(3\theta)$

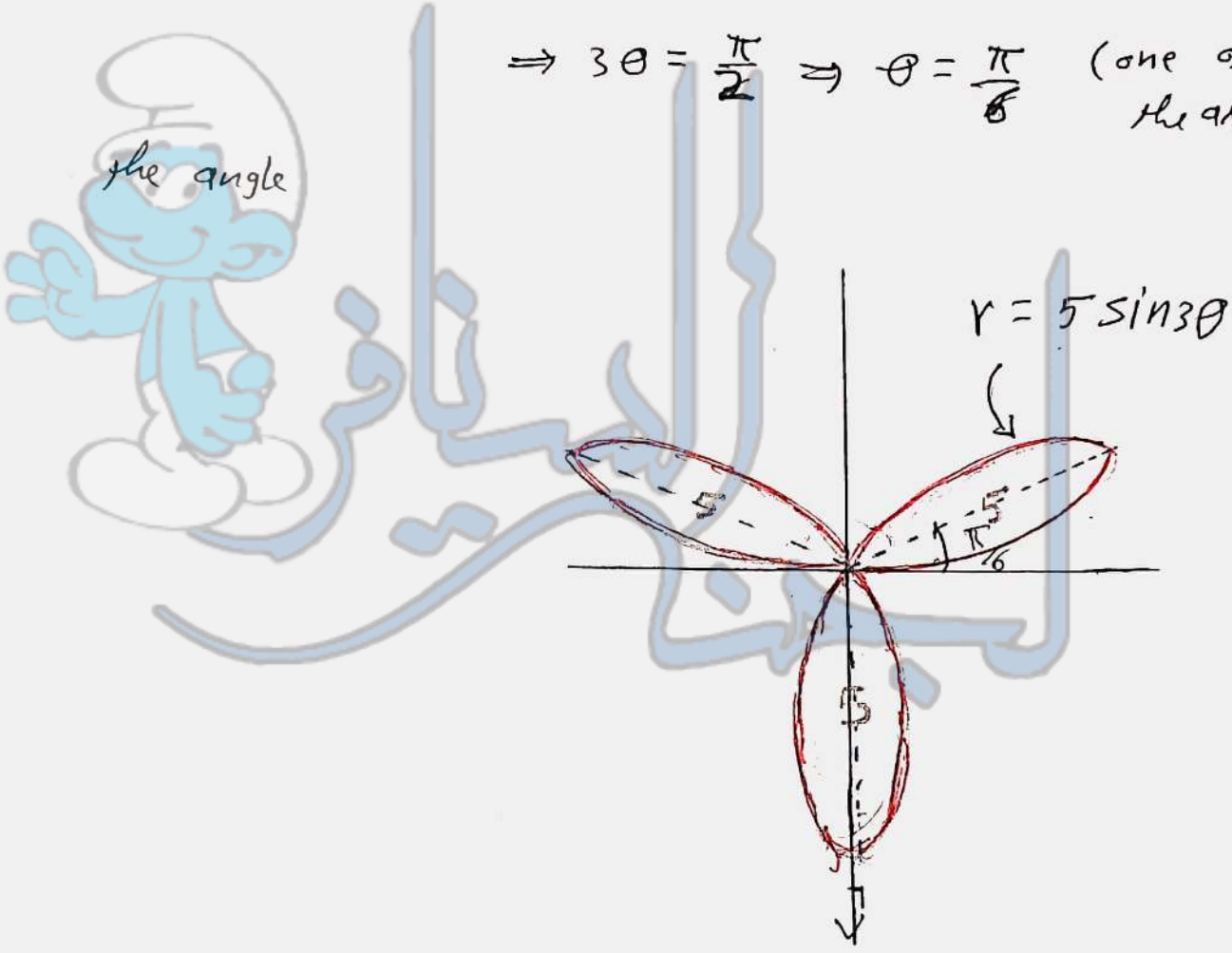
$r = 5\sin(3\theta)$ rose with 3-petals

the angle between any two adjacent axes of the petals is $\frac{360^\circ}{3} = 120^\circ$

Every axis of the petals occurs when $\sin(3\theta) = 1$

$\Rightarrow 3\theta = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{6}$ (one of the axes)

the angle



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Arc length in Polar Coordinates:

* The arc length of the polar curve $r = f(\theta)$, $\theta_1 \le \theta \le \theta_2$ is given by

$$L = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

e.g: Find the arc length of

1) $r = e^{2\theta}$, from $\theta = 0$ to $\theta = \pi$

Sol:

$$L = \int_0^{\pi} \sqrt{(e^{2\theta})^2 + (2e^{2\theta})^2}\, d\theta$$

$$= \int_0^{\pi} \sqrt{e^{4\theta} + 4e^{4\theta}}\, d\theta = \int_0^{\pi} \sqrt{5e^{4\theta}}\, d\theta$$

$$= \sqrt{5} \int_0^{\pi} e^{2\theta}\, d\theta$$

$$= \frac{\sqrt{5}}{2} e^{2\theta} \Big|_0^{\pi} = \frac{\sqrt{5}}{2}(e^{2\pi} - 1) \quad \text{L.U.}$$

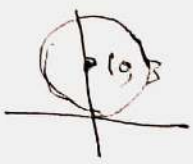
2) $r = 6\sin\theta$, $0 \le \theta \le \pi$

Sol:

$$L = \int_0^{\pi} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

$$= \int_0^{\pi} \sqrt{36\sin^2\theta + 36\cos^2\theta}\, d\theta$$

$$= \int_0^{\pi} 6\, d\theta = 6\pi \quad \text{L.U}$$



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Area in Polar Coordinates:

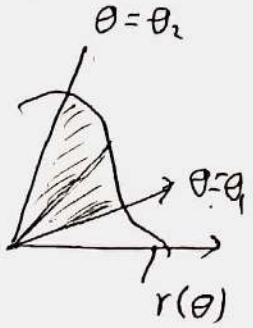
* If $r = f(\theta)$, $\theta_1 \le \theta \le \theta_2$ $(\theta_2 - \theta_1 \le 2\pi)$

and $f(\theta) \ge 0 \quad \forall \theta \in [\theta_1, \theta_2]$ or $(f(\theta) \le 0 \quad \forall \theta \in [\theta_1, \theta_2])$

then the area of the region bounded by

$r = f(\theta)$, $\theta = \theta_1$ and $\theta = \theta_2$ is given by

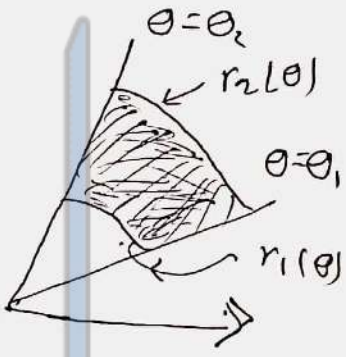
$$A = \frac{1}{2} \int_{\theta_1}^{\theta_2} r^2 \, d\theta$$



* If $r_1(\theta) \le r_2(\theta) \quad \theta \in [\theta_1, \theta_2]$

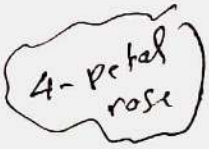
then the area between $r_1(\theta)$ and $r_2(\theta)$ $\theta \in [\theta_1, \theta_2]$ is given by

$$A = \frac{1}{2} \int_{\theta_1}^{\theta_2} [r_2^2(\theta) - r_1^2(\theta)] \, d\theta$$

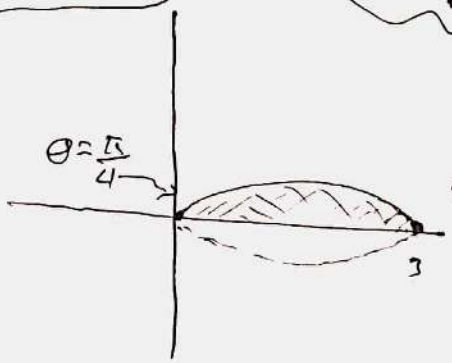


e.g: Find the area of one-petal of the rose $r = 3\cos 2\theta$.

Sol: (4-petal rose) (A: area of 1 petal)



$$A = \left[\frac{1}{2} \int_0^{\pi/4} r^2(\theta) \, d\theta \right] * 2$$



$$A = \int_0^{\pi/4} 9\cos^2 2\theta \, d\theta$$

$$= \frac{9}{2} \int_0^{\pi/4} (1 + \cos 4\theta) \, d\theta$$

$$= \frac{9}{8}\pi \quad \text{S.U}$$

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e.g: Find the area of region in the first quadrant that is inside the cardioid $r = 2 + 2\cos\theta$ and outside the circle $r = 3$.

Sol:

$$A = \frac{1}{2}\int_0^{\pi/3} [(2+2\cos\theta)^2 - 3^2]\,d\theta$$

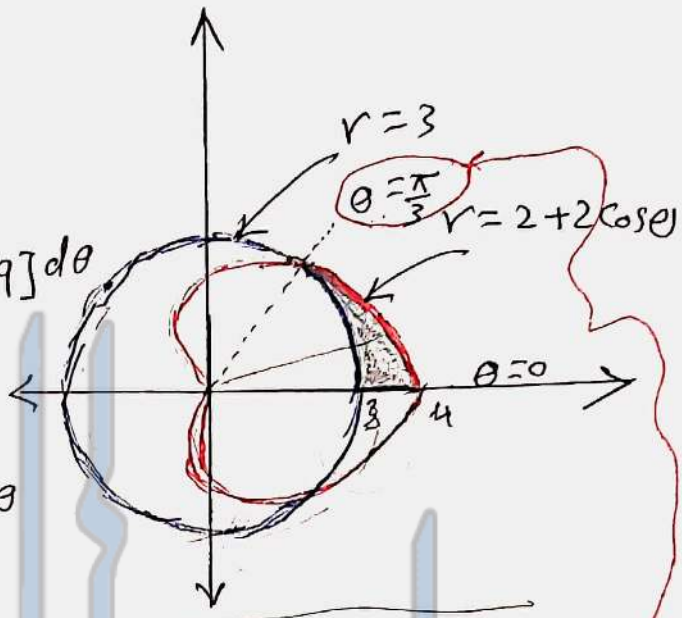
$$= \frac{1}{2}\int_0^{\pi/3} [4 + 8\cos\theta + 4\cos^2\theta - 9]\,d\theta$$

$$= \frac{1}{2}\int_0^{\pi/3} [8\cos\theta + 4\cos^2\theta - 5]\,d\theta$$

$$= \frac{1}{2}\int_0^{\pi/3} [8\cos\theta + 2(1+\cos 2\theta) - 5]\,d\theta$$

$$= \frac{1}{2}\int_0^{\pi/3} [8\cos\theta + 2\cos 2\theta - 3]\,d\theta$$

$$= \frac{1}{2}\left[\frac{9\sqrt{3}}{2} - \pi\right] \quad \text{S.U.}$$



$2 + 2\cos\theta = 3$
$\Rightarrow 2\cos\theta = 1$
$\Rightarrow \cos\theta = \frac{1}{2}$
$\Rightarrow \theta = \frac{\pi}{3}$

END

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